Canadian Health Measures Survey

Because your health matters!

Canadian Health Measures Survey (CHMS) Data User Guide: Cycle 6

December 2021

Statistique Canada Statistics Canada

Canada

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1. Introduction

The Canadian Health Measures Survey (CHMS) is a comprehensive, direct health measures survey developed to address important data gaps and limitations in existing health information. It is conducted by Statistics Canada in partnership with Health Canada (HC) and the Public Health Agency of Canada (PHAC). The results will provide comprehensive health information to advance health surveillance and research in Canada.

This document will help users work with and understand data from cycle 6 of the CHMS, which was obtained through the collection of directly measured indicators of health and wellness from January 2018 to December 2019, on a representative sample of 5,797 Canadians aged 3 to 79 years. The survey consisted of an in-home general health interview followed by a visit to a mobile examination centre (MEC), sometimes referred to as a mobile clinic. Reference laboratories and the MEC laboratory analyzed biological specimens for indicators of general health, chronic disease, infectious disease, nutritional status and environmental biomarkers.

This document also provides data users with information on the complexity of the data and any limitations that could have an impact on their use. It explains the methods and concepts used to collect the data at the household and the MEC. Subsequent sections of the document contain information about data processing and the creation of derived variables. Content regarding sampling and weighting methodology, and guidelines for the creation of tabulations have also been included to assist the data user. Quality assurance and quality control information is provided to describe characteristics of the data which might limit their usefulness or interpretation. The document concludes with a series of appendices which provide supporting information that will be helpful to users of the CHMS data files.

2. Important notes related to this document

2.1 Acronyms and abbreviations

Throughout this document, acronyms and abbreviations for terms associated with the Canadian Health Measures Survey (CHMS) are spelled out the first time they appear in a chapter, with the acronym/abbreviation put in brackets immediately afterwards. The next time that term appears in the chapter, only the acronym/abbreviation is used. A full list of these acronyms and abbreviations can be found in Appendix 1.

2.2 References and end notes

Background information such as references for research articles, definitions and procedural information is cited frequently throughout this document so that the user can find additional information related to the text. When this occurs, a small superscript number is put at the end of the text and users can consult the corresponding number in Chapter 13 – References and end notes, for additional information.

2.3 Survey documentation

Extensive documentation on the CHMS is available to all data users and the general public on the Statistics Canada website. A description of the CHMS, basic methodological information, links to the household and mobile examination centre (MEC) questionnaires and the CHMS bibliography can be accessed through the following link:
www23.statcan.gc.ca/imdb-bmdi/document/5071_D6_T9_V1-eng.htm.

In addition, Appendix 2 provides a list of documents that are available upon request. References to the documents in this list are made throughout the User Guide in order to help users identify which documents are related to a particular topic.

Chapters of significant importance in this User Guide are Chapter 11 – File usage, which outlines when each data file and its corresponding documentation are released as well as describes how to work with the data files, and Chapter 12 – Guidelines for tabulation, analysis and release.

Users wanting to obtain copies of the documents in the list or to obtain further information about the survey can contact Statistics Canada's Statistical Information Service (toll-free 1-800-263-1136; 514-283-8300; statcan.infostats-infostats.statcan@canada.ca; teletypewriter (TTY) 1-800-363-7629).

2.4 Updates to CHMS documentation

2.4.1 User guide

There are expected to be several different versions of the User Guide, corresponding to each of the main data release dates. In order to keep track of content changes among the different versions, a

summary of the type of information changed and location where the change was made will be provided in a table below.

Type of Change	Section #	Content Description	Release Date (where change first appears)
Added section	6.3.4	Blood and urine subsamples	June 2021
Added section	6.3.4.1	Fasted subsample	June 2021
Added section	6.3.4.2	Acrylamide, chromium and perfluoroalkyl substances (ACP) in blood subsample	December 2021
Added section	6.3.4.3	Trace elements (cadmium, lead, selenium, and mercuries) in blood subsample	December 2021
Added section	6.3.4.4	Environmental urine main subsample	December 2021
Added section	6.3.4.5	Tobacco metabolites subsample	December 2021
Added section	7.4.5	Bone health	September 2021
Added section	7.4.6	Vision health	December 2020
Added section	7.4.6.1	Visual acuity	December 2020
Added section	7.4.6.2	Visual field	December 2020
Added section	7.4.6.3	Retinal photography	December 2020
Added section	7.4.6.4	Intraocular pressure	December 2020
Added section	8.6	Adjustments for urine creatinine and specific gravity	December 2021
Updated section content	8.7	Added information on guidelines analysis concerning lab data	June 2021
Updated section content	8.7	Added information on conservative limit of detection (LODc)	December 2021
Added section	9.12	Weighting for selected subsamples	December 2020
Added section	9.12.1	Weighting for vision subsample data	December 2020
Added section	9.12.2	Weighting for blood and urine data	June 2021
Added section	9.12.2.1	Weighting for the fasted subsample	June 2021
Added section	9.12.2.2	Weighting for acrylamide, chromium and perfluoroalkyl substances in blood subsample	December 2021
Added section	9.12.2.3	Weighting for trace elements (cadmium, lead, selenium , and mercuries) in blood subsample	December 2021
Added section	9.12.2.4	Weighting for environmental urine main subsample	December 2021
Added section	9.12.2.5	Weighting for tobacco metabolites subsample	December 2021
Added section	9.12.3	Weighting for activity monitor subsample data	September 2021
Added section	9.12.4	Weighting for jumping	October 2021

		mechanography subsample data	
Added section	10.1.2	Blood draw and urine response rates	June 2021
Added section	10.1.3	Response rates for blood and urine subsamples	June 2021
Added section	10.1.3.1	Fasted subsample response rates	June 2021
Added section	10.1.3.2	Acrylamide, chromium and perfluoroalkyl substances in blood subsample response rates	December 2021
Added section	10.1.3.3	Trace elements (cadmium, lead, selenium, and mercuries) in blood subsample response rates	December 2021
Added section	10.1.3.4	Environmental urine main subsample response rates	December 2021
Added section	10.1.3.5	Tobacco metabolites subsample response rates	December 2021
Added section	10.1.4	Activity monitor response rates	September 2021
Added section	10.3.3.10	Vision data review	December 2020
Added section	10.3.3.11	Activity monitor data review	September 2021
Added section	10.3.3.12	Peripheral quantitative computed tomography data review	October 2021
Added section	10.3.3.13	Jumping mechanography data review	October 2021
Updated section content	Table in section 11.1	Updated the list of data files released and date of release	December 2021
Added section	11.1.4	Postal code file (pc)	December 2020
Added section	11.1.5	Vision data full sample (vis_full)	December 2020
Added section	11.1.6	Vision data subsample (vis_sub)	December 2020
Added section	11.1.7	Relationship (DYAD) data file (rel)	December 2020
Added section	11.1.8	Non-environmental lab data full sample (nel_full)	June 2021
Added section	11.1.9	Fasted subsample (fast_sub)	June 2021
Added section	11.1.10	Medication full sample - including dosage derived variables (med_full)	June 2021
Added section	11.1.11	Activity monitor data full sample (am_full)	September 2021
Added section	11.1.12	Activity monitor data subsample (am_sub)	September 2021
Added section	11.1.13	Activity monitor data per minute files (am_permin)	September 2021
Added section	11.1.14	pQCT data full sample (pqct_full)	October 2021
Added section	11.1.15	Jumping data subsample (jump_full)	October 2021
Added section	11.1.16	Environmental lab data full sample (el_full)	December 2021
Added section	11.1.17	Blood acrylamide, chromium and perfluoroalkyl substances in blood subsample file (acp_sub)	December 2021

Added section	11.1.18	Trace elements (cadmium, lead, selenium, and mercuries) in blood (environmental lab subsample) file (el2_sub)	December 2021
Added section	11.1.19	Environmental urine main subsample file (eu_sub)	December 2021
Added section	11.1.20	Tobacco metabolites subsample file (tob_sub)	December 2021
Added section	12.1.4	Medication data and derived variables	June 2021
Added section	12.2.2.2	Chemistry and lipid panels	June 2021
Added section	12.2.2.3	Analytical intervals	June 2021
Added section	12.2.2.4	Environmental data	September 2021
Added section	12.2.2.5	Activity monitor data for 3 to 5 years old	September 2021
Added section	12.2.2.6	Vitamin D standardization	September 2021
Updated section content	12.2.4	Updated information on SAS and SUDAAN	June 2021
Updated section content	Appendix 2	Added information on a new document available: Relationship (DYAD) data files – Documentation	December 2020
Updated section content	Appendix 5	Added exclusion information	October 2021
Added section	Appendix 6	Medication Classification System	June 2021
Added section	Appendix 7B	CHMS Combined Cycles 5 and 6 (sites 1 to 10) Vision Subsample Response Rates by age group and sex	December 2020
Added section	Appendix 7C	CHMS Cycle 6 Blood Draw and Urine Response Rates by age group and sex	June 2021
Added section	Appendix 7D	CHMS Cycle 6 Fasted Subsample Response Rates by age group and sex	June 2021
Added section	Appendix 7E	CHMS Cycle 6 Activity Monitor Response Rates by age group and sex	September 2021
Added section	Appendix 7F	CHMS Combined Cycles 5 (all) and 6 (sites 1-11) Jumping Subsample Response Rates by age group and sex	Octobre 2021
Added section	Appendix 7G	CHMS Cycle 6 Acrylamide, Chromium and Perfluoroalkyl Substances (ACP) in Blood Subsample Response Rates by age group and sex	December 2021
Added section	Appendix 7H	CHMS Cycle 6 Trace Elements (Cadmium, Lead, Selenium and Mercuries) in Blood Subsample Response Rates by age group and sex	December 2021

Added section	Appendix 7I	CHMS Cycle 6 Environmental Urine Main Subsample Response Rates by age group and sex	December 2021
Added section	Appendix 7J	CHMS Cycle 6 Tobacco Metabolites Subsample Response Rates by age group and sex	December 2021
Added section	Appendix 8	Activity Monitor Reseach	September 2021

2.4.2 Data files

If major errors are discovered in the data files after they have been disseminated, the files will be corrected and re-released. An errata is also available to document other inconsistencies discovered in the data files and documentation after dissemination.

3. Canadian Health Measures Survey (CHMS) background and objectives

3.1 CHMS background

Policy makers, researchers and health professionals from many fields have expressed a need for an on-going national and comprehensive source of accurate health measures to assist them in addressing the health needs of all Canadians.

In 2003, Health Canada (HC) and the Public Health Agency of Canada (PHAC) supported Statistics Canada in obtaining funding for a direct measures health survey to address longstanding limitations within Canada's health information system. This support was announced in the 2003 Budget as part of an extension of the Health Information Roadmap Initiative and permanent funding was secured in the 2008 Budget.

Collection for cycle 1 of the survey took place from March 2007 to February 2009, with dissemination of the main results occurring from January 2010 until April 2011. Collection for cycle 2 of the survey took place from August 2009 to November 2011, with dissemination of the main results occurring from September 2012 until April 2013. Cycle 3 collection occurred from January 2012 until December 2013 with dissemination of the main results occurring from October 2014 to October 2015. Cycle 4 collection occurred from January 2014 to December 2015 with dissemination of the main results occurring from October 2016 to August 2017. Cycle 5 collection occurred from January 2016 to December 2017 with dissemination of the main results occurring from October 2018 to June 2020. Cycle 6 collection occurred from January 2018 to December 2019 with dissemination of the main results occurring from October 2019 to December 2021.

The information collected will create national baseline data on the extent of such major health concerns as obesity, hypertension, cardiovascular disease, exposure to infectious diseases, and exposure to environmental contaminants. In addition, the survey will provide indications about illness and the extent to which many diseases may be undiagnosed among Canadians. Data from the CHMS will enable us to determine relationships between health status and risk factors, and to explore emerging public health issues.

A large amount of data is collected and disseminated from each cycle of the survey, as summarized in the CHMS bibliography. In addition, researchers can apply to the CHMS Biobank to obtain blood, urine and DNA samples to do their own research.

3.2 CHMS objectives

The main objectives of the CHMS are to:

- explore emerging public health issues and new measurement technologies
- establish national baseline data on major health concerns
- determine relationships among risk factors, protection practices and health status
- assess the validity of self- and proxy-reported information

- assemble a nationally representative sample for storage in a biobank

4. Survey approval

4.1 Authority

The 2008 Budget provided on-going funding for the Canadian Health Measures Survey (CHMS) to be conducted by Statistics Canada in partnership with Health Canada (HC) and the Public Health Agency of Canada (PHAC).

The survey falls under the authority of the federal *Statistics Act*. Statistics Canada may only collect health information for statistical research purposes and may not use or disclose individual participant information for any other purpose without the written consent of participants.

4.2 Ethical protocols and privacy standards

All processes of the CHMS were reviewed and approved by the HC and PHAC Research Ethics Board to ensure that internationally recognized ethical standards for human research were met and maintained. In addition, protocols were developed through extensive consultation with recognized experts and were performed by accredited health professionals in conformance with universal precautions.

Several meetings were also held with the Office of the Privacy Commissioner of Canada and with provincial privacy commissioners regarding CHMS protocols to ensure that participants' privacy rights were protected. A full Privacy Impact Assessment (PIA) was completed for the CHMS and reviewed through the Office of the Privacy Commissioner of Canada – the authority that continues to provide oversight to the CHMS as well as a complaint route and redress mechanism for CHMS participants. This assessment is updated regularly to reflect any changes in the CHMS.

Participation in this survey was voluntary. The voluntary nature of the survey was stated in the introductory letter, brochure, video, *Information and Consent Booklet*, and on the Statistics Canada website (www.statcan.gc.ca\chms). The documents also emphasized the safety and standards used in all tests. CHMS staff answered any questions respondents had regarding the risks of participating in the tests and the use of their data in an interactive consent process throughout the household health interview and the visit to the mobile examination centre.

5. Survey content

Consultation on proposed content, data requirements and operational considerations has been on-going with many groups, individuals and agencies such as Health Canada (HC), the Public Health Agency of Canada (PHAC), and several internal and external expert advisory groups and committees.

Cycle 6 was made up of a household interview and a visit to a mobile examination centre (MEC). The household interview included general demographic information and an in-depth health questionnaire. The MEC visit included not only physical measures tests but also the collection of blood and urine samples from respondents. Some samples were analyzed in a laboratory at the MEC, such as the complete blood count (CBC), which includes platelets, red blood cell count and white blood cell count. The remaining samples were analyzed at seven external reference laboratories. Respondents were also asked to wear an activity monitor for the seven days following their visit to the MEC and to provide a second urine sample sometime within the seven days following their visit to the MEC.

The Canadian Health Measures Survey (CHMS) aims to keep content fairly consistent between a pair of cycles to allow for more opportunities to combine the data, providing a larger sample size to work with and more precise estimates for small prevalence. As such, the content for cycles 1 and 2 is quite similar as well as the content for cycles 3 and 4 and the following cycles 5 and 6. More significant changes were made to the survey content between cycles 2 and 3 and between cycles 4 and 5 in order to obtain data on some new measures of public health interest.

The main changes between cycles 4 and 5 were the removal of spirometry, hearing tests, skin pigmentation, tap water sample and indoor air sample and the addition of physical fitness tests (modified Canadian Aerobic Fitness Test, sit and reach), musculoskeletal health tests (jumping mechanography, peripheral quantitative computed tomography or pQCT), vision tests (visual acuity, intraocular pressure, visual field and retinal photography), hair sample collection, saliva sample collection as well as a home urine sample. Several changes were made to the questionnaires where relevant questions were removed or added as needed to accompany the physical measures tests. Finally, there were also a large number of new and changed laboratory measures, particularly in relation to the home urine sample and environmental contaminant exposure in hair, blood and urine.

In cycle 6, the hair and saliva samples collection were removed. Toxoplasmosis test was also removed. The vision prescription questionnaire was removed from the household interview and collected only at the MEC. Also, a new questionnaire on recreational cannabis use (CAN block) was added to the screening component at the MEC.

A content summary document for cycles 1 to 8 of the CHMS is available upon request.

6. Sample design

6.1 Target population

The Canadian Health Measures Survey (CHMS) covers the population 3 to 79 years of age living in the ten provinces. Excluded from the survey's coverage are: persons living in the three territories; persons living on reserves and other Indigenous settlements in the provinces; full-time members of the Canadian Forces; the institutionalized population and residents of certain remote regions. Altogether these exclusions represent approximately 3% of the target population.

6.2 Sample size and allocation

To produce reliable estimates at the national level by age group and sex, it was determined that this survey must be carried out on a sample of at least 5,700 persons over a two-year period to allow estimates for the following 11 groups: ages 3 to 5 (both sexes combined), ages 6 to 11 (males and females), 12 to 19 (males and females), 20 to 39 (males and females), 40 to 59 (males and females) and 60 to 79 (males and females).

6.3 Sampling frames and sampling strategy

To meet the requirements of the CHMS, a multistage sampling strategy was used. An overview of the sampling strategy is provided below; for further information about the sampling strategy refer to Mather (2020)[1].

6.3.1 Sampling of collection sites

The sample of dwellings is selected within collection sites in such a way that respondents are able to travel to the MEC within a reasonable period of time. A collection site is a geographic area with a population of at least 10,000 and a maximum respondent travel distance of 50 kilometres in urban areas and 75 kilometres in rural areas. Areas not meeting these criteria were excluded. Using Census geography, 425 collection sites were created. The geographic units used to define the sites were also grouped with respect to provincial boundaries, census metropolitan-area boundaries, health regions and population density criteria.

Although only national estimates were required, the collection sites were stratified into five regions to ensure that the allocation of the sample was spread across the country. The regions identified, based on Statistics Canada’s standard regional boundaries, were British Columbia, the Prairies (Alberta, Manitoba and Saskatchewan), Ontario, Quebec and the Atlantic provinces (Newfoundland and Labrador, Prince Edward Island, Nova Scotia and New Brunswick).

A large number of collection sites (with few respondents) would be the recommended sampling strategy because it would help to optimize the precision of the estimates. However, the logistical and cost constraints associated with the use of a MEC restricted the number of collection sites to 16. The number of sites selected by region is provided in Table 6.1.

Table 6.1 Selection of collection sites for the CHMS, by region – Cycle 6

Region	Estimated target population, all ages based on 2011 Census	Number of sites in region	Number of sites allocated
Atlantic	**2,152,533**	58	2
Quebec	**7,535,464**	91	3
Ontario	**12,666,189**	106	6
Prairies	**5,963,543**	114	3
British Columbia	**4,247,924**	56	2
Total	**32,565,653**	**425**	**16**

The method of sorting the collection sites prior to sampling was modified for cycle 6 to improve the selection of collection sites that belong to census metropolitan area (CMA) or not. A CMA is an area consisting of one or more adjacent municipalities centering on a large urban area (known as an urban core). The urban core must have a population of at least 100,000 to form a CMA. Changes were carried out to the sorting method for all regions except for the Atlantic region. The collection sites for the Atlantic region used the same sorting method as for previous cycles where the collection sites were first sorted by province, then CMA and finally size of the population. For the Prairies, the collection sites of Saskatchewan and Manitoba were grouped before sorting. And then to ensure that at least one CMA site was selected, the collection sites in Alberta were sorted by CMA and then non-CMA and the collection sites in Manitoba and Saskatchewan were sorted in the reverse order – by non-CMA and then CMA. In the Quebec region (for the CMA of Montreal) and in the Ontario region (for Toronto and all CMAs that had 2 or more collection sites), the collection sites within their given CMA were grouped together during the sorting. The collection sites were then sampled systematically with a probability of selection proportional to the size of their population in each of the 5 regions of Canada. While not every province would have a collection site, the CHMS sites were chosen to represent the Canadian population, east to west, with larger and smaller population densities.

Data collection at the 16 sites was carried out sequentially over two years. The sites were ordered to take into account seasonality and the temporal effect, subject to operational and logistical constraints. The temporal effect was corrected by distributing uniformly the number of sites per region between the first and the second year.

The specific collection sites by city/municipality are listed in Appendix 3. Note that as each cycle of the CHMS was designed to produce national estimates only, it is not recommended to do analysis at lower geographic levels for a single survey cycle as it could result in either extreme sampling variability or unstable estimates of the sampling variability.

6.3.2 Dwelling sampling

The most recent version of the dwelling universe file (DUF) of the Household Survey Frame (HSF) was used to select dwellings. Using the date of birth of household members from the most recent

version of the socio-economic indicators file (SEF) of the HSF, as well as more current information from other administrative sources, within each collection site, dwellings with known household composition at the time of sample selection were stratified by the occupants' age at the time of the survey. Age was determined based on the starting date of data collection at each site. Six age-group strata[2] were created, corresponding to the six CHMS age groups (3 to 5, 6 to 11, 12 to 19, 20 to 39, 40 to 59 and 60 to 79 years) as follows:

- 3 to 5 stratum dwellings where at least one 3-to-5-year-old is present, else,
- 6 to 11 stratum: dwellings where at least one 6-to-11-year-old is present, else,
- 12 to 19 stratum: dwellings where at least one 12-to-19-year-old is present, else,
- 60 to 79 stratum: dwellings where at least one 60-to-79-year-old is present, else,
- 20 to 39 stratum: dwellings where at least one 20-to-39-year-old is present, else,
- 40 to 59 stratum: dwellings where at least one 40-to-59-year-old is present, else,
- "Other" strata: dwellings not included in the above-mentioned strata, such as 1) vacant dwellings at the time of sampling, 2) dwellings with people outside the CHMS age ranges based on household composition at the time of sampling or 3) had missing stratum information (missing dates of birth of the members living in the household) .

Each stratum had a high probability of having dwellings inhabited by persons in the desired age groups, whether they were the same occupants or were replaced by a similar household. Within each site, a simple random sample of dwellings was selected in each stratum. The sample size was allocated in each stratum so that, combined with the strategy for sampling participants in the survey, an equal number of respondents by age group would be obtained. Each selected dwelling was contacted to draw up a current list of the members of the household, and this list was then used to select survey respondents.

Table 6.2 shows the distribution of the number of dwellings selected per site. In all, 9,960 dwellings were selected, with an average of 623 dwellings per site.

Table 6.2 CHMS cycle 6 — Number of dwellings selected per site

Site	Number of
1	650
2	599
3	649
4	699
5	625
6	650
7	600
8	710
9	650
10	575
11	585
12	600
13	574

14	675
15	549
16	570
TOTAL	**9 960**

6.3.3 Respondent sampling

Different selection probabilities by age group and sex were used to ensure that the sampling targets at the MEC were attained. The sampling factors shown in Table 6.3 were used for the first site in cycle 6. Changes were made to these sampling factors throughout the cycle to modify the number of people selected by age group and sex. It was necessary to update the factors to react to how collection was going for each group. The dwellings selected were contacted to obtain a list of current household members. In each dwelling, one or two people were selected, depending on the household composition. Two people were selected from households with children aged 3 to 11: one child randomly selected from those aged 3 to 11 and a second person aged 12 to 79. If no 3 to 11 year olds were living in the household, only one person was selected from the household members aged 12 to 79. The sampling factors for the selection of people aged 12 to 79 were designed to avoid large person-sampling weights. Since some age groups have a weight that is up to 10.5 times higher than that of other age groups, it is possible that a selected person would have a very high sampling weight when there are many household members in a dwelling. Therefore when a specified minimum number of people aged 12 to 79 are living in a household, the weight for each person is reset to 1. In such cases, each household member has an equal chance of being selected. A careful balance of the parameters required for the respondent sampling strategy was obtained through studies and simulations.

Table 6.3 CHMS cycle 6 – Selection weight multiplicative factors for the person-level sampling strategy, by sex and age group

Sex	Sampling factors for 3 to 11		Sampling factors for 12 to 79			
	3 to 5	6 to 11	12 to 19	20 to 39	40 to 59	60 to 79
Male	4.50	2.80	6.50	1.20	0.95	5.00
Female	4.50	2.80	6.80	0.65	0.95	3.20

6.3.4 Blood and urine subsamples

Subsamples of the survey respondents are selected for different laboratory tests on the blood and urine specimens collected. As some blood lipid tests require that the respondents fast, approximately half of the respondents are asked to fast prior to their MEC appointment and are usually scheduled to attend the MEC in the morning. Some environmental chemicals are also measured on subsamples of survey respondents due to the high cost of performing these tests on the entire sample. All of the blood and urine subsamples were selected independently; that is, without consideration of who was selected for the other subsamples. This means that a specific respondent can be selected for any number of subsamples. The CHMS content summary document indicates which tests have been done on each subsample.

6.3.4.1 Fasted subsample

Each sampled dwelling was randomly flagged to indicate whether a respondent should fast prior to the MEC appointment. It required that respondents fast for at least 10 hours, whereas shorter eating restrictions were imposed on those with non-fasted appointments. Pregnant women, people with diabetes, youth less than 6 years old and other special cases were not asked to fast, even if the dwelling was flagged to be fasted. This random allocation reduced the potential for bias, which could occur if respondents were given the option to fast. During collection, the sampling rates were adjusted to obtain approximately half of the sample where respondents were selected to fast and were actually fasted prior to the MEC appointment.

6.3.4.2 Acrylamide, chromium and perfluoroalkyl substances (ACP) in blood subsample

Respondents aged 3 to 79 attending the MEC were randomly selected to be included or excluded from the ACP subsample. The targeted subsample sizes were 250 respondents by sex for the following age groups: 3 to 5, 6 to 11, 12 to 19 as well as 500 respondents by sex for the 20 to 79 years old, for a total of 2,500 respondents.

6.3.4.3 Trace elements (cadmium, lead, selenium, and mercuries) in blood subsample

Respondents aged 3 to 5 and 20 to 79 attending the mobile examination centre (MEC) were selected to be included from the cadmium, lead, selenium and mercury (total, inorganic and methyl) blood subsample. Half of the 6 to19 years old were selected for the test. The targeted subsample sizes were 250 respondents by sex for the following age groups: 3 to 5, 6 to 11, 12 to 19 as well as 500 respondents by sex for the 20 to 79 years old, for a total of 4,500 respondents.

6.3.4.4 Environmental urine main subsample

Respondents aged 3 to 79 attending the MEC were randomly selected to be included or excluded from the environmental urine main subsample. The targeted subsample sizes were 250 respondents by sex for the following age groups: 3 to 5, 6 to 11, 12 to 19 as well as 500 respondents by sex for the 20 to 79 years old, for a total of 2,500 respondents.

6.3.4.5 Tobacco metabolites subsample

Amongst MEC respondents aged 6 to 79, all those who were identified as a smoker or as a non-smoker exposed to smoke, as well as some of those identified as a non-smoker non-exposed to smoke, were selected for inclusion in the urine tobacco metabolites subsample. The goal was to have a total of 2,800 respondents for the subsample, with non-smoker non-exposed to smoke respondents making up the difference after selecting all those identified as smoker or as non-smoker exposed to smoke.

7. Data collection

7.1 Preparation for collection

7.1.1 The Canadian Health Measures Survey (CHMS) team

The CHMS team is a diverse, well-trained, experienced group of individuals. The group can be subdivided into three sub-teams: field team, mobile examination centre (MEC) team, and head office staff. Each of these three sub-teams was responsible for specific portions of the survey.

7.1.1.1 Field team

Unlike previous CHMS cycles, the field team assigned to CHMS collection was recruited, trained and managed through the nearest Statistics Canada Regional Office. It was comprised of Statistics Canada household interviewers and a senior interviewer. The household interviewers were primarily responsible for contacting selected households, conducting the household interview, explaining the MEC portion of the survey to respondents and attempting to secure their participation at the MEC. The senior interviewer was responsible for conducting data quality assurance activities for the household component, overseeing the non-response follow-up and monitoring the household collection rates.

7.1.1.2 Mobile examination centre (MEC) team

The CHMS MEC team consisted of health professionals responsible for various components of the physical measures testing and a site manager who oversaw the day-to-day operation of the MEC. The health measures specialists (HMSs) were responsible for performing the majority of the physical health measures (e.g., blood pressure, anthropometry, fitness testing, jumping mechanography, and vision testing). The medical radiation technologists (MRTs) were responsible for performing the bone mineral density test (pQCT) along with blood pressure, anthropometry, and vision testing. The laboratory technologists/phlebotomists conducted the specimen collection (blood and urine), performed the complete blood count (CBC) analysis and processed the biological samples for storage and shipment to the reference labs. In addition to the health professionals, the MEC team also consisted of administrative staff who worked at the reception desk and the appointment booking desk, as well as a site logistics officer who took care of the maintenance of the trailers.

7.1.1.3 Head office staff

The CHMS staff at head office processed data, monitored data collection response rates and data quality, provided human resource support and conducted periodic site visits to ensure staff were following correct protocols. In addition, head office staff prepared and mailed the respondent's report of selected laboratory tests, and provided information about the survey to respondents, the public and the media. Finally, a medical advisor followed-up with respondents about critical or sensitive results.

7.1.2 The mobile examination centre (MEC)

Two sets of trailers were used to conduct the physical measures and laboratory components of the CHMS. Each MEC was comprised of three trailers, the administrative trailer, the clinic trailer and the laboratory trailer. The trailers had several different rooms including a reception area, restrooms, an administrative office, two fitness testing areas, a vision testing area, a pQCT testing area, a screening area, an anthropometry area, a phlebotomy area and a laboratory.

Using MECs provided several benefits over a fixed examination centre site (e.g., an examination centre set up in an office building or hospital). They provided a standardized collection environment (equipment set up, room size, etc.) that was designed to meet Statistics Canada security and confidentiality policies as well as the flexibility of locating the MEC near the selected respondents' homes.

7.1.3 Informatics environment

Computer assisted interviewing (CAI) was used to capture the responses for the household, physical measure and laboratory components of the CHMS. CAI allowed for custom interviews for every respondent based on their individual characteristics and survey responses. This included:

- Questions that were not applicable to the respondent (valid skip), were skipped automatically.
- Edits to check for inconsistent answers or out-of-range responses were applied automatically and on-screen prompts were shown when an invalid entry was recorded. Immediate feedback was given to the respondent and the interviewer was able to correct any inconsistencies.
- Question text, including reference periods and pronouns, was customised automatically based on factors such as the age and sex of the respondent, the date of the interview and answers to previous questions.

In order to perform computerized data capture within the MEC, a unique data capture architecture had to be developed as there was a requirement for multiple users in different MEC rooms to access a single respondent's case file. This required the development of a complex, fully customized data capture application that used components of the computer assisted telephone interview (CATI) environment.

To reduce data entry errors, increase efficiency of data collection and reduce the need for double entry and data entry verification, the CHMS MEC data capture system was developed to accept direct input from electronic testing equipment. This included communication (both one and two-way) between the application and several of the measurement devices (e.g. automated blood pressure cuff, weight scale). In cases where the direct input was not functioning or not possible and manual entry was required, the data were entered twice. In several cases (e.g. visual field test, retinal photography, pQCT, jumping mechanography), the data was captured and saved separately using specialized software. This data was later linked back to the data captured in the CATI environment by the data processing team at head office.

In order to support the electronic capture of physical measures data and to support the operational and administrative needs, the MEC was equipped with its own computer server. At the end of every day, encrypted data were transmitted from the trailer server via a dedicated out-going phone line to Statistics Canada headquarters. Encryption software was used to ensure the confidentiality of the data during transmissions between the MEC and headquarters.

7.1.4 Questionnaire design

Respondents were asked health-related questions at both the household and the mobile examination centre (MEC). These questions provide contextual information for the physical measures data (e.g. fitness information, blood and urine analyses). Users only interested in questionnaire data should consider using the Canadian Community Health Survey (CCHS), which has a much larger sample size than the CHMS and therefore provides more statistical power and possibilities for analysis at lower geographical areas.

7.1.4.1 Household questionnaire

Household questionnaire

The household questionnaire content was developed with input from stakeholders (Health Canada and the Public Health Agency of Canada) and from external experts who participated as members of various advisory committees.

Prior to finalizing the questions, one-on-one qualitative test interviews were conducted to look at specific questionnaire content, particularly the content new to cycle 5. As a result of this testing, improvements were made to questionnaire wording and instructions and to the flow of questions.

Some of the main modifications to the household questionnaire between cycles 4 and 5 of the survey included the following:

- Addition of questions related to the new physical measures, including vision health, bone health and sleep apnea.
- Addition of questions on toxoplasmosis.
- Addition of questions on the use of electronic cigarettes.
- Addition of questions on the neighbourhood environment.
- Addition of questions on maternal breastfeeding.
- Questions on phlegm, hepatitis, hobbies, sleep, and strengths and difficulties were removed.
- Addition of questions on steroid medication use.
- The statement informing respondents that Statistics Canada would be linking their household interview information with tax data in cycle 5 was replaced by questions asking for household and personal income in cycle 6.
- Questions on the time spent outdoors during the past month were reworked into a new block of questions focusing on the 7 days prior the interview.

7.1.4.2 Clinic Questionnaire

Clinic Questionnaire

Development of the clinic questionnaire proceed in much the same way as that of the household questionnaire. Content was determined through a comprehensive consultation process and multiple iterations of the collection application were generated. Each iteration was assessed on flow within the MEC for both the respondent and staff. Quantity and quality of data collected was also assessed.

The clinic questionnaire includes a set of self-reported health questions similar to the type of questions asked within the household questionnaire. In cycle 6, the questions included at the MEC were related to medication use, vision and fish and shellfish consumption.

In addition, the clinic questionnaire includes introductory text/instructions and screening and administrative questions related to the physical measures tests conducted at the MEC. This included questions on physical and health conditions as well as health restrictions related to women's health, physical activity, bone mineral density tests, and vision tests.

7.2 Collection

Cycle 6 data collection took place between January 2018 and December 2019, and included 16 collection sites spread across Canada from Lower Sackville, Nova Scotia to Nanaimo, British Columbia (see Appendix 3). While one set of trailers was being used for collection, the second set would be moved to the next location to begin the rigorous set of procedures required to prepare for collection.

7.2.1 Collection – Household interview

One to two weeks prior to the start of household interviews at each collection site, introductory material was mailed to the dwellings selected for the survey informing them that they would be contacted to participate in the survey. Interviewers called or drove to each dwelling to book an interview. Once contact was made with the household, the interviewer introduced the survey by outlining the basic steps of the survey and informing the person that participation was voluntary and that any information provided would be kept confidential under the authority of the *Statistics Act*.

Based on the demographic information collected, one or two persons in the household were selected to participate in the survey. A selection algorithm was used to try and reach an equal distribution of people among the different age groups. If two persons were selected in a household, one person was always 3 to 11 years old and the other 12 to 79 years old. (also see Chapter 6 – Sample Design).

Prior to commencing the interview, the respondent was informed about the survey and was shown a brief four minute introductory video. For respondents between the ages of 3 and 11, a parent/guardian was present during the interview to answer questions with assistance from the

child. All respondents aged 12 to 79 years who were able to answer questions on their own were asked to do so.

At the end of the interview the interviewer provided the respondent with an information package, explaining the MEC portion of the survey, information about the tests performed at the MEC and general information about the survey. The interviewer briefly reviewed the material in the information package and answered any questions. At that time, the interviewer informed the respondent that they had been assigned to a fasting or non-fasting appointment at the MEC. The fasting appointments required respondents 6 to 79 years of age to fast for 12 hours in preparation for specific laboratory tests. See Appendix 4 for a list of the pre-testing guidelines provided to respondents during the household interview.

At the end of the household interview, if possible, the interviewer set up an appointment at the MEC for the respondent. In some cases, the respondent or parent of the respondent set up an appointment on their own at a later time. At the end of each day, the interviewers transmitted all completed cases back to Statistics Canada using encryption software to ensure the confidentiality of the data during the transmission.

7.2.2 Collection – MEC

Upon arrival at the MEC, the respondent's information was logged into the database at the reception desk. The MEC staff verified that the respondent's name, sex, date of birth and official language (collected during the household interview) were correct.

Prior to beginning the physical measures tests, the respondent had to provide their consent to participate in the MEC portion of the survey. Parents or guardians gave consent on behalf of children aged 3 to 13. Children aged 6 to 13 provided their assent to participate.

After the consent module, the respondent (or parent/guardian of younger respondents) was asked some screening questions. Parents or guardians completed the screening questions on behalf of children aged 3 to 13. Depending on the fasting status, the age of the respondent and the responses to the screening questions, some respondents were excluded from certain measures or laboratory tests (see Appendix 5 – Exclusion Criteria). Adherence to the pre-testing guidelines was verified and documented within the application.

Respondents then performed all measures or provided biospecimens for laboratory tests for which they were eligible. For the activity monitor and home urine sample, respondents were provided instructions to follow when they returned home.

Every respondent could, at any time, refuse to participate in a measure or test. The order of the measures and blood tests was set in such a way that the effects of a certain measure (e.g., increased blood pressure from the fitness tests) did not affect the results of another measure (e.g., resting blood pressure, intraocular pressure test). In some cases, the results of one test (e.g. resting blood pressure) could influence whether a person was deemed eligible to participate in another test (e.g. fitness tests) (see Appendix 5 – Exclusion Criteria).

After the blood and urine samples were collected from survey respondents, they were processed, analyzed for the CBC and temporarily stored in fridges and freezers within the MEC laboratory. An initial DNA extraction step was carried out at the MEC prior to shipping. Stored samples were sent weekly to reference laboratories in Ottawa, Quebec City, Burlington, Edmonton, Toronto and Winnipeg for additional analyses related to general health, nutrition, chronic disease, volatile organic compounds (VOCs), and environmental exposure as well as infectious diseases and for storage in the CHMS biobank in Winnipeg.

Prior to leaving the MEC, the respondent received a report of their measurements, and a letter for their health care provider if it was required (e.g., extremely high or low blood pressure). A few months after the visit to the MEC, a report of laboratory tests containing many of the respondent's blood and urine test results, as well as jumping mechanography and pQCT and results from their activity monitor was sent to them. Respondents 14 years of age or older received their own reports, while parents of respondents 3 to 13 years old received their child's reports. Test result with no established reference intervals, were not interpreted. The values of these tests were available to respondents on demand.

In order to maximize response rates to the CHMS, respondents who were unable to attend the MEC, but were willing to have a selected number of measures done in their home were offered the option of a home visit.

The home visit was conducted by a minimum of two CHMS staff members (most often a health measures specialist (HMS) and a laboratory technologist) using paper questionnaires for recording data. There were no differences in the procedures used to conduct the measurements in the home, however, there were minor differences in the equipment used (e.g., scale, stadiometer) because all home visit equipment needed to be portable. As with the visit to the MEC, the respondent could, at any time, refuse to participate in any measure or laboratory test. Once back at the MEC, the HMS entered the data from the home visit into the application and coded it as a home visit.

There were 2 home visits during cycle 6.

7.3 Minimizing non-response

7.3.1 Minimizing non-response – Household interview

To ensure the best possible response rate at the household, many practices were used to minimize non-response.

7.3.1.1 Introductory material

Before the start of each collection period, introductory materials were sent to the selected households, explaining the different steps of the survey and emphasizing the importance of the survey by providing examples of how CHMS data would be used.

7.3.1.2 Initiating contact

Interviewers were instructed to make all reasonable attempts to obtain interviews. When the timing of the interviewer's visit was inconvenient, an appointment was made to come back at a more convenient time. If no one was home on first visit, a notice that a CHMS interviewer had stopped by their home was left at the door. Numerous personal visits were made at different times on different days until potential respondents were home and available to do the interview. If interviewers were unable to make contact with anyone at the household, they tried phoning to arrange for a personal visit. In some cases a phone number was not available for the case and the interviewers had to research to find a name and number, using directory listings, neighbours, superintendents, and management/rental offices.

7.3.1.3 Refusal procedures

The interviewer tried to convince the respondent of the importance and potential benefits of participating in the survey. If the individual refused to participate in the survey, a refusal letter was sent and they were then contacted a second time by either another interviewer or the senior interviewer who, again, stressed the importance of the survey and the household's participation in it.

7.3.1.4 Language barriers

The introductory materials (the introductory letter and the CHMS brochure) were available in English and French. Respondents were interviewed in the official language of their choice. To remove language as a barrier to conducting interviews, where possible, the CHMS team recruited interviewers with some competencies in a language other than the two official languages. When necessary, an interviewer or an external interpreter with the appropriate language competency joined the interviewer so that questions/instructions could be translated for the respondent. If, no one with the desired language competency could be found, it was also acceptable for a household member who was willing and able to translate for the respondent to do so. Note that this was not considered a proxy interview. The household member was simply translating the questions and the respondent’s answers directly to the interviewer, not answering for the respondent.

7.3.1.5 Youth respondents

Interviewers were obligated to obtain verbal permission from parents/guardians to interview youths between the ages of 12 and 17 who were selected for the survey. Several measures were taken to alleviate potential parental concerns and to ensure a completed interview. Interviewers provided the parent or guardian with a copy of the *Information and Consent Booklet* which contained a section dedicated to parents and guardians. This document explained the purpose of collecting information from youth, listed the subjects to be covered in the survey and explained the need to respect a youth's right to privacy and confidentiality.

When interviewing respondents 12 to 17 years of age, interviewers ensured that the parent was in the home but that the interview took place outside of parents/siblings hearing range, unless permission was obtained from the youth for a parent to be present. If the selected youth could not

be interviewed in a private setting, the interviewer read the questions out loud with a parent or guardian in the room. The youth then answered the questions directly onto the computer. If privacy and confidentiality could not be respected, the case was coded as a refusal with a permanent note indicating that privacy/confidentiality could not be respected.

If parents asked to know more about the type of questions asked in the survey, interviewers first directed them to the topics listed in the Information and Consent Booklet. If they asked to see the actual questions, interviewers showed them the content section of the Interviewer's Manual. For those parents who requested a copy of the questions, a copy was available through the Data Collection Manager, as well as at Statistics Canada's head office in Ottawa.

7.3.1.6 Proxy interviews

In the CHMS, parents/guardians answered questions about their children aged 3 to 11. This included all household modules that were applicable to children. Children assisted in responding to some questions for which the parents may not have known the answers (e.g., participation in activity during school hours).

In cases where the selected respondent 12 years of age or over was, for reasons of physical or mental limitations, incapable of completing an interview at the household, another knowledgeable member of the household supplied information about the selected respondent. While these proxy respondents were able to provide accurate answers to most of the survey questions, the more sensitive or personal questions were beyond their scope of knowledge. This resulted in some questions from the proxy interview being unanswered. Every effort was taken to keep proxy interviews to a minimum. The variable "PROXY" on the data set indicates whether or not a household interview was completed by proxy.

In cycle 6, 28.4% of the interviews were proxy. Of the proxy interviews, 97.1% of the respondents were under 12 years of age and 2.9% were 12 years of age or older.

7.3.2 Minimizing non-response – MEC

Approximately 70% of respondents who completed a household interview in cycle 6 agreed to go to the MEC. Some of the main practices used to obtain this high level of participation are described below.

7.3.2.1 Non-response follow-up

MEC staff was responsible for following up with any respondents who did not book an appointment at the end of their household interview and did not call the MEC booking desk to set up an appointment within a few days after their household interview. The staff members followed similar refusal procedures as household interviewers.

7.3.2.2 Flexible MEC hours

Strategies specific to the MEC included the creation of MEC opening hours and appointment times that provided maximum flexibility to the respondent. The MEC staff tried to accommodate as many respondents as possible at each site.

7.3.2.3 Refusal procedures

To minimize the non-response to the CHMS clinic component, the MEC staff was instructed to make all reasonable attempts to convince respondents who participated in the household interview to attend the MEC. The appointment booking desk staff, who had received specific training in handling refusal conversions, followed-up with respondents who refused to participates in the MEC portion of the survey. If they were unsuccessful in booking an appointment, the MEC site manager or clinical team leader would call one final time to attempt to book an appointment. Respondents who could not be contacted (e.g., no answer at the home phone number) were sent a "No Contact Letter" asking them to phone the MEC to book an appointment.

7.3.2.4 Language barriers

Mobile examination centre (MEC) staff handled language barriers in the same way as household interviewers. CHMS staff, external interpreters or family members with knowledge of the third language were used to help the respondent understand instructions and forms in order to complete the visit at the MEC.

7.3.2.5 Youth respondents

As with the household interview, parents/guardians answered all questions about their children aged 3 to 11. Since the age of consent for the MEC portion of the CHMS was 14 years of age, parents/guardians also answered these questions for their 12 and 13 year old youths, though the youths usually assisted. Youths aged 14 and over were responsible for signing their own consent form and answering all questions although parents/guardians were in some cases present during their visit at the MEC and able to assist on difficult questions. To maximize efficiency at the MEC, the selected child or youth usually did the physical measures tests with one CHMS staff member while their selected parent was doing tests with another CHMS staff member.

In cases where the selected respondent 14 years of age or over was, for reasons of physical or mental limitations, incapable of answering questions and completing the consent form at the MEC, the parent/guardian assisted.

7.4 Physical measures protocols

All physical measures protocols were done at the MEC except for the activity monitor which was worn by respondents once they returned home, and the home urine sample which was provided within the seven days following the MEC visit. A brief description of the physical measures protocols can be found below, with information on exclusion criteria provided in Appendix 5.

7.4.1 Anthropometry

The anthropometry component consisted of four main physical measures tests: standing height, weight as well as waist and neck circumference. All tests were done on eligible respondents aged 3 to 79 years old. Respondents having one or more acute or chronic conditions which could affect the results of an anthropometry test result were excluded from participating in that particular test (e.g. a cast would affect a respondent's weight measurement).

7.4.1.1 Standing height

Standing height is an assessment of the maximum vertical size of the respondent. This measure was taken on all the respondents who were able to stand unassisted. Standing height was measured with a fixed stadiometer with a vertical backboard and a moveable headboard using a procedure based on the Canadian Society for Exercise Physiology-Canadian Physical Training for Health (CSEP-PATH)[3]. A self-reported height was captured for respondents who were not eligible because of an acute or chronic condition (e.g., in a wheelchair) or who refused to have their height measured.

7.4.1.2 Weight

The respondent's weight was taken on a Mettler Toledo digital scale, following the CSEP-PATH protocol[3].

7.4.1.3 Waist circumference

Waist circumference provides an indication of abdominal fat distribution and is an important indicator of the health risks associated with obesity. Waist measurements were taken on respondents using the National Institutes of Health (NIH) protocol[4].

Note: In cycle 1, the World Health Organization (WHO) measurement protocol was used. Prediction equations can be applied in datasets in which the WHO protocol was used. The document on the use of equations to predict NIH waist circumference measures[5] is available upon request.

7.4.2 Heart rate and blood pressure

Heart rate and blood pressure measurements were taken on all eligible respondents aged 6 to 79 years old using an oscillometric blood pressure measurement device. A series of blood pressure and heart rate measurements were taken at one minute intervals following a five minute rest period. The last five measurements were used to determine the average resting heart rate and blood pressure. A 2nd series of measurements was performed when required.

Note: In cycle 6, the need to select whether the blood pressure data was electronically transferred or manually entered was removed. The blood pressure was always measured electronically therefore the variable AFT_24 was not release.

7.4.3 Physical fitness tests

7.4.3.1 Modified Canadian Aerobic Fitness Test (mCAFT)

Aerobic fitness is a measure of the combined efficiency of the lungs, heart, vasculature, and exercising muscles. The eligible age group for the mCAFT was 8 to 69 years of age. The protocol for the mCAFT component is based on the CSEP-PATH protocol for adults 15-69 years of age[3], but was applied to all age groups.

7.4.3.2 Sit and reach

The sit and reach test is a commonly used field test to assess low back and hip-joint flexibility. The protocol for the Sit and Reach Component was based on the CSEP-PATH[3].

7.4.3.3 Grip strength

The grip strength component provided information on the muscular strength of all eligible respondents aged 6 to 79 years old. Grip strength is heavily linked to general health and independence. The protocol for the grip strength component was based on the CSEP-PATH[3].

7.4.4 Activity monitor

An Actical physical activity monitor was given to all eligible respondents aged 3 to 79, along with an adjustable belt, an XPRESSPOST envelope, and an information sheet. Respondents were asked to wear an activity monitor for 7 days following their visit to the MEC. Activity monitors provide key information on the wearer's level of physical activity, including intensity, timing (day and time), frequency and duration as well as information on sedentary behaviour (excluding sleep). The HMS could assess if respondents in wheelchairs were excluded from this measure based on the respondent's condition.

7.4.5 Bone health

7.4.5.1 Peripheral quantitative computed tomography (pQCT)

The objective of this physical measure is to determine the muscle and bone health of the Canadian population by analyzing characteristics of lower leg muscles and bone mass, size and density at the tibia. The Peripheral Quantitative Computed Tomography (pQCT) is a fully automated measuring system used to quantify the bone, muscle and fat of a limb. It is the key methodology for analyzing muscle-bone relationships. All eligible respondents aged 6 to 79 completed a pQCT scan. The duration of the test was approximately 15 minutes. In partnership with Dr. Frank Rauch and his associates at the Shriners Hospitals for Children, the following protocol was developed. At first, the MRT measured the respondent's left tibia (unless exclusion criteria for the left leg were met, see Appendix 5 for more information). Then the respondent was comfortably seated on the examination table with the leg to be measured properly positioned in the pQCT device for the measurement. The first scan of the leg called a scout scan was performed in order to precisely place a reference line. The second scan was measuring the leg muscles and bone mass, size and

density at 3 specific sites that correspond to 4%, 38% and 66% of the distal tibia length. The respondent's effective dose of radiation was expected to be less than 0.003 mSv per scan series, which corresponds to one day of natural background radiation a Canadian would normally receive.

7.4.5.2 Jumping mechanography

The jumping mechanography is a practical and reproducible technique performed to quickly assess the dynamic leg muscle function of the Canadian population by analyzing dynamic muscle performance via natural motor movements. This technique, developed in partnership with Dr. Frank Rauch and his associates at the Shriners Hospitals for Children, uses hopping and jumping force plate tests to evaluate ground reaction forces during the tests and derives key descriptors of muscle performance (e.g., peak force, peak power). All eligible respondents aged 6 to 79 performed two different tests: the Multiple Two-Leg Hopping (M2LH) and the Single Two-Leg Jump (S2LJ). During the M2LH, for each trial, the respondent had to perform 10 hops on both forefeet with stiff knees. While for the S2LJ, each trial consisted of a single jump performed in a fluid movement with the feet leaving the plate at the same time and then landing at the same time. For each of the two tests, the respondent repeated the test until three valid trials had been completed or a maximum of five trials had been attempted.

7.4.6 Vision health

The vision health assessment consisted of an introduction questionnaire and four visual screening tests: visual acuity, visual field, retinal photography and intraocular pressure.

7.4.6.1 Visual acuity

The visual acuity test is a standard measurement used in ophthalmic examinations to measure the acuteness or clearness one can see of an object at a distance and is thus a good indicator of distance vision and a good predictor of near vision. The visual acuity test is also an indication of the proper functioning of the entire visual system.

Eligible respondents aged 6 to19 years old, 40 to 79 years old, and 20 to 39 years old with reported Type 1 or Type 2 diabetes, completed the visual acuity test. The chart employed is the Early Treatment Diabetic Retinopathy Study (ETDRS) acuity test. Initially developed to test visual acuity in patients with diabetic retinopathy, it is now the world-wide standard for visual acuity testing considering its high degree of comparative consistency (same number of letters per row and equal spacing between lines). The results obtained were recorded as a visual acuity result (20/20, 20/40, etc.) and a letter count score. To complete the test the respondent had to read the letters displayed on a monitor, one line at a time, from a standardized sitting position of two metres. The test was performed for both eyes together and then each eye separately.

It is important to note that the visual acuity test completed in the CHMS was a "best-corrected visual acuity" test, meaning that a respondent reporting using distance viewing glasses or contact lenses had to wear them to perform the test, otherwise they were tested with pinhole occluders. Also, a respondent obtaining a result equal or lower to 20/40 during one of the 3 testing conditions (right/left eye, both eyes) had to repeat that same condition with pinhole occluder(s).

7.4.6.2 Visual field

The purpose of the visual field test is to detect dysfunction in both central and peripheral vision which can be caused by various medical conditions such as glaucoma, stroke, brain tumors, neurological deficits, central or peripheral retinal disease, eyelid conditions (ptosis or drooping), optic nerve disease, and diseases that affect the visual pathway to the brain.

The visual field examination used a Humphrey Matrix 800 Frequency Doubling Technology to assess the field of vision of each eye separately and 30 degrees around the macula on eligible respondents 40 to 79 years old, and those aged 20 to 39 years old with reported Type 1 or Type 2 diabetes. Respondents were tested with the (N-30-1) screening test. The flickering or striped patterns that the respondents were seeing on the screen of the testing device assessed the sensitivity of the detection of motion in their visual field, specifically in the central and peripheral fields.

7.4.6.3 Retinal photography

Retinal photography was used as part of the assessment of the overall health of the eye for eligible respondents aged 40 to 79 years old, and those 20 to 39 years old with reported Type 1 or Type 2 diabetes. Respondents did not receive dilation drops prior to capturing the images. The images obtained from this test can help detect conditions such as glaucoma, diabetic retinopathy, macular degeneration and cancer. The retinal camera (Nidek AFC 330 Non-mydriatic fundus camera) functions by having a low power microscope with a high-resolution camera attached. The unit takes pictures of two different views of the internal surface of each eye in order to get a clear view of the retina, optic disc, macula and posterior pole that can be analyzed later.

7.4.6.4 Intraocular pressure

The objective of the intraocular pressure (IOP) test is to measure the firmness of the eye, created by the fluid pressure inside it. The IOP test is standard in optometrist and ophthalmologist clinics and provides the specialist with a representation of the ocular health of the respondent and helps to identify any signs of glaucoma. The IOP was measured on eligible respondents aged 40 to 79 years old, and those aged 20 to 39 years old with reported Type 1 or Type 2 diabetes, by non-contact applanation tonometry using an air-puff tonometer (Reichert 7-CR) that applies a standardized puff of air to gently flatten the cornea and provide an intraocular pressure reading in millimeters of mercury. IOP values correlated to Goldmann's Applanation and corneal compensated IOP values were calculated by the machine.

7.5 Laboratory measures protocols (blood and urine)

A brief description of the laboratory measures protocols used for the CHMS can be found below. See Chapter 5 for a summary of the changes to laboratory tests from cycles 5 to 6. Additional supporting documentation is available upon request.

7.5.1 Sample collection

Blood and urine samples were collected from all eligible respondents at the MEC in order to obtain nationally representative information on a variety of biomarkers (nutrition, chronic and infectious diseases, environmental exposure, etc.). A second urine sample was also completed by the respondent at their home.

7.5.1.1 Blood collection

The blood was collected by a phlebotomist using a standardized venipuncture technique. The amount of blood taken from respondents depended on their age and consent to storage:

With consent to storage:

- 3 to 5 years: 25.0 ml
- 6 to 11 years: 40.0 ml
- 12 to 13 years: 58.0 ml
- 14 to 19 years: 72.0 ml
- 20 to 79 years: 78.0 ml

Without consent to storage:

- 3 to 5 years: 22.0 ml
- 6 to 11 years: 37.0 ml
- 12 to 13 years: 38.0 ml
- 14 to 19 years: 44.0 ml
- 20 to 79 years: 48.0 ml

A deviation from the order of blood collection tubes typically done in a clinical setting was made to accommodate the priority of the assay(s).

7.5.1.2 Urine collection

Respondents were asked to provide urine to conduct all the tests for which they were eligible.

7.5.1.3 Home urine collection

In order to simulate a 24-hour urine collection, respondents who provided urine during the MEC visit were asked to provide another urine sample at home within a 7-day period following their MEC visit. Respondents sent their urine samples directly to the Health Canada – Nutrition Research Division laboratory in Ottawa for sodium, potassium and/or iodine analysis. Health Canada also created another two aliquots for creatinine analysis at the Centre de Toxicologie du Québec (CTQ) and for storage at the National Microbiology Laboratory (NML). A pre-addressed envelope was provided to all respondents receiving the home urine sampling kit.

Note: There are several reasons for the large amount of missing results of the home urine collection such as the kit not being available, the respondent refusing it or it did not get returned to the laboratory.

7.5.2 Analysis of CBC performed at the mobile examination centre (MEC)

A Complete Blood Count (CBC) was performed in the MEC laboratory by a technologist for all respondents from whom a blood sample was collected. Results from any unsuitable samples (e.g., severe lipemia or clot in tube) were not reported.

7.5.3 Processing and storage of the blood and urine samples

It was important to process the specimens as soon as possible because the quality and integrity of the blood and urine specimens deteriorates over time. Blood was centrifuged at 18°C for 15 minutes at 3,400 revolutions per minute to separate plasma and serum and to allow for the aliquoting into smaller tubes. The urine was also aliquoted into smaller tubes. These tubes were placed in shipping trays and stored in the MEC laboratory in either the refrigerator or the freezer, depending on the test. All specimens were stored as soon as processing was complete to ensure the quality of the samples' integrity. A four hour time limit from the point of collection was set for blood samples to be processed and stored. For most samples, however, processing and storage was achieved within two hours from the time of collection. Given specific pre-analytical requirements for homocysteine and RBC chromium, a time limit of 2 and 3 hours, respectively, was set for the processing and storage of these samples.

7.5.3.1 Urine specific gravity

Before the urine was aliquoted, the specific gravity was measured for every respondent that provided a urine sample. The specific gravity was taken with two consecutive readings, from the same drop of urine. The two specific gravity readings had to be the same or within 0.001 of each other. If there was a difference of 0.001, the lower of the two values was recorded in the data capture application. If the difference was > 0.001 then the measurement was repeated with a fresh drop of urine.

7.5.4 Shipment of the blood and urine

The shipping of the blood and urine aliquots was done once a week to the reference laboratories on pre-assigned shipping days. All packages were sent to one of the nine reference laboratories: 1. Dynacare in London for biomarkers related to general health (chemistry and lipid panels); 2. Health Canada-Nutrition in Ottawa for biomarkers related to nutrition and vitamins; 3. Centre de toxicologie du Québec (CTQ) in Quebec City for environmental biomarkers and urinary creatinine (urinary output adjustment and kidney health); 4. National Microbiology Laboratories (NML) in Winnipeg for DNA analysis and the CHMS biobank; 5. Ontario Region Health Canada in Toronto for acrylamide and organohalogens; 6. Western Region Health Canada (WRHC) in Burnaby for parabens; 7. Shriners Hospital for Children in Montreal for biomarkers related to bone health; and 8. Laboratoire de santé publique du Québec (LSPQ) in Quebec City for biomarkers related to infectious diseases.

Shipments were packaged according to the International Air Transport Association (IATA) and Transportation of Dangerous Goods regulations for biological specimens. All shipments were sent by overnight delivery using a courier company certified to handle dangerous goods and were scheduled to arrive at the reference laboratory only on weekdays. A specimen tracking system was also used in order to allow staff to determine the status of every tube shipped to the reference laboratory. In addition, the temperature of refrigerated shipments was monitored using pre-programmed devices that took the temperature every 30 minutes during shipping. This provided assurance that the samples received by the reference labs were maintained at an adequate temperature to preserve the sample integrity during shipping.

There were some occurrences where shipments were delayed and the condition of the samples required assessment. If it was determined that the shipment conditions had been inadequate, the affected results were removed to ensure that only the highest quality of data was reported to the CHMS by the reference laboratories.

8. Data processing

8.1 Verification

One of the unique features of the Canadian Health Measures Survey (CHMS) is that three different sets of data are collected for the same respondent: household interview data, physical measures data, and laboratory results data. Each set of data has to be processed on its own, yet they cannot be completely separated from each other because at various points during processing the three sets of data have to be used together.

The processing of the household interview data was performed in a manner similar to that of other health surveys at Statistics Canada. The data are validated first at the record level and then at the individual variable level, followed by detailed top-down editing. During data collection, processing takes place on a daily basis. The household interview responses have to be processed on a daily basis in order for the data to be available at the mobile examination centre (MEC) in time for the respondent's visit to the MEC.

Similarly, the processing of the physical measures data begins with the data being validated first at the record level and then at the individual variable level, followed by detailed top-down editing. Also, because the laboratory tests are determined based on responses received at the MEC, the MEC data are used to generate a file containing a list of the tests for which laboratory results are expected to be received. This laboratory "control" file is used in processing the laboratory results data.

The processing of the laboratory data involves significant file manipulation due to the fact that several different file types are received from the MEC and the various reference laboratories. As with the household and the physical measures data, the laboratory data are validated at the record level and at the individual variable level, and several new variables are then derived. The laboratory data are processed on a day-to-day basis so that any results that have been identified as outside of a normal range at the reference laboratories and the MEC are available in a timely fashion for reporting to respondents.

8.2 Mark-all-that-apply questions

During the initial phase of data processing, mark-all-that-apply questions are expanded, with each response category in the original question becoming a series of separate questions with a yes or no response. In the example below, the respondent selected both 2 and 3 as answers to the original question. The answers to the new questions are based on answers from the original question.

Original question:

CCC_Q96 **What type of hepatitis do you have?**

INTERVIEWER: Read categories to respondent. Mark all that apply.

1 **Hepatitis A**

2 **Hepatitis B**

3 **Hepatitis C**

Final questions / responses:

CCC_96A **What type of hepatitis do you have? - Hepatitis A**

1 Yes

2 No

Don't Know, Refused

CCC_96B **What type of hepatitis do you have? - Hepatitis B**

1 Yes

2 No

Don't Know, Refused

CCC_96C **What type of hepatitis do you have? - Hepatitis C**

1 Yes

2 No

Don't Know, Refused

8.3 Coding

As in previous cycles, pre-coded answer categories were supplied for all suitable variables, and the interviewers and clinic staff were trained to assign a respondent's answer to the appropriate category. In the event that a respondent's answer could not be easily assigned to an existing category, several questions also allowed the interviewer to enter a long-answer text in the "Other-specify" category. All such questions were closely examined at head office during processing. For

some household questions, the long answers were coded into one of the existing listed categories. If this was not possible, the response was coded as 'Other'. For the MEC responses, long answers were reviewed and some responses were coded to existing categories. For the remaining 'Other-specify' answers at both the household and the MEC, some new categories were created when there was a sufficient number of responses. The remaining responses were coded as 'Other'. For all questions, the "Other-specify" responses will be taken into account when refining the answer categories for future cycles.

As in previous cycles, certain data were collected as long answers and had codes assigned. This included medication and other health product names, and job description information.

For medications, databases of standard descriptions were available in the computer-assisted interviewing (CAI) applications, and a code could be assigned at the time of collection based on a search of the appropriate database. Any description without a code was extracted during processing and coded manually. Over the course of the full survey, there were over 500 medications and other health products that were manually assigned a Drug Identification Number (DIN), Natural Health Product Number (NPHN), Homeopathic Medicine Number (DIN-IM) or Exemption Number (EN). As Statistics Canada is not permitted to release drug names, the specific drug identifiers are not available on the data files. Instead, coded variables representing information about the type of medication are placed on the data file. Codes representing specific drug names were put on the medication file (see Appendix 6 for information on the medical classification systems used).

For the assignment of industry and occupation codes, over 3,000 records with data in the job description fields were extracted and sent for coding. The North American Industry Classification System (NAICS) 2017 version 2.0 and National Occupational Classification – Statistics (NOC) 2016 version 1.1 codes were used. To maintain consistency with the coding systems used in previous cycles, the job description responses were also coded using the North American Industry Classification System (NAICS) 2012 and National Occupational Classification – Statistics (NOC) 2011 codes, in cycle 6.

More information about the classification systems for job descriptions can be found at:

https://www.statcan.gc.ca/eng/subjects/standard/naics/2017v2/index
https://www.statcan.gc.ca/eng/subjects/standard/noc/2016/indexV1.1
https://www.statcan.gc.ca/eng/subjects/standard/naics/2012/index
https://www.statcan.gc.ca/eng/subjects/standard/noc/2011/index

8.4 Editing

Most editing of the data was performed at the time of the interview by the CAI application. It was not possible for interviewers/HMS to enter out-of-range values and flow errors were controlled through programmed skip patterns. For example, CAI ensured that questions that did not apply to the respondent were not asked. Edits requiring corrective action were incorporated in the CAI application to deal with inconsistent responses. In addition, warnings not requiring corrective action were also included to identify unusual (i.e., improbable rather than impossible) values as a means of catching potential errors and allowing correction at source.

At head-office, the data underwent a series of processing steps that resulted in some of the data being adjusted. As a final validation step, the CAI edits were re-applied to the processed data. As a result, the final data were complete and contained reserve codes for responses of "less than limit of detection", "valid skip", "don't know", "refusal" and "not stated".

Table 8.1 Reserve code of responses

Reserve Code label	Reserve code
Greater than upper analytical range	9.4, 94, 99.4, etc.
Less than limit of detection	9.5, 95, 99.5 etc.
Valid Skip	6, 96, 99.6 etc.
Don't Know	7, 97, 99.7 etc.
Refusal	8, 98, 99.8 etc.
Not Stated	9, 99, 99.9 etc.

Information on imputation of household income and the treatment of other missing data can be found in Sections 12.1.2 and 12.1.3.

8.5 Creation of derived variables

Derived variables (DV) are created to facilitate data analysis and to minimize the risk of errors. The three most common types of DVs are created by: collapsing data from one variable into groups; combining the data from one or more survey questions for a single respondent; or combining data from more than one respondent. DVs generally have a "D" (derived) or "G" (grouped) in the fourth character of the variable name.

Two other types of DVs can be found on the CHMS data file: converting responses given in various time units (days, weeks, months or a year) into one type of unit such as yearly and calculation of body mass index (BMI).

Specifications were received providing details on how to create each derived variable. Cycle 6 of the CHMS has 413 DVs. Many of these DVs were created within the capture application in order to be able to provide initial results to the respondents at the end of their visit to the MEC, while others were created after the completion of data processing.

All derived variables underwent a validation process after creation in order to ensure that the output provided the requested data. Modifications to the specifications were necessary if, during validation by subject matter experts, the output from the derived variables was determined to be flawed. Complete documentation for all derived variables is available upon request.

8.6 Adjustments for urine creatinine and specific gravity

<u>Adjustment for creatinine</u>

Derived variables providing an adjustment for creatinine for the specific urinary biomarker concentrations are not provided in their respective environmental laboratory data files. Thus, the

following equation should be used in order to adjust the concentrations of those urinary biomarkers for creatinine:

AdjustedForCreatinine = *BiomarkerVariable* / (LAB_UCRE / 8.84)

For example, free cotinine (LAB_COT) adjusted for creatinine, the formula should be:

AdjustedForCreatinine = LAB_COT / (LAB_UCRE / 8.84)

Please note that the variable LAB_UCRE is located on the nel_full data file.

All chemical variables measured in urine and reported in µg/L can be adjusted for creatinine. This includes relevant chemical variables in EU_SUB and TOB_SUB files (see the Data Dictionary specific to each data files for further details on the variables).

Specific gravity

Derived variables providing an adjustment for specific gravity for the specific urinary biomarker concentrations are not always provided. Therefore, the following equation should be used in order to adjust the concentrations of those urinary biomarkers for specific gravity:

AdjustedForSpecificGravity(nmol/L) = *Variable(mg/L)* / ((LAB_USPG - 1) / 0.024))

For example,
AdjustedForSpecificGravity (nmol/L) = LAB_PHEN (mg/L) / ((LAB_USPG - 1) / 0.024))

This represents a re-arrangement of this formula[6 7]:

[Analyte]normalized = [Analyte]measured × ((SGavg – 1) / (SG − 1))

[Analyte]normalized – normalized analyte concentration
[Analyte]measured – measured analyte concentration
SGavg – average specific gravity of all samples in the study (1.024)
SG – measured specific gravity of the sample

8.7 Analytical range

Laboratory data are obtained from the analysis of blood and urine samples. The pre-analytical conditions (e.g., sample volume, specimen processing procedures, etc.) and methods used by laboratories determine the analytical range within which an analyte can reliably be measured.
The low end of the analytical range is identified as the limit of detection (LOD). The LOD is defined as the lowest concentration at which the analyte's analytical response is measured to be greater than the noise. The LOD for the environmental measures were evaluated using the United States Environmental Protection Agency's methodology[8] or a similar method. The upper limit of the analytical range (ULOD) is determined by the highest standard concentration that produces a linear analytical response. Another value that is of interest is the limit of quantification (LOQ). The LOQ is the minimal concentration that can be quantified to a pre-defined level of reliability

for a specific method. For some CHMS laboratory data, the LOD and the LOQ have the same values.

Laboratories are unable to determine the concentration of the analyte outside the analytical range. When laboratory results for analytes measured in blood and urine fall below or above the analytical range, the laboratory does not report a value and therefore the data are assigned a reserve code ending in 5 for results less than LOD (e.g., 9.5, 9.95, 999.5, etc.) or ending in 4 for results above the upper analytical range (e.g., 9.4, 9.94, 999.4, etc.). A full listing of the analytical ranges including LOD and LOQ values is available upon request.

Special considerations concerning lab data:

Many of the environmental contaminants have data that is highly skewed – as such, it is recommended to produce the geometric mean in addition to the arithmetic mean.

Further, many of the analytes (particularly the environmental contaminants) contain a proportion of results "below the limit of detection" (<LOD). Even though laboratory instrumentation cannot detect residual amounts of these analytes, a residue may be present at some level <LOD, and may still present a potential concern to human health. The CHMS has agreed that since a result <LOD is still considered to be a valid result, for the purpose of analysis, these cases are to be replaced with the value of the LOD (as specified in the laboratory Standard Operating Procedure (SOP) document) divided by 2, as recommended by the Environmental Protection Agency (EPA)[8]. In many cases the proportion of the sample <LOD is significant and can have an impact on the calculation of the estimate. For example, if 30% of the sample is <LOD, any percentile up to the 30th cannot be accurately measured since all respondent values up to and around that percentile will be equal to the value of the LOD divided by 2. If the LOD value was 4, all percentiles up to the 30th percentile would be calculated to be 2, but this similarity within the percentile spread is unlikely to be true in the population.

Caution should be used to determine if the calculation of means are appropriate if >40% of the un-weighted sample is <LOD. Statistics Canada generally does not publish a mean if the proportion of the sample <LOD is greater than 40 percent.

The values greater than the upper limit of the analytical range/upper limit of detection (ULOD) are increasing at an exponential rate, with no theoretical ceiling value. It is recommended to set the ULOD cases (9.4, 9.94, 999.4, etc.) to missing (.) during analysis. If the user plans to impute the ULOD, they will need to determine if these numbers are useful or an outlier in the population. If calculating means instead of medians, these imputed values may have a significant impact on a population average if left in the analysis.

Conservative LOD (LODc):

As mentioned above, it is recommended imputing half the analytical LOD for non-detects for analysis. However, LODs might differed across cycles for most environmental contaminants. For consistency across cycles and to ensure meaningful trend analyses, a conservative LOD (LODc) has to be determined. The LODc is defined with the highest LOD across cycles. All chemical

concentrations below the LODc will be replaced with the value equal to the LODc divided by two[9]. For example, the LODs for the mono (2-ethylhexyl) phthalate (MEHP) are 0.2, 0.08, 0.11 and 0.077 µg/L in cycles 1, 2, 5 and 6 respectively. The LODc for MEHP is therefore 0.2 µg/L; all non-detects and all detects with reported concentrations across cycles below 0.2 µg/L will be replaced with half the LODc, 0.1 µg/L.

9. Weighting

In order for estimates produced from survey data to be representative of the population covered and not merely of the sample itself, users must incorporate weighting factors (survey weights) into their calculations. A survey weight is assigned to each person included on the final dataset, that is, in the sample of persons who responded to the entire survey. This weight corresponds to the number of people represented by the respondent in the population as a whole.

The survey weight is calculated as the inverse of the probability that the respondent was selected for the survey. As described in Chapter 6 (Sample design), the Canadian Health Measures Survey (CHMS) is a multi-stage sample that uses two sampling frames for selecting its sample; an area frame of geographic units (clusters) for constructing and selecting collection sites, and a list frame of the dwellings within each site. The probability of selection for the survey is determined by multiplying the probability of selection at each stage.

In accordance with the weighting strategy, the selection weights for collection sites are multiplied by the selection weights for dwellings (households) and adjusted for non-response. Following the conversion of household weights into person weights, the latter are adjusted for non-response at the interview stage and the mobile examination centre (MEC) stage, and with several other adjustments, this weight becomes the final survey weight. The steps of the weighting process are outlined in Sections 9.1 to 9.10.

9.1 Selection weights for collection sites

The first step is to calculate the selection weight for each collection site. For each site, this weight is calculated as follows within each region [10]:

Selection weight of a collection site within a given region[11] =

$$\frac{Sum\ of\ persons\ in\ all\ sites\ contained\ within\ the\ region}{(Number\ of\ persons\ in\ the\ site) \times (Number\ of\ sites\ selected\ within\ the\ region)}$$

There is no adjustment for non-response at the level of the collection sites, since all sites participated in the survey.

9.2 Selection weights for dwellings

For each collection site selected, a list of all dwellings was obtained from the sample frame which is based on the 2011 and 2016 censuses and updated with other administrative files. These dwellings were stratified into nine groups corresponding to the six age group strata and three “other” strata, using household composition as specified in Section 6.3.2 (Dwelling sampling). The sample of dwellings was allocated among these strata.

For a given dwelling, the selection weight is equal to the inverse of the probability of selection of the dwelling within the stratum to which it belongs.

$$\frac{Count\ of\ all\ dwellings\ in\ the\ strata}{Count\ of\ all\ dwellings\ in\ the\ sample\ from\ the\ strata}$$

This weight is then multiplied by the collection site selection weight.

9.3 Removal of out-of-scope units

Among all the dwellings sampled, a proportion is identified during collection as being outside the scope of the survey. Examples of out-of-scope cases for the CHMS are: demolished dwellings, dwellings under construction, vacant, seasonal or secondary dwellings, institutions, and dwellings in which all household members are under 3 or over 79 years of age or are full-time members of the Canadian Forces. These dwellings are simply removed from the sample, leaving only dwellings in the scope of the survey. The dwellings that remain in the sample retain the same weight as at the previous stage.

9.4 Household non-response

During collection, a proportion of the households sampled inevitably resulted in a non-response. This usually occurs when the household refuses to participate in the survey, provides unusable data or cannot be reached to conduct the interview. The weight of non-respondent households is redistributed to respondents within homogeneous response groups (HRGs). In order to create these HRGs, the score method based on logistic regression is used[12]. First a logistic regression model is created to estimate the response probability, and then these probabilities are used to divide the sample into groups with similar response propensities. The logistic regression models are created from the limited amount of information available for all households. This includes data from the frame such as the strata and geographic location, and paradata about the data collection such as the number of attempts to contact the household and the elapsed time between the first and last contact. An adjustment factor is then calculated within each HRG as follows:

$$\frac{Sum\ of\ weights\ for\ all\ dwellings\ (households)}{Sum\ of\ weights\ for\ all\ respondent\ households}$$

The weight of respondent households is multiplied by this adjustment factor to produce the adjusted household weight. Non-respondent households are eliminated from the weighting process starting at this point.

9.5 Creation of the person weight

Since the final sampling unit for the CHMS is the person, the adjusted household weight up to this point must be converted into a person weight. This is obtained by multiplying the adjusted household weight by the inverse of the probability of selection of the person selected in the household. It should be kept in mind that the probability of a person being selected changes depending on the number of persons in the household and their ages (see Section 6.3.3, Respondent sampling for more details).

9.6 Non-response at the questionnaire level

The CHMS has a three-stage collection process. First, the interviewer obtains the complete list of persons living in the household, then he or she interviews the person(s) selected in the household, and finally, the selected person or persons report to the CHMS MEC.

In some cases, interviewers succeed in completing only the first step, either because they cannot contact the person(s) selected, or because the person or persons selected refuse to be interviewed. Such cases are defined as non-responses at the questionnaire level, and an adjustment factor must be applied to the weights of respondent persons to compensate for this non-response. Just as for non-response at the dwelling (household) level, the adjustment is applied within classes defined by the score method using response probabilities from a logistic regression model. The model is based on the characteristics available for all respondents and non-respondents, which includes all the characteristics collected when the members of the household are listed, such as the number of persons in the household, in addition to frame information, geographic information and paradata. An adjustment factor is calculated within each class, as follows:

$$\frac{Sum\ of\ weights\ for\ all\ selected\ persons}{Sum\ of\ weights\ for\ all\ selected\ persons\ responding\ to\ the\ questionnaire}$$

Thus, the weight of respondent persons is multiplied by this adjustment factor. Persons not responding to the questionnaires are removed from weighting starting at this point.

9.7 Non-response at the MEC level

Respondents to the questionnaire are then invited to go to the CHMS MEC for physical measurements. In some cases, people refuse to participate or do not keep their appointment at the MEC. Such cases are defined as non-responses at the MEC level, and to compensate for this non-response, an adjustment factor must be applied to the weights of the MEC participants. Just as for non-response at the dwelling (household) and questionnaire levels, the adjustment is applied within classes defined by their probability of attending the MEC. This probability is obtained from a logistic model using the characteristics available for respondents and non-respondents. All the characteristics collected on the questionnaire during the interview (such as income class, whether or not the respondent is employed, general health status, and frequency of smoking), in addition to frame information, geographic information and paradata, were made available to create the non-response models. An adjustment factor is calculated within each class as follows:

$$\frac{Sum\ of\ weights\ for\ all\ persons\ responding\ to\ the\ questionnaire}{Sum\ of\ weights\ for\ all\ persons\ participating\ at\ the\ MEC}$$

The weights of the persons participating at the MEC were accordingly multiplied by this adjustment factor. Persons who did not report to the MEC are removed from the weighting process starting at this point.

9.8 Calibration

This procedure is applied to ensure that the sum of the final weights corresponds to the estimates of populations defined at the scale of the five Canadian geographic regions[10], for each of the 11 age-sex groups of interest, the six age groups are 3 to 5, 6 to 11, 12 to 19, 20 to 39, 40 to 59 and 60 to 79 where the latter five groups are further broken down by sex. An additional criterion was used to calibrate the 20 to 39 and the 40 to 59 age groups to compensate for the fact that persons in these age groups living with kids have a greater chance of being selected than those living without kids. In households where there was at least one person aged 3 to 11, a second person aged 12 to 79 was selected for the survey. The second person selected was usually a parent aged 20 to 39 years old or 40 to 59 years old. To compensate for any potential bias caused by the selection method, these two age groups were split into those living with and without children aged 3 to 11.

The population estimates are based on the most recent census counts, as well as on counts of births, deaths, immigration and emigration since then. The calibration was carried out using the mean of the monthly estimates (covering the survey period) for each cross-tabulation of standard regional boundaries and age-sex groups. The population estimates for the 20 to 39 and the 40 to 59 age groups in each region were split into those living with and without kids aged 3 to 11 based on the estimated ratios of women and men in these two age groups living with and without kids from the sampling frame.

9.9 Winsorization

Note that following a series of adjustments applied to the weights, it is possible that some units will have weights that stand out from the other weights to the point of being aberrant. Some respondents may actually represent an abnormally high proportion in their group and therefore strongly influence both the estimates and the variance. To avoid this situation, a respondent weight that contributes aberrantly to the age-sex group is adjusted downward using a method known as "winsorization." In this process, respondent weights that are considered to be outliers are replaced by the highest non-outlier weight for that age and sex group. All of the weights are then adjusted to redistribute the surplus weight (the part of the weight that is higher than the highest non-outlier weight). This is done by multiplying the non-outlier weights by an adjustment factor to create the winsorized adjusted weights. The adjustment factor is calculated as:

$$\frac{Sum\ of\ original\ final\ weights}{Sum\ of\ non\text{-}outlier\ weights}$$

9.10 Second calibration

This is an exact repetition of step 9.8 applied to the winsorized adjusted weights. After the second calibration, the weight adjustment was obtained. The resulting weight is the final CHMS weight found in the data file bearing the variable name WGT_FULL.

9.11 Bootstrap weights

The CHMS uses a complex sampling design to select the sample and there are no simple formulas that can be used to calculate the variance of the survey estimates. Instead, a re-sampling approach known as the bootstrap method is used to approximate the sample variance. The bootstrap method involves creating subsamples of the full sample by randomly selecting "n-1" collection sites with replacement among the "n" collection sites in each region[13]. An adjusted weight is then calculated for each respondent in the selected subsample. This is repeated 500 times to create the bootstrap weights. To calculate the variance of a point estimate (such as the mean), the estimate for each of the 500 replicates is calculated using the bootstrap weight. The variability among the 500 estimates gives the variance estimate. The bootstrap weights are provided on the final CHMS weight file with the names BSW1-BSW500. Refer to Section 11.5 on the use of the weight variable and Section 12.2.4 on available software for more information.

9.12 Weighting for selected subsamples

9.12.1 Weighting for vision data subsample

A separate weight was created for the analysis of the subsample of the vision data related to the Retinal Photography Component (RTP) and the Visual Field Component (FDT) (wgt_vis), even though all eligible respondents were asked to participate in the RTP and FDT vision tests, they were stopped after site 10 in cycle 6. For this reason, the vision data is not a true subsample. To compensate for the loss of data from sites 11-16 in cycle 6, the collected cycle 6 vision data was combined with the cycle 5 vision data.

The weighting steps described in Sections 9.1 to 9.7 were carried out separately on the cycle 6 vision data. In brief, non-response adjustments factors were applied within homogeneous response classes at the collection site, household and MEC levels on the respondents from the first 10 sites. The classes are created based on the probability of response from a logistic model using the characteristics available for all respondents and non-respondents in sites 1-10. The "score" method is then used to define the classes. The final four steps, first calibration, winsorization, second calibration, and bootstrap weights (see Sections 9.8 to 9.11) were then applied to the respondents from the first 10 sites. After calibration, the weight adjustment was obtained for the cycle 6 data. Then, the cycle 6 vision weights were combined with the cycle 5 weights resulting in the vision subsample weight called WGT_VIS and the corresponding bootstrap weights are labelled as BSW1-BSW500. They are available on the vision subsample combined weight file.

9.12.2 Weighting for blood and urine data

Subsamples were selected during the sample collection. For each of the subsamples, a sample weight was created that accounts for this extra step in the sample selection process. As the selection for each subsample was done independently, a separate weight is calculated for each of these subsamples.

9.12.2.1 Weighting for the fasted subsample

The fasted subsample was selected when the sample of dwellings were selected, and thus occurred prior to completion of the household questionnaire. To create the fasted subsample weights, the steps described in Sections 9.1 to 9.4 remain the same and then at the person weight creation step (see Section 9.5) the subsample flags that were assigned to the dwellings were attributed to the selected person(s). Before adjusting for non-response at the questionnaire level (see Section 9.6), the person weight of those selected for the fasted subsample was adjusted to incorporate the subsample sampling weight. An adjustment factor was derived by collection site and stratum as follows:

$$\frac{\textit{Sum of weights for selected persons}}{\textit{Sum of weights of persons selected for the fasted subsample}}$$

The weights of the persons selected for the subsample were accordingly multiplied by this adjustment factor. Persons who were not selected for the fasted subsample were removed from the weighting process.

An additional step was required to adjust for persons who were selected for the subsample but who did not fast or did not provide blood. Such cases were defined as non-respondents to the fasted subsample and to compensate for this non-response an adjustment factor was applied to the weights of the persons with a valid measure. Just as for non-response at the collection site, dwelling (household) and MEC levels, the adjustment is applied within homogeneous response classes. The classes are defined using the score method with logistic regression based on the characteristics available for all respondents and non-respondents. All the characteristics collected on the questionnaire during the interview and measures taken at the MEC, in addition to geographic information and paradata, were available for creating the homogeneous response classes. An adjustment factor was calculated within each class as follows for the fasted subsample:

$$\frac{\textit{Sum of weights for all persons selected to fast and participated at the MEC}}{\textit{Sum of weights for all persons selected who were fasted and had a valid measure}}$$

The weight of the persons who were selected for the subsample who had a valid measure was accordingly multiplied by this adjustment factor. Persons who did not have a valid measure were removed from the weighting process.

The final four steps, first calibration, winsorization, second calibration, and generating bootstrap weights (see Sections 9.8 to 9.11) were then applied to the fasted subsample. After the second calibration, the weight adjustment was obtained. The resulting fasted subsample weight is called WGT_FAST and the corresponding bootstrap weights are labelled BSW1-BSW500. They are available on the fasted subsample weight file.

9.12.2.2 Weighting for the acrylamide, chromium and perfluoroalkyl substances (ACP) in blood subsample

The weighting for all these subsamples was produced by considering as respondents all those who have valid data for at least one of the subsamples. *(note : for analyse on the treatment of missing data as partial non-response can be found in Section **12.1.3 Other missing data** from user guide).*

Respondents were selected for the ACP subsample when they attended the MEC, therefore the non-response adjustments applied to the full sample weights remain the same at the roster, household questionnaire and MEC levels (steps 9.1 to 9.7). Two additional adjustments were applied to the weights following step 9.7 to adjust for respondents not selected for the subsample and to account for non-response in those selected. To adjust for those not selected for the subsample, the weight of each respondent not selected was redistributed to the weights of the selected respondents using the following adjustment factor within each combination of site, age group and sex:

$$\frac{\textit{Sum of weights for all persons participating at the MEC}}{\textit{Sum of weights for all persons participating at the MEC and selected for the subsample}}$$

The weight of each respondent selected for the subsample was then multiplied by this adjustment factor. Respondents not selected for the ACP subsample were removed from the weighting process.

A second adjustment was applied to the weights of the respondents selected for the subsample to account for non-response, which occurred when a respondent did not provide blood or a valid measure could not be obtained for at least one of the laboratory tests.

The adjustment is applied within homogeneous non-response classes, similar to the other non-response adjustments at the collection site, household and MEC levels. The classes are defined using the score method with logistic regression based on the characteristics available for all respondents and non-respondents. Characteristics collected during the household interview and at the MEC, in addition to geographic information and paradata were available to create the homogeneous response classes. An adjustment factor for the ACP subsample was calculated within each class as follows:

$$\frac{\textit{Sum of weights for all persons selected for the subsample}}{\textit{Sum of weights for all persons selected for the subsample and had a valid measure}}$$

The weight of each respondent selected for the subsample and having a valid result for this measure was multiplied by this adjustment factor. Respondents that did not have any valid measures were removed from the weighting process.

The weights are then winsorized and calibrated and the bootstrap weights are generated following the steps described in Sections 9.8 to 9.11. The resulting ACP subsample weight is called WGT_ACP and the corresponding bootstrap weights are labelled BSW1-BSW500. They are available on the ACP subsample weight file.

9.12.2.3 Weighting for the trace elements (cadmium, lead, selenium, and mercuries) in blood subsample

All respondents 3 to 5 years old and 20 to 79 years old were selected for the trace elements in blood when they attended the mobile examination centre (MEC). 50% of 6 to 19 years old were selected when they attended the MEC. Therefore the non-response adjustments to the full sample weights remain the same for the roster, household questionnaire and MEC levels (steps 9.1 to 9.7). Two additional adjustments were applied to the weights from step 9.7 to adjust for respondents not selected for the 6 to 19 years old subsample and to account for non-response in those selected. To adjust for those not selected for the subsample, the weight of each respondent not selected was redistributed to the weights of the selected respondents using the following adjustment factor within each combination of site, age group and sex:

$$\frac{\textit{Sum of weights for all 6 to 19 years old persons participating at the MEC}}{\textit{Sum of weights for all 6 to 19 years old persons participating at the MEC and selected for the subsample}}$$

The weight of each 6 to 19 years old respondent selected for the subsample was then multiplied by this adjustment factor. Respondents not selected for the subsample were removed from the weighting process.

A second adjustment was applied on the weights of the respondents selected for the subsample to account for non-response, which occurred when a respondent did not provide blood or a valid measure could not be obtained on at least one of the laboratory tests.

The adjustment is applied within homogeneous non-response classes, similar to the other non-response adjustments at the collection site, household and MEC levels. The classes are defined using the score method with logistic regression based on the characteristics available for all respondents and non-respondents. Characteristics collected during the household interview and at the MEC, in addition to geographic information and paradata were available to create the homogeneous response classes. An adjustment factor was calculated within each class as follows for the trace elements in blood subsample:

$$\frac{\textit{Sum of weights of persons selected for the subsample}}{\textit{Sum of weights of persons selected for the subsample and had a valid measure}}$$

The weight of each respondent selected for the subsample and having a valid result for this measure was multiplied by this adjustment factor. Respondents that did not have a valid measure were removed from the weighting process.

The weights are then winsorized and calibrated and the bootstrap weights are generated following the steps described in Sections 9.8 to 9.11. The resulting of trace elements in blood weight is called WGT_EL2 and the corresponding bootstrap weights are labelled BSW1-BSW500. They are available on the trace elements in blood subsample weight file. There are supplementary adjustments for 20 to 39 and 40 to 59 years old living with or without kids.

9.12.2.4 Weighting for the environmental urine main subsample

Respondents were selected for the environmental urine main subsample when they attended the MEC, therefore the non-response adjustments to the full sample weights remain the same for the roster, household questionnaire and MEC levels (steps 9.1 to 9.7). Two additional adjustments were applied to the weights from step 9.7 to adjust for respondents not selected for the subsample and to account for non-response in those selected. To adjust for those not selected for the subsample, the weight of each respondent not selected was redistributed to the weights of the selected respondents using the following adjustment factor within each combination of site, age group and sex:

$$\frac{\textit{Sum of weights for all persons participating at the MEC}}{\textit{Sum of weights of persons participating at the MEC and selected for the subsample}}$$

The weight of each respondent selected for the subsample was then multiplied by this adjustment factor. Respondents not selected for the subsample were removed from the weighting process.

A second adjustment was applied on the weights of the respondents selected for the subsample to account for non-response, which occurred when a respondent did not provide urine or a valid measure could not be obtained on at least one of the laboratory tests.

The adjustment is applied within homogeneous non-response classes, similar to the other non-response adjustments at the collection site, household and MEC levels. The classes are defined using the score method with logistic regression based on the characteristics available for all respondents and non-respondents. Characteristics collected during the household interview and at the MEC, in addition to geographic information and paradata were available to create the homogeneous response classes. An adjustment factor for the environmental urine main subsample was calculated within each class as follows:

$$\frac{\textit{Sum of weights of persons selected for the subsample}}{\textit{Sum of weights of persons selected for the subsample and had a valid measure}}$$

The weight of each respondent selected for the subsample and having a valid result for this measure was multiplied by this adjustment factor. Respondents that did not have a valid measure were removed from the weighting process.

The weights are then winsorized and calibrated and the bootstrap weights are generated following the steps described in Sections 9.8 to 9.11. The resulting urine environmental contaminants subsample weight is called WGT_EU and the corresponding bootstrap weights are labelled BSW1-BSW500. They are available on the environmental urine main subsample weight file.

The calibration of the weights is done differently for this subsample. Calibration for each age group is done by sex but there are no supplementary adjustments for 20 to 39 years old living with or without kids.

9.12.2.5 Weighting for the tobacco metabolites subsample

In order to create the tobacco metabolites subsample weights, the steps described in Sections 9.1 to 9.7 are followed. The weights of the respondents selected for the tobacco metabolites subsample were adjusted to incorporate the subsample sampling weight. An adjustment factor was derived by collection site, age, sex and smoker type as follows:

$$\frac{Sum\ of\ weights\ of\ MEC\ respondents\ for\ the\ CHMS}{Sum\ of\ weights\ of\ selected\ persons\ for\ the\ tobacco\ metabolites\ subsample}$$

The weights were then adjusted for non-response which occurred when a respondent did not provide urine or a valid measure could not be obtained for at least one of the tobacco related laboratory tests. The adjustment is applied within homogeneous non-response classes, similar to the other non-response adjustments. The classes are defined using the score method with logistic regression based on the characteristics available for all respondents and non-respondents. All characteristics collected during the interview and measures taken at the clinic, in addition to geographic information and paradata, were available for creating the homogeneous response classes. An adjustment factor was calculated within each class as follows:

$$\frac{Sum\ of\ weights\ of\ selected\ persons\ for\ the\ tobacco\ metabolites\ subsample}{Sum\ of\ weights\ of\ persons\ with\ a\ valid\ measure\ for\ at\ least\ one\ of\ the\ tobacco\ related\ tests}$$

The weights are then winsorized and calibrated and the bootstrap weights are generated following the steps described in sections 9.8 to 9.10. The resulting tobacco subsample weight is called WGT_TOB and the corresponding bootstrap weights are labelled BSW1-BSW500. They are available on the tobacco metabolites subsample weight file.

9.12.3 Weighting for activity monitor data subsample

A separate weight was created for the analysis of the subsample of the activity monitor data (wgt_am), even though all respondents were asked to wear the activity monitor for one week following the MEC appointment. For this reason the activity monitor data is not a true subsample. Analysis could only be performed for 6 to 79 year old respondents who had at least 4 days with at least 10 hours of valid data and 3 to 5 year old respondents who had at least 3 days with at least 5 hours of valid data. To compensate for any bias due to a difference in the respondents who had the required number of valid days of data and those that did not, a separate weight was created for the analysis of activity monitor data.

The weighting steps described in Sections 9.1 to 9.7 were carried out and then an additional adjustment was made to account for respondents who did not have the required number of valid days of data. Cases where there were not at least 4 valid days of data (3 days of valid data for youths 3 to 5 years old) were treated as non-respondents to the activity monitor. The weights of the persons with the required number of valid days were multiplied by an adjustment factor to account for non-response. Similar to the non-response adjustment at the collection site, household and MEC levels, the adjustment is applied within homogeneous response classes. The classes are created based on the probability of response from a logistic model using the characteristics

available for all respondents and non-respondents. All of the characteristics collected during the interview and at the MEC, in addition to geographic information and paradata, were available to create the non-response models. The "score" method is then used to define the classes. The adjustment factor within each class is calculated as follows:

$$\frac{\textit{Sum of weights for all persons particpating at the MEC}}{\textit{Sum of weights for all persons with the required number of valid days}}$$

The weights of the persons who have the required number of days of valid entries were multiplied by this adjustment factor. Persons who did not have the required number of days of valid entry were removed from the weighting process. The final four steps, first calibration, winsorization, second calibration, and bootstrap weights (see Sections 9.8 to 9.11) were then applied to the respondents. After calibration, the weight adjustment was obtained. The resulting activity monitor subsample weight is called WGT_AM and the corresponding bootstrap weights are labelled as BSW1-BSW500. They are available on the activity monitor subsample weight file.

9.12.4 Weighting for jumping mechanography data subsample

A separate weight was created for the analysis of the subsample of the jumping data related to the Jumping Force Measurement (JFM) and Hoping Force Measurement (HFM) (wgt_jump), even though all eligible respondents were asked to participate in the jumping tests. For this reason the jumping data is not a true subsample. However, the JFM and HFM tests were stopped after site 11 in cycle 6. To compensate for the loss of data from sites 12-16 in cycle 6, the collected cycle 6 jumping data was combined with the cycle 5 jumping data.

The weighting steps described in Sections 9.1 to 9.7 were carried out separately on the cycle 6 jumping data. In brief, non-response adjustments factors were applied within homogeneous response classes at the collection site, household and MEC levels on the respondents from the first 11 sites. The classes are created based on the probability of response from a logistic model using the characteristics available for all respondents and non-respondents in sites 1-11. The "score" method is then used to define the classes. The final four steps, first calibration, winsorization, second calibration, and bootstrap weights (see Sections 9.8 to 9.11) were then applied to the respondents from the first 11 sites. After calibration, the weight adjustment was obtained for the cycle 6 data. Then, the cycle 6 jumping weights were combined with the cycle 5 weights resulting in the jumping subsample weight called WGT_JUMP and the corresponding bootstrap weights are labelled as BSW1-BSW500. They are available on the jumping subsample combined weight file.

10. Data quality

10.1 Response rates

One of the most important ways to ensure that the survey data collected are truly representative of the Canadian population is to maximize participation of selected respondents in all parts of the survey. Response rates are very useful data quality indicators that help to measure success in achieving this goal. The following section describes how response rates are derived for the full sample and subsamples.

10.1.1 Household and MEC response rates

In all, 9,143 dwellings were selected within the scope of the Canadian Health Measures Survey (CHMS)[14]. Of these dwellings, 6,737 agreed to provide information on the composition of the household, for a household response rate of 73.7 %. From these respondent households, 9,306 persons were selected (one or two persons per household) to participate in the survey, of whom 8,286 responded to the questionnaire, for a response rate of 89.0%. Of these persons, 5,797 then reported to the CHMS mobile examination centre (MEC) for physical measurements, for a response rate of 70.0 %. At the Canadian level, a combined response rate of 45.9 % was observed for cycle 6 of the CHMS. It is important to note that the combined response rate is not obtained by multiplying the response rates at the person and household levels (or questionnaire level and the MEC level), since two persons were selected in some households. Appendix 7A shows the combined response rates and the relevant information for calculating them for the given age groups and gender.

Below is a description of how the different components of the equation must be used to calculate combined response rates correctly.

Response rate at the household level (HR)

$$HR = \frac{R}{S} = \frac{\textit{\# of respondent households}}{\textit{\# of households within the scope of the survey}}$$

Response rate at the person level among households where <u>one person</u> was selected (questionnaire)

$$PQ1 = \frac{Q1}{PS1} = \frac{\textit{\# of respondents to the questionnaire among households where one person was selected}}{\textit{\# of persons selected among households where one person was selected}}$$

Response rate at the person level among households where <u>two persons</u> were selected (questionnaire)

$$PQ2 = \frac{Q2}{PS2} = \frac{\textit{\# of respondents to the questionnaire among households where two persons were selected}}{\textit{\# of persons selected among households where two persons were selected}}$$

Response rate at the person level among households where <u>one person</u> was selected (MEC)

$$PC1 = \frac{C1}{Q1} = \frac{\textit{\# of participants at the MEC among households where one person was selected}}{\textit{\# of respondents to the questionnaire among households where one person was selected}}$$

Response rate at the person level among households where <u>two persons</u> were selected (MEC)

$$PC2 = \frac{C2}{Q2} = \frac{\textit{\# of participants at the MEC among households where two persons were selected}}{\textit{\# of respondents to the questionnaire among households where two persons were selected}}$$

Ratio for households where <u>one person</u> was selected

$$RR1 = \frac{R1}{R} = \frac{\textit{\# of respondent households among those where one person was selected}}{\textit{\# of respondent households}}$$

Ratio for households where <u>two persons</u> were selected

$$RR2 = \frac{R2}{R} = \frac{\textit{\# of respondent households among those where two persons were selected}}{\textit{\# of respondent households}}$$

Note: The "*# of respondent households"* (R) is the sum of *"# of respondent households among those where one person was selected"* (R1) and the "*# of respondent households among those where two persons were selected"* (R2).

Once all the above components have been calculated, a user can calculate the **combined response rate** (COMBRR) using the following formula:

$$COMBRR = HR * [(RR1 * PQ1 * PC1) + (RR2 * PQ2 * PC2)]$$

The household response rate (HR) is determined by the total number of respondent households (6,737) and the households in scope of the survey (9,143). The households in scope cannot be

calculated separately for each age group and sex. For this reason, the household response rate is not included in Appendix 7A.

$$HR = \frac{R}{S} = \frac{\textit{\# of respondent households}}{\textit{\# of households within the scope of the survey}}$$

$$HR = \frac{6{,}737}{9{,}143} = 0.737$$

$$= \mathbf{73.7\%}$$

Below you will find an example of calculating the combined response rate for Canada using the information provided in Appendix 7A.

$$PQ1 = \frac{Q1}{PS1} = \frac{3{,}708}{4{,}168} = 0.890$$

$$PC1 = \frac{C1}{Q1} = \frac{2{,}596}{3{,}708} = 0.700$$

$$PQ2 = \frac{Q2}{PS2} = \frac{4{,}578}{5{,}138} = 0.891$$

$$PC2 = \frac{C2}{Q2} = \frac{3{,}201}{4{,}578} = 0.699$$

$$R = R1 + R2 = 4{,}168 + 2{,}569 = 6{,}737$$

$$RR1 = \frac{R1}{R} = \frac{4{,}168}{6{,}737} = 0.619$$

$$RR2 = \frac{R2}{R} = \frac{2{,}569}{6{,}737} = 0.381$$

$$COMBRR = 0.737 * [(0.619 * 0.890 * 0.700) + (0.381 * 0.891 * 0.699)]$$

$$= 0.459$$

$$= \mathbf{45.9\%}$$

10.1.2 Blood draw and urine response rates

Although the file with the laboratory data includes all MEC participants, some laboratory analysis could not be performed because the respondents did not or could not provide blood or urine. Of the 5,797 participants who reported to the CHMS MEC for physical measurements, 5,471

participants provided blood and 5,651 provided urine. The combined response rate for blood draw was 43.7% whereas the combined response rate for urine was 44.9%. It is important to note that the combined response rate is not obtained by multiplying the response rates at the person and household levels, since two persons were selected in some households. Appendix 7B shows the combined response rates and the relevant information for calculating them for the given age groups by gender.

Below is a description of how the different components of the equation must be manipulated to calculate combined response rates correctly. Four additional response rates are required to derive the blood draw and the urine combined response rates:

Response rate at the person level among households where <u>one person</u> was selected (blood draw)

$$BC1 = \frac{B1}{C1} = \frac{\textit{\# of persons who provided blood among households where one person was selected}}{\textit{\# of participants at the MEC among households where one person was selected}}$$

Response rate at the person level among households where <u>two persons</u> were selected (blood draw)

$$BC2 = \frac{B2}{C2} = \frac{\textit{\# of persons who provided blood among households where two persons were selected}}{\textit{\# of participants at the MEC among households where two persons were selected}}$$

Response rate at the person level among households where <u>one person</u> was selected (urine)

$$UC1 = \frac{U1}{C1} = \frac{\textit{\# of persons who provided urine among households where one person was selected}}{\textit{\# of participants at the MEC among households where one person was selected}}$$

Response rate at the person level among households where <u>two persons</u> were selected (urine)

$$UC2 = \frac{U2}{C2} = \frac{\textit{\# of persons who provided urine among households where two persons were selected}}{\textit{\# of participants at the MEC among households where two persons were selected}}$$

Blood draw combined response rate (BCOMBRR)

$$BCOMBRR = HR * [(RR1 * PQ1 * PC1 * BC1) + (RR2 * PQ2 * PC2 * BC2)]$$

Urine combined response rate (UCOMBRR)

$$UCOMBRR = HR * [(RR1 * PQ1 * PC1 * UC1) + (RR2 * PQ2 * PC2 * UC2)]$$

where HR, RR1, PQ1, PC1, RR2, PQ2 and PC2 are described in Section 10.1.1.

10.1.3 Response rates for blood and urine subsamples

As described in Section 6.3.4, subsamples of the survey respondents were selected. A combined response rate for each of the subsamples can be obtained by modifying the response rate calculations shown in Section 10.1.1. The fasting subsample response rate is calculated differently to account for the difference in selection for the subsamples. Dwellings are selected for the fasted subsample prior to the questionnaire, whereas individual respondents are selected at the MEC for most of the other subsamples.

10.1.3.1 Fasted subsample response rates

From the 6,737 respondent households, 4,021 were selected for the fasted subsample. In these households, 4,979 persons 6 to 79 years of age were selected (one or two persons per household) to participate in the fasted subsample, of whom 4,389 responded to the questionnaire, for a response rate of 88.2%. Of these persons, 3,043 then reported to the CHMS MEC for physical measurements, for a response rate of 69.3%. Of these persons, 2,432 were actually fasted and had a valid measure on at least one of the laboratory tests. At the Canadian scale, a combined response rate of 36.8% was observed, using the formula given below. It is important to note that the combined response rate is not obtained by multiplying the response rates at the person and household level, since two persons were selected in some households. Appendix 7C shows the fasted subsample combined response rates and the relevant information for calculating them for the given age groups by gender.

Response rate to the questionnaire at the person level among households where <u>one person</u> was selected for the fasted subsample

$$SSQ1 = \frac{SQ1}{SPS1} = \frac{\textit{\# of respondents to the questionnaire among households where one person was selected for the subsample}}{\textit{\# of persons selected for the subsample among households where one person was selected for the subsample}}$$

Response rate to the questionnaire at the person level among households where <u>two persons</u> were selected for the fasted subsample

$$SSQ2 = \frac{SQ2}{SPS2} = \frac{\textit{\# of respondents to the questionnaire among households where two persons were selected for the subsample}}{\textit{\# of persons selected for the subsample among households where two persons were selected for the subsample}}$$

Response rate to the MEC at the person level among households where <u>one person</u> was selected for the fasted subsample

$$SSC1 = \frac{SC1}{SQ1} = \frac{\textit{\# of participants at the MEC among households where one person was selected for the subsample}}{\textit{\# of respondents to the questionnaire among households where one person was selected for the subsample}}$$

Response rate to the MEC at the person level among households where <u>two persons</u> were selected for the fasted subsample

$$SSC2 = \frac{SC2}{SQ2} = \frac{\textit{\# of participants at the MEC among households where two persons were selected for the subsample}}{\textit{\# of respondents to the questionnaire among households where two persons were selected for the subsample}}$$

Response rate to the fasted subsample at the person level among households where <u>one person</u> was selected for the subsample

$$SSF1 = \frac{SF1}{SC1} = \frac{\textit{\# of respondents with a valid measure for the subsample among households where one person was selected for the subsample}}{\textit{\# of participants at the MEC among households where one person was selected for the subsample}}$$

Response rate to the fasted subsample at the person level among households where <u>two persons</u> were selected for the subsample

$$SSF2 = \frac{SF2}{SC2} = \frac{\textit{\# of respondents with a valid measure for the subsample among households where two persons were selected for the subsample}}{\textit{\# of participants at the clinic among households where two persons were selected for the subsample}}$$

Fasted subsample combined response rate

FSCOMBRR = HR * [(RR1 * SSQ1 * SSC1 * SSF1) + (RR2 * SSQ2 * SSC2 * SSF2)]

where HR, RR1 and RR2 are described in Section 10.1.1.

10.1.3.2 Acrylamide, chromium, and perfluoroalkyl substances in blood subsample response rates

Of the 5,797 participants who reported to the CHMS mobile examination centre (MEC) for physical measurements, 2,762 people 3 to 79 years of age were selected for the acrylamide, chromium and perfluoroalkyl substances subsample. Of these persons 2,519 had a valid laboratory results for at least one of the acrylamide, chromium, or perflouroalkyl substances tests. The combined response rate was 42.8%. It is important to note that the combined response rate is not

obtained by multiplying the response rates at the person and household level, since two persons were selected in some households. Appendix 7G shows the combined response rates and the relevant information for calculating them for the given age groups by gender.

Below is a description of how the different components of the equation must be manipulated to calculate combined response rates correctly. Two additional response rates are required to derive the acrylamide, chromium, and perfluoroalkyl substances response rates:

Response rate at the person level among households where <u>one person</u> was selected for the acrylamide, chromium and perfluoroalkyl substances subsample

$$\mathrm{AC1} = \frac{\mathrm{ACR1}}{\mathrm{ACS1}} = \frac{\begin{array}{c}\textit{Number of persons who had a valid laboratory measure}\\ \textit{for at least one of the acrylamide, chromium, or}\\ \textit{perfluoroalkyl substances tests among MEC}\\ \textit{participants where one person was selected}\end{array}}{\begin{array}{c}\textit{Number of participants at the MEC who were selected}\\ \textit{for the acrylamide, chromium and perfluoroalkyl substances}\\ \textit{subsample among MEC participants where one person was selected}\end{array}}$$

Response rate at the person level among households where <u>two persons</u> were selected for the acrylamide, chromium and perfluoroalkyl substances subsample

$$\mathrm{AC2} = \frac{\mathrm{ACR2}}{\mathrm{ACS2}} = \frac{\begin{array}{c}\textit{Number of persons who had a valid laboratory measure}\\ \textit{for at least one of the acrylamide, chromium, or}\\ \textit{perfluoroalkyl substances tests among MEC}\\ \textit{participants where two persons were selected}\end{array}}{\begin{array}{c}\textit{Number of participants at the MEC who were selected}\\ \textit{for the acrylamide, chromium or perfluoroalkyl substances}\\ \textit{subsample among households where two persons were selected}\end{array}}$$

Acrylamide, chromium, perfluoroalkyl substances subsample combined response rate

ACCBRR = HR * [(RR1 * PQ1 * PC1 * AC1) + (RR2 * PQ2 * PC2 * AC2)]

where HR, RR1, PQ1, PC1, RR2, PQ2 and PC2 are described in Section 10.1.1.

10.1.3.3 Trace elements (cadmium, lead, selenium, and mercuries) in blood subsample response rates

Of the 5,797 participants who reported to the CHMS mobile examination centre (MEC) for physical measurements, 4,918 people 3 to 79 year old were selected for the trace elements in blood subsample. Of these persons 4,597 had valid laboratory results. The combined response rate for the trace elements in blood subsample was 43.3%. It is important to note that the combined response rate is not obtained by multiplying the response rates at the person and household level, since two persons were selected in some households. Appendix 7H shows the combined response rates and the relevant information for calculating them for the given age groups by gender.

Below is a description of how the different components of the equation must be manipulated to calculate combined response rates correctly. Two additional response rates are required to derive the trace elements in blood response rates:

Response rate at the person level among households where <u>one person</u> was selected for the trace elements in blood subsample

$$\mathrm{SSU1} = \frac{\mathrm{SU1}}{\mathrm{SC1}} = \frac{\begin{array}{c}\textit{Number persons who had a valid laboratory measure for the trace}\\ \textit{elements in blood subsample among MEC participants}\\ \textit{where one person was selected}\end{array}}{\begin{array}{c}\textit{Number of participants at the MEC who were selected for the trace}\\ \textit{elements in blood subsample among households}\\ \textit{where one person was selected}\end{array}}$$

Response rate at the person level among households where <u>two persons</u> were selected for the trace elements in blood subsample

$$\mathrm{SSU2} = \frac{\mathrm{SU2}}{\mathrm{SC2}} = \frac{\begin{array}{c}\textit{Number persons who had a valid laboratory measure for the trace}\\ \textit{elements in blood subsample among MEC participants}\\ \textit{where two persons were selected}\end{array}}{\begin{array}{c}\textit{Number of participants at the MEC who were selected for the trace}\\ \textit{elements in blood subsample among households}\\ \textit{where two persons were selected}\end{array}}$$

Trace elements in blood subsample combined response rate

EUCBRR = HR * [(RR1 * PQ1 * PC1 * SSU1) + (RR2 * PQ2 * PC2 * SSU2)]

where HR, RR1, PQ1, PC1, RR2, PQ2 and PC2 are described in Section 10.1.1.

10.1.3.4 Environmental urine main subsample response rates

Of the 5,797 participants who reported to the CHMS mobile examination centre (MEC) for physical measurements, 2,631 people 3 to 79 year old were selected for the environmental urine main subsample. Of these persons 2,541 had valid laboratory results. The combined response rate for the environmental urine main subsample was 44.7%. It is important to note that the combined response rate is not obtained by multiplying the response rates at the person and household level, since two persons were selected in some households. Appendix 7I shows the combined response rates and the relevant information for calculating them for the given age groups by gender.

Below is a description of how the different components of the equation must be manipulated to calculate combined response rates correctly. Two additional response rates are required to derive the environmental urine main subsample response rates:

Response rate at the person level among households where <u>one person</u> was selected for the environmental urine main subsample

$$\mathrm{SSU1} = \frac{\mathrm{SU1}}{\mathrm{SC1}} = \frac{\begin{array}{c}Number\ persons\ who\ had\ a\ valid\ laboratory\ measure\ for\ the \\ environmental\ urine\ main\ subsample\ among\ MEC\ participants \\ where\ one\ person\ was\ selected\end{array}}{\begin{array}{c}Number\ of\ participants\ at\ the\ MEC\ who\ were\ selected\ for\ the \\ environmental\ urine\ main\ subsample\ among\ households \\ where\ one\ person\ was\ selected\end{array}}$$

Response rate at the person level among households where <u>two persons</u> were selected for the environmental urine main subsample

$$\mathrm{SSU2} = \frac{\mathrm{SU2}}{\mathrm{SC2}} = \frac{\begin{array}{c}Number\ persons\ who\ had\ a\ valid\ laboratory\ measure\ for\ the \\ environmental\ urine\ main\ subsample\ among\ MEC\ participants \\ where\ two\ persons\ were\ selected\end{array}}{\begin{array}{c}Number\ of\ participants\ at\ the\ MEC\ who\ were\ selected\ for\ the \\ environmental\ urine\ main\ subsample\ among\ households \\ where\ two\ persons\ were\ selected\end{array}}$$

Environmental urine main subsample combined response rate

EUCBRR = HR * [(RR1 * PQ1 * PC1 * SSU1) + (RR2 * PQ2 * PC2 * SSU2)]

where HR, RR1, PQ1, PC1, RR2, PQ2 and PC2 are described in Section 10.1.1.

10.1.3.5 Tobacco metabolites subsample response rates

Of the 5,797 participants who reported to the CHMS mobile examination centre (MEC) for physical measurements, 2,839 people 6 to 79 year old were selected for the urine tobacco metabolites subsample and 2,780 of them provided urine and had a valid measure for at least one of the tobacco measures. At the Canadian scale, a combined response rate of 45.2% was observed, using the formula given below. It is important to note that the combined response rate is not obtained by multiplying the response rates at the person and household level, since two persons were selected in some households. Appendix 7J shows the combined response rates and the relevant information for calculating them for the given age groups by gender.

Below is a description of how the different components of the equation must be manipulated to calculate combined response rates correctly. Two additional response rates are required to derive the tobacco response rates:

Response rate for the tobacco measures among households where <u>one person</u> was selected for the CHMS

$$\text{TOB1} = \frac{\text{TOBR1}}{\text{TOBS1}} = \frac{\begin{array}{c}\textit{Number of persons with valid data}\\ \textit{among households where one person was selected}\end{array}}{\begin{array}{c}\textit{Number of persons selected for the subsample}\\ \textit{among households where one person was selected}\end{array}}$$

Response rate for the tobacco measures among households where <u>two persons</u> were selected for the CHMS

$$\text{TOB2} = \frac{\text{TOBR2}}{\text{TOBS2}} = \frac{\begin{array}{c}\textit{Number of persons with valid data}\\ \textit{among households where two persons were selected}\end{array}}{\begin{array}{c}\textit{Number of persons selected for the subsample}\\ \textit{among households where two persons were selected}\end{array}}$$

Tobacco metabolites subsample combined response rate

$$\text{TOBCRR} = \text{HR} * [(\text{RR1} * \text{PQ1} * \text{PC1} * \text{TOB1}) + (\text{RR2} * \text{PQ2} * \text{PC2} * \text{TOB2})]$$

The CHMS household response rate (HR), ratios of households where one and two people are selected (RR1 and RR2), and questionnaire and clinic response rates among households where one and two people are selected (PQ1, PQ2, PC1, and PC2) are described in Section 10.1.1.

10.1.4 Activity monitor response rates

Of the 5,797 participants who reported to the CHMS mobile examination centre (MEC) for physical measurements, 5,764 participants were offered an activity monitor. Of these persons, 4,050 returned the activity monitor with at least four days of valid entries for 6-79 year olds or at least 3 days of valid entries for 3-5 year olds. The combined response rate (AMCBRR) for the activity monitor was 31.5%. It is important to note that the combined response rate is not obtained by multiplying the response rates by the person and household scales, since two persons were selected in some households. Appendix 7E shows the combined response rates and the relevant information for calculating them for the given age groups by gender.

Below is a description of how the different components of the equation must be manipulated to calculate combined response rates correctly. Two additional response rates are required to derive the activity monitor combined response rates:

Response rate at the person level among households where <u>one person</u> was selected (activity monitor)

$$\text{AM1} = \frac{\text{V1}}{\text{OF1}} = \frac{\begin{array}{c}\textit{\# of persons who returned the activity monitor with the required}\\ \textit{number of days of valid entries among MEC participants}\\ \textit{where one person was selected}\end{array}}{\begin{array}{c}\textit{\# of participants at the MEC who were offered an activity monitor}\\ \textit{among households where one person was selected}\end{array}}$$

Response rate at the person level among households where <u>two persons</u> were selected (activity monitor)

$$AM2 = \frac{V2}{OF2} = \frac{\textit{\# of persons who returned the activity monitor with the required number of days of valid entries among MEC participants where two persons were selected}}{\textit{\# of participants at the MEC who were offered an activity monitor among households where two persons were selected}}$$

Activity monitor combined response rate (AMCBRR)

AMCBRR = HR * [(RR1 * PQ1 * PC1 * AM1) + (RR2 * PQ2 * PC2 * AM2)]

where HR, RR1, PQ1, PC1, RR2, PQ2 and PC2 are described in Section 10.1.1.

10.2 Errors in surveys

A survey yields estimates based on the information collected from a sample of persons. Somewhat different estimates may have been obtained if a complete census had been conducted using the same questionnaire, the same interviewers, the same measurement experts, the same supervisors, the same processing methods, etc. as used for the survey. The difference between the estimates based on the sample and those resulting from a complete enumeration conducted under similar conditions, is called the sampling error of the estimates.

In addition, errors that are not related to sampling may be made at almost any stage of a survey. Interviewers may have misunderstood the instructions, respondents may have made errors when completing the questionnaire, responses may have been incorrectly captured, measurements may have been made incorrectly, and errors may have crept in when the data were processed and totalled. These are all examples of non-sampling errors.

10.2.1 Non-sampling errors

Over a great number of observations, random errors will have little effect on the estimates drawn from the survey. However, errors that occur systematically will contribute to biases in the estimates from the survey. Much time and effort was devoted to reducing non-sampling errors in the survey. Quality assurance measures were applied at each stage of the data collection and processing cycle to control the quality of the data. Further details on the quality assurance procedures for each stage of the survey are provided in Section 10.3 - Quality assurance and control.

The effect of non-response on survey results is a major source of non-sampling error in surveys. The scope of non-response varies from partial non-response (where the respondent does not respond to one or more questions) to total non-response. In cycle 6 of the CHMS, there was little partial non-response, since once the questionnaire began, respondents tended to complete it. There was total non-response when the person selected to participate in the survey refused to do so or could not be contacted by the interviewer. Cases of total non-response were taken into account during weighting by correcting the weights of persons who responded to the survey in order to

compensate for those who did not respond. See Chapter 9 for more information on how the survey weights were adjusted to account for non-response.

10.2.2 Sampling errors

Since the estimates from a sample survey inevitably include sampling errors, good statistical methods require researchers to provide users with some indication of the scope of this error.

Among these indicators are the variance, the standard error, the coefficient of variation and confidence intervals.

For CHMS, calculation of a precise variance is not an easy matter, since there is no simple mathematical formula that can take into account all aspects of the sample design and the selection probabilities. It is therefore necessary to turn to other methods to estimate these measures of precision, such as re-sampling methods. Among these, the bootstrap method is the one recommended for analysing CHMS data. 500 bootstrap replicates have been created for the analysis of the full sample data (see Section 9.11) and for each of the subsamples. The bootstrap replicates are available on each weight file, labelled as BSW1-BSW500 and are used by calculating the value of the desired estimate for each of the bootstrap replicates and then measuring the variability between the bootstrap estimates to come up with the variance. There are several different software packages available that will use the bootstrap replicates to create an estimate of the variance (see section 12.2.4).

The standard error is calculated by taking the square root of the variance. To get a better indication of the size of the standard error, it is often more useful to express the standard error in terms of the estimate being measured. The resulting measure, called the coefficient of variation (CV), is obtained by dividing the standard error of the estimate by the estimate itself, and it is expressed as a percentage of the estimate.

For example, assume that a person estimates that 20% of Canadians aged 12 to 79 smoke regularly and this estimate has a standard error of 0.005. The CV of this estimate is then calculated as follows:

$$(0.005 / 0.20) \times 100\% = 2.5\%$$

Statistics Canada often uses the CV results for data analysis. Table 12.1 in section 12.3.2 provides the Statistics Canada guidelines for releasing estimates based on their CV.

The standard error can also be used to derive confidence intervals. A confidence interval provides a range between a minimum and maximum value that are derived by substracting or adding to the estimate, the standard error (s.e.) multiplied by a parameter associated to the probability that the true value of the population estimate is contained in that range. Typically, the parameter is chosen to provide a probability of 95% but other values can be chosen. The formula for a two-sided confidence interval is in the form:

$\hat{Y} \pm t_{1-\alpha/2,d}$ multiplied by s.e. where $\hat{Y}$ is the weighted estimate
t is the Student distribution

1-α is the chosen probability
d is the number of degrees of freedom[15]

Using the same example as above, a 95% confidence interval for the proportion of smokers would be derived in the following manner:

estimate: 20%,
standard error: 0.005
Student-t value for a two-sided 95% confidence interval with 11 degrees of freedom: 2.2

Minimum value: 0.20 – (0.005 * 2.2) = 0.189
Maximum value: 0.20 + (0.005 * 2.2) = 0.211

Confidence interval: (18.9%, 21.1%).

10.3 Quality assurance and control

There are many problems that can introduce errors in a direct measures survey. These errors can significantly affect the integrity of survey results. To ensure the success of the CHMS in meeting its objectives, quality assurance (QA) and quality control (QC) measures were implemented in all processes including those described below and previously in Chapter 7 (Data Collection).

QA anticipates problems and therefore consists of those activities that take place before data collection or in improving and refining data collection. QC responds to observed problems and thus consists of those activities that take place during and after data collection. The goal of QA and QC is to ensure the reliability and validity of the data and to reduce systematic bias to the lowest possible level.

10.3.1 Training of household interviewers and Mobile Examination Centre (MEC) staff

10.3.1.1 Initial training

Training of all staff emphasized the goals and objectives of the survey, survey methodology, and quality control guidelines. Training also included questionnaire/application content and functionality, standardization of survey procedures, data transmission, refusal conversion techniques, and administrative procedures.

Training involved both classroom training and mandatory reading of procedures and training manuals. The core of the position specific training involved hands-on practice with instructors. Experts from various fields related to the survey measures (e.g., jumping mechanography and vision testing) and biospecimen collection/processing protocols provided seminar sessions to the appropriate staff and participated in aspects of the hands-on training.

<u>Household interviewers:</u>

Prior to collection, regional senior interviewers provided three days of training to the household interviewers. They held mock interviews to familiarize themselves with household questionnaire content, to simulate difficult situations and to practice potential non-response situations. They also discussed techniques for dealing with sensitive questions. Head Office staff provided support in the initial Cycle 6 sites, then any time after where collection was taking place in a new region which had not yet had any training.

MEC staff:

The physical measures for cycle 6 were returning from previous cycles (grip strength, mCAFT, sit and reach). One week of training, conducted at the headquarters, was provided to the MEC staff prior to collection. The administrative staff received specific training on refusal conversion techniques and telephone skills.

The laboratory technologists received supplementary training on blood and urine collection, processing, storage, transport of dangerous goods and running of laboratory tests, as well as re-enforcement training on laboratory protocols.

10.3.1.2 Ongoing training – Dry run day

Prior to the start of collection at each site, one day was set aside for community volunteers recruited by an external placement agency to participate in a visit to the MEC. These days were referred to as dry runs. The purpose of the dry runs was to ensure that all the equipment was functioning correctly and to test the precision of the analytical laboratory tests. It also provided the MEC staff with the opportunity to practice their skills before the beginning of the collection sites, as well as perform on-going training.

10.3.2 Household component

10.3.2.1 Monitoring – Household interview

A rotating schedule of observations ensured that each interviewer was observed two or three times per year. Additional observations were conducted for new interviewers. Formal debriefing sessions were held in two of the sixteen sites where factors affecting data quality were discussed.

In addition to monitoring the work of the interviewers, staff from head office performed interview observations to monitor the functionality of the household interviewing system, the respondents' understanding of the household survey content and the usefulness of the communications tools. Observers provided feedback on these items to the content development and respondent relations teams at head office, and problems were addressed as required.

10.3.2.2 Household questionnaire response rates

Monitoring of the household collection response rates were conducted throughout the cycle by staff at head office preparing collection progress reports. Staff monitored the reasons for non-response by age and sex, the number of contact attempts, the distribution of contact attempts by

time of day, the refusal rates and refusal conversions attempts, and the distribution of the fasting/non-fasting flag to ensure that the target number of respondents per age and sex group would be achieved.

10.3.2.3 Validation of questionnaire responses

At the end of each site, notes and remarks made by interviewers within a respondent's case file were reviewed and adjustments to the data were made when required. Feedback from this review was provided to the household interviewers as required. In addition, the frequency of answer categories within a question was determined for "other-specify", "don't know", and "refusal". Questions with a high rate of non-response were monitored according to expected rates from other Statistics Canada health surveys and investigated if the rate was higher than expected.

Validation of questionnaire responses was also performed on questions that are included in other Statistics Canada health surveys with similar content, in particular, the Canadian Community Health Survey – Annual Component.

10.3.3 Mobile examination centre (MEC) component

The following section will provide information concerning the quality assurance (QA) and quality control (QC) procedures that were put in place and are specific to collection of the physical measures and laboratory data during the visit to the MEC.

10.3.3.1 Equipment selection

The quality of the equipment used for collection was essential to ensure data accuracy and validity. In selecting the appropriate equipment, a combination of consulting, researching, testing, and evaluation was completed. This was done considering industry standards and in conjunction with partners (Health Canada, Public Health Agency of Canada), experts from other physical measures surveys (e.g., National Health and Nutrition Examination Survey (NHANES) in the United States) and CHMS advisory committees.

When determining the MEC laboratory equipment needs, many considerations were taken into account such as the size of the equipment (due to space constraints), the cost of the equipment, the accuracy and precision of the testing equipment, as well as the comparability to other international surveys. The reliability of the instrumentation, including frequency of breakdowns, repairs and maintenance, were also examined. Other items considered when selecting the equipment used for data collection include the infrastructure needs (e.g., use of water, energy consumption, waste disposal), the ease of operation and maintenance, training courses included, the availability and timeliness of service throughout the country, the laboratory biosafety guidelines, and the test throughput.

10.3.3.2 Protocols and procedures

To ensure consistency between MEC staff on all measurement techniques, procedure manuals containing detailed protocols for each measure were developed. These protocols were developed

in consultation with, and reviewed by experts in each measurement field (when appropriate), ensuring the highest quality and least biased data collection. For standardization purposes, these protocols were covered thoroughly during staff training and scripted within the data capture application. Staff members were required to review these protocols periodically during collection so as to keep themselves up to date. When changes to protocols were made, all staff members were informed and provided with the updated protocols, re-training was provided if necessary. All changes to protocols were documented (date of update, reason for updating, process that was followed).

An Equipment Verification, Calibration and Maintenance Manual was developed to ensure that appropriate calibration and maintenance of all testing equipment was performed during collection. The calibration and maintenance was performed to meet or exceed the standards established by the equipment manufacturers.
Experts were consulted for the MEC laboratory standard operating procedures (SOP). The SOP includes information on pre-analytical functions (biospecimen treatment time, storage and shipping conditions), testing protocols (e.g., complete blood count (CBC)), quality control procedures and equipment use, calibration, and maintenance.

All CHMS reference laboratories also followed standard operating procedures that were developed for every test and technique performed in these laboratories. These provided uniform assay protocols that laboratory staff used to ensure similar results and consistent performance. The majority of the reference laboratories are also ISO certified (CTQ - ISO 17025 and ISO 17043; WRHC - ISO 17025; LSPQ - ISO 15189:2012; Dynacare - ISO15189 Plus) and Shriners have an Accreditation Canada Certification.

10.3.3.3 Mobile examination centre (MEC) environment

All efforts were made to ensure measurements were carried out under controlled and standardized conditions and according to specified procedures. Due to the fact that certain measures and equipment were highly sensitive to changes in room temperature (e.g., mCAFT, blood pressure), every effort was made to keep the MEC at a comfortable and constant room temperature (21°C ± 2°C). The environmental conditions of the MEC testing rooms (temperature, humidity and barometric pressure) were recorded at a minimum once per shift, as well as anytime the temperature went outside of the ± 2°C range. In addition, careful monitoring of the conditions within the MEC laboratory were undertaken to ensure that the collection, analysis, storage and shipment of samples were performed under the appropriate conditions.

10.3.3.4 Adherence to pre-testing guidelines

At the beginning of the visit to the MEC, adherence to the pre-testing guidelines (see Appendix 4) was verified and documented within the data capture application and adherence rates were assessed at head office. The purpose of these guidelines was to ensure testing standardization by minimizing the potential that external factors would affect the results of certain tests. Pre-testing standardization was done to enhance the quality of the data collected.

10.3.3.5 Equipment monitoring

Regular verification, calibration, and maintenance of all the equipment used for data collection during the CHMS was performed to ensure data accuracy and validity. This testing was performed to meet or exceed the standards established by the equipment manufacturers and experts in the field.

10.3.3.6 Data entry verification

All paper forms such as the respondent verification sheet and the consent form were manually entered and subsequently verified by a manager to ensure data entry accuracy.

10.3.3.7 MEC – data collection monitoring

Senior staff at the MEC, staff from head office, and certain subject matter experts took turns observing data collection at the MEC. A rotating schedule of observations ensured that each MEC staff member was observed every second site. Additional observations were conducted for new MEC staff.

Observers provided feedback on these items to the MEC staff and to the content development and quality control teams at head office, and problems were addressed as required.

Replicate measurements were done regularly at the MEC on the anthropometry component. This allowed an assessment of the intra-rater reliability of measurements performed by the MEC staff.

10.3.3.8 Data validation

Data validation was performed to ensure that the CHMS data were consistent with similar data sources including other Statistics Canada surveys and international surveys. Data validation was done to compare various physical and laboratory measures, and to compare self-reported data from the household interview to directly measured data from the MEC visit (e.g., height, weight), by site and overall. The physical measures data were also compared against results from previous cycles as well as data sources that contained directly measured data, such as the National Health and Nutrition Examination Survey (NHANES) in the United States.

10.3.3.9 Replicate testing

10.3.3.9.1 Anthropometry replicate testing

Anthropometry replicates were performed between the same HMS or MRT and referred to as intra-tester replicates. Intra-tester replicates are a means of comparing consistency of results amongst the same HMS or MRT. Intra-tester replicate measurements of all the measures of the anthropometry component were completed on 10.7% of respondents that attended the MEC.

10.3.3.10 Vision data review

All visual field and retinal photography tests went through a two stage review process by external reviewers. The first stage involved a double blind review (each case was reviewed by two different individuals without communicating their results) performed by junior staff at the Kensington Eye Institute. If the two reviewers' findings did not agree, issues were found with the data, or pathologies were flagged then the data was sent to the second stage of the review process. The second stage of the review process involved a senior eye institute staff (doctors specializing in glaucoma and retinal photography). The doctor specializing in the appropriate measure would review the data (blind to the results of the first review process) and make a final adjudication on the results.

All visual field test were evaluated for reliability for their inclusion into analysis. All reliable tests are then graded for whether they 'pass' or 'fail'. A pass indicates a normal test, while a fail would indicate a possible pathology. Retinal photos are graded for overall image quality to determine their inclusion. Any images deemed inadequate are not graded. All graded images are graded for diabetic retinopathy, macular edema, macular degeneration, and cup to disc ratios.

10.3.3.11 Activity monitor data review

A data review process was developed to clean and process the activity monitor data. The review process was broken down into four distinct steps: Step 1 involved the downloading and saving of respondent data to ensure that no malfunctioning monitors were returned to the field, as well as following up with respondents who had not returned the monitor. Step 2 involved stacking all respondent data into a single file, and dropping non-valid/bad data (such as initialization errors, or spurious data); Step 3 involved accepting only those with at least 1 day of viable wear-time (at least 5 hours of data for 3 to 5 year olds and at least 10 hours of data for 6 to 79 year olds), and calculating the activity intensity per minute; finally, Step 4 involved applying reserve codes to missing data and generating the derived variables. See Section 12.2.2.5 and Appendix 8 for more information on the quality of the activity monitor data, and on research being done.

10.3.3.12 Peripheral quantitative computed tomography data review

All the pQCT scans underwent an external review process completed by two different reviewers from the Shriners Hospitals for Children. A clinical biochemist specialized in bone health completed a preliminary review to assess the pQCT scans for quality, perform quantitative analyses and generate a report. Then a medical interpretation and final adjudication of the data was performed by Dr. Frank Rauch, pediatrician and researcher specialized in bone health and rare bone diseases at Shriners Hospital for Children. The final report content is detailed in Table 10.3 Quality assessment of the pQCT images. As described in Section 7.4.5.1 – Peripheral quantitative computerized tomography (pQCT), the three sites where the measurements were taken correspond to 4%, 38% and 66% of the distal tibia length of the left leg (unless exclusion criteria for the left leg were met, see Appendix 5 for more details). This external review process was made possible by a grant from the Canadian Institutes of Health Research (Operating Grant, Application number 325809).

Table 10.3 Quality assessment of the pQCT images

Processed	Side-Corr. Eval	4% Mvmnt	T4 Usable	T38 Usable	T66 Usable	Comment
The date when the scan was processed, in the format YYYYMMDD	Evaluation of the limb side information entered at the time of the pQCT scanning. 1= limb side information correct; 2= limb side information incorrect	Qualitative measure of movement at the 4% site. Scale 1-5: 1= none/very minimal; 2= minimal; 3 = moderate; 4= severe; 5= extreme	Usability of the scan at site 4% 1= usable; any other number = not usable	Usability of the scan at site 38% 1= usable; any other number = not usable	Usability of the scan at site 66% 1= usable; any other number = not usable	Text comment

10.3.3.13 Jumping mechanography data review

As described in Section 7.4.5.2, the jumping mechanography assessment is comprised of two tests: the Multiple two-leg hopping (M2LH) test and the Single two-leg jump (S2LJ). All results automatically deemed valid by the collection application, or manually by the HMS administering the test, were reviewed under the supervision of Dr. Frank Rauch, pediatrician and researcher specialized in bone health and rare bone diseases at Shriners Hospitals for Children. This process was made possible by a grant from the Canadian Institutes of Health Research (Operating Grant, Application number 325809). The reviewer's role is to do a final validation of the trials and identify one trial for each test meeting the following criteria:

1- The test was performed correctly according to the output signals in the software;
2- Trial that gives the highest jump of all the valid trials performed for the S2LJ;
3- Trial that gives the hop with the highest force of all valid hops performed for the M2LH.

Table 10.4 Trial invalidation factors for jumping mechanography tests

Test(s)	Description
M2LH & S2LJ	Respondent does not stand still for the required two seconds prior to starting the test. Excessive movement before take-off.
	Respondent does not initiate the test within the first 5 seconds the test is started.
	Respondent does not stand still for the required two seconds after landing from the jump. Excessive movement after landing.
M2LH	Respondent's heels were touching the platform for one or multiple hops.
	Respondent's knees bend beyond 20 degrees continuously throughout the M2LH.

	Respondent uses their arms to gain momentum.
	Selected first hop or last hop in the series.
	Several hops in the series invalidated by the system indicating that the respondent did not perform the test correctly (e.g.: heel touches or feet landing separately).
S2LJ	Feet not leaving or landing at the same time.
	Respondent does not stop jumping after the maximum of one jump is performed during the test. Multiple jumps.

For each test the respondent will usually perform several trials (the target is three trials for the M2LH and three trials for the S2LJ, but they can perform up to five trials for each test). Only the results from the 'best' trial per test will be used in the final report data set.

During the collection, the signal from the force plate sensors was sampled at a frequency of either 400 or 800 Hz. Analysis was performed internally to determine if a low pass filter should be applied to the data before performing a decimation (down sampling) at a factor of 2/1 of the signal sampled at 800 Hz in order to estimate the sequence that would have been obtained at 400 Hz. The conclusion reached by the analysis of 400 vs 800 sampling speeds was that there was no difference between the means of the data at 400 versus the data of 800. As such the data will not be adjusted or decimated.

10.3.4 Head office

10.3.4.1 Correcting for bias

The CHMS has experienced several means of non-response, such as:

1. The selected dwelling may have refused to provide the household composition.
2. The person(s) selected amongst the household members refused to answer the questionnaire.
3. The person refused to participate in the MEC component.
4. The person refused to provide blood and/or urine for the laboratory tests for current or future analysis.
5. The participants refused to use or did not return the activity monitor provided at the MEC.

At each level of non-response, characteristics available for respondents and non-respondents were used in a logistic regression model to identify variables which explain most non-response. The variables which were highly correlated with response or non-response and which were used in the logistic regression models included collection site, age group, sex, household size, education, income, travel distance to the MEC, and the number of days and weeks to contact the dwelling and complete the household questionnaire. Based on the results of each regression, homogeneous response groups are created. The non-response adjustments are applied within these groups to adjust the survey weights (see Chapter 9 - Weighting). Using the survey weights to create estimates

will minimize non-response bias due to differences in the survey variables between respondents and non-respondents.

Several studies were done on the cycle 1 data to test for bias in the respondents to the MEC, because it was believed that less healthy people were less likely to go to the MEC, and for bias caused by the oversampling of respondents aged 20 to 39 who were living with children aged 6 to 11[16]. The results showed that MEC respondents, with the adjustment for MEC non-response, are similar to the household questionnaire respondents. Although the cycle 1 data did not show a significant bias due to the 2-person selection strategy which favoured the selection of a 20 to 39 year old living with a child aged 6 to 11, it was anticipated that a greater bias would be observed in cycle 2 with the addition of the 3 to 5 age group. To compensate for any potential bias, the survey weights and bootstrap weights since cycle 2 are created using a post-stratification by age group and sex, with an additional adjustment for 20 to 39 year olds and 40 to 59 year olds living with and without children aged 3 to 11.

Information on the treatment of other missing data such as partial non-response and lab results with values outside the analytical range can be found in Sections 12.1.2 and 12.1.3.

10.3.4.2 Errata

An errata is available containing inconsistencies discovered in the data files and documentation after dissemination. It is updated regularly and available upon request.

11. File Usage

11.1 Description of data files

The Canadian Health Measures Survey (CHMS) dissemination files only contain data from respondents who attended the mobile examination centre (MEC) and agreed to share their responses with Health Canada (HC) and the Public Health Agency of Canada (PHAC). For cycle 6 of the CHMS, after the completion of processing, and with the removal of non-share records [17], 5,797 records remain on the full sample data files.

There will be 21 data files released for CHMS cycle 6, with the releases occurring in several releases.

See Section 12.2.3 for information on combining data from multiple cycles of the survey.

Date of release	Files Released	Data File Name*	Corresponding Bootstrap Weights File*
October 14, 2020	• Household full sample (excludes medication)	hhd_full	wgt_full
	• Clinic full sample (excludes: vision, bone health, activity monitor and medication)	clc_full	wgt_full
December 14, 2020	• Vision full sample	vis_full	wgt_full
	• Vision subsample - Cycles 5 & 6 combined (visual field and retinal photography)	vis_sub	wgt_vis
	• Relationship file (two person HHLD)	rel	wgt_full
	• Postal code file - Contains postal code information for each respondent in the survey (special restrictions are in place for the use of this file - available in RDCs only)	pc	n/a

Date of release	Files Released	Data File Name*	Corresponding Bootstrap Weights File*
June 28, 2021	• Non-environmental lab data full sample (includes iodine)	nel_full	wgt_full
	• Fasted subsample	fast_sub	wgt_fast
	• Medication full sample (includes dosage DVs)	med_full	n/a
September 1, 2021	• Activity monitor data full sample	am_full	wgt_full
	• Activity monitor data subsample	am_sub	wgt_am
	• Activity monitor per minute data	am_permin	n/a
October 27, 2021	• Jumping data subsample (cycles 5 & 6)	jump_sub	wgt_jump
	• Peripheral quantitative computed tomography (pQCT) data full sample	pqct_full	wgt_full
December 14, 2021	• Environmental lab blood full sample	el_full	wgt_full
	• Acrylamide, chromium and perfluoroalkyl substances - blood subsample	acp_sub	wgt_acp
	• Trace elements (cadmium, lead, selenium, and mercuries) in blood - Environmental lab subsample	el2_sub	wgt_el2
	• Environmental urine - main subsample	eu_sub	wgt_eu
	• Tobacco metabolites - urine subsample	tob_sub	wgt_tob
Fall 2022	• Organohalogens in pooled serum	oh_pooled	n/a

*All files are available in .txt and .sas7bdat formats.

Most of the data files listed in the table above contain a weight variable. The bootstrap weight files contain more detailed information (i.e. 500 bootstrap weight variables for variance calculation purposes). More information on bootstrap weights and weighting can be found in Chapter 9.

The survey includes respondents 3 to 79 years of age; however, some of the measures or tests were only done for certain age groups, for one sex, on fasting respondents, or on a random subgroup of ages. Refer to the CHMS Content summary (available upon request) for more information.

For information regarding the actual questions asked of the respondents, users can consult the household and clinic questionnaires at the following link: Household and clinic questionnaires.

11.1.1 Household full sample file (hhd_full)

This data file contains the responses collected during the household interview, except for the medication data and derived variables which will be released separately at a later date.

The file also contains derived variables (DVs) created by the collection applications and other DVs created after data processing. A household/MEC DV document is available that contains information on how these DVs are created/calculated.

Each of the 5,797 records represents one respondent who went to the MEC, participated in at least some of the measures, and then agreed to share their household and MEC results with HC and PHAC.

The data contained in this file covers household questionnaire content on the following topics:

- Alcohol use
- Anthropometry
- Chronic conditions
- General health
- Health utility index
- Illicit drug use
- Mental health
- Musculoskeletal fitness
- Neighbourhood environment
- Nutrition (dietary fat, grains, fruit and vegetable, milk and dairy, meat, salt, water and soft drink consumption)
- Physical activity and sedentary behaviours
- Pregnancy/Birth
- Sexual health
- Sociodemographic characteristics
- Sleep apnea
- Smoking, exposure to second-hand smoke and vapour, and electronic cigarettes
- Steroids and osteoporosis medication
- Time spent outdoors
- Toxoplasmosis

11.1.2 Clinic full sample file (clc_full)

This data file contains the responses collected during the MEC visit (questionnaire responses and physical measures results), with the exception of medication, vision, jumping mechanography and pQCT which will be released separately at a later date.

The file also contains derived variables (DVs) created by the collection applications and other DVs created after data processing. A household/MEC DV document is available that contains information on how these DVs are created/calculated.

Each of the 5,797 records represents one respondent who went to the MEC and participated in at least some of the measures and then agreed to share their household and MEC results with HC and PHAC.

The data contained in this file covers clinic questionnaire and/or physical measures content on the following topics:

- Adherence to guidelines and screening questions
- Anthropometry
- Cardiovascular health and fitness
- Musculoskeletal fitness (excluding pQCT and jumping mechanography)
- Nutrition (fish and shellfish consumption)

11.1.3 Income files

In cycle 5, respondents were asked permission to link to and use administrative files to obtain the income. However, the share income statement did not include the permission to use the information from these files. Therefore, two sets of income data were produced: one that can be used in the Research Data Centers (linked income file) and one that can be used by the share partners (share income file). See the *CHMS Data User Guide: Cycle 5* for details about these files. In cycle 6, questions on the personal and household income were asked as in previous cycles (cycles 1 to 4) and data are included in the household full sample file

11.1.4 Postal code file (pc)

This file contains the six character postal code (DHH_DPC) for all 5,797 persons who participated in the survey, along with their CLINICID (see Section 11.2 - Key variables for linking data files). The file is only available in Statistics Canada Research Data Centers (RDCs). Users will need to link the postal code file to another data file, such as the full sample household or full sample clinic data file, as well as to the corresponding bootstrap weights file in order to produce estimates for the total population (see Section 11.5 - Use of weight variables). Researchers must specifically state the need for the file in their RDC application (as it is separate from other CHMS data files) and must clearly explain what analysis will be done using postal codes.

Users need to be aware of the restrictions on the use of this file. The file can only be used for deriving variables that are to be created and analyzed at the national level. It cannot be used to try

to produce small area estimates, as the survey is designed only for national level estimates. The file must be used at the person level with the appropriate weights. It cannot be used to compare members of the same household or for producing household estimates since household weights are not available.

As with all other data, the postal codes of respondents may not be removed from the RDCs; however, a public document is available for each cycle of the survey that lists all CHMS collection sites along with the forward sortation area (first three characters of the postal codes) that are covered off geographically by each site.

11.1.5 Vision data full sample (vis_full)

This file contains data for 5,797 respondents and includes 87 variables specific to the vision health questionnaire and screening tests (visual acuity and intraocular pressure) as well as the full sample weight. A complete list of variables can be found by referring to the vision file data dictionary.

11.1.6 Vision data subsample (C5C6_vis_sub)

This file contains data for 5,125 respondents from both cycle 5 and cycle 6 and includes 186 variables specific to the vision health screening tests (visual field and retinal photography) as well as the combined subsample weight. A complete list of variables can be found by referring to the vision file data dictionary.

NOTE: the visual field (FDT) and the retinal photography (RTP) data collections were stopped after site 10 in cycle 6. Data from these sites (11 to 16) were not included on the file.

Due to there being only one site in the Atlantic region that collected FDT and RTP data in cycle 6, the Atlantic stratum had to be collapsed with the Quebec stratum to draw the bootstrap replicates. The variable STRATUM is added to the bootstrap weight file (VIS_WGT) to ensure the ability to calculate degrees of freedom.

15 – Atlantic (Cycle 5)
24 – Quebec (Cycle 5) / Atlantic & Quebec (Cycle 6)
35 – Ontario
49 – Prairies
59 – British Columbia

11.1.7 Relationship (DYAD) data file (rel)

The relationship data file contains information linking individuals of participating 2-person households in the Canadian Health Measures Survey (CHMS). The data file contains four variables: the household ID (HHLDID) which links individuals in a given household, the respondent CLINICID, the relationship type (REL) and the full weight variable (WGT_FULL).

There were 1,020 dyads (2-person households with child 3-11 years old at clinic) in Cycle 1, 1,601 in Cycle 2, 1,505 in Cycle 3, 1,518 in Cycle 4, 1,549 in Cycle 5 and 1,514 in Cycle 6.

The CLINICID uniquely identifies each respondent and is the key linking variable found on all CHMS files. It is used to link each relationship file to another data file, such as the Cycle 1, Cycle 2, Cycle 3, Cycle 4, Cycle 5, or Cycle 6 master file, as well as to the corresponding bootstrap weights file in order to produce estimates for the total population (see Section 11.5 – Use of weight variables).

Users need to be aware of the restrictions on the use of the six relationship files. The files can only be used for deriving variables that are to be created and analyzed at the national level. They cannot be used to try to produce smaller estimates, as the survey is designed only for national level analysis.

The files may be used with full sample or subsample weights. As a very rough guideline, data users should have at least 500 pairs in each subgroup for their analysis. In other words, a sex-specific analysis on 6-11-year-olds would require 500 pairs of boys and 500 pairs of girls. However, if the outcome of interest is very rare in the sample (e.g., <10% of the sample has X condition of interest), greater sample sizes will be required.

The birth parent/birth child dyad is the most prevalent relationship in the data and may be analyzed. Users may also analyze the child/sibling relationship but must use caution in their analysis: due to the sampling strategy of 2 person households in the CHMS in Cycles 1 to 6, one sibling will always be between 3 and 11 years old (6 to 11 years old in Cycle 1) and the other sibling will always be 12 or older. This may not be representative of the average sibling age relationship in the Canadian population. All other dyads are of insufficient sample size for analysis, even if combining the maximum number of cycles of data. A feasibility report on the use of the relationship files is available upon request.

Birth Parent/Birth Child dyad breakdown by age group of child:

	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6
Biological parent and child 3 to 5 years old	N/A	471	410	452	459	472
Biological parent and child 6 to 11 years old	675	643	581	597	597	611

11.1.8 Non-environmental lab data full sample (nel_full)

This file includes data for the 5,797 respondents and contains 59 variables including the appropriate weight variable. Specifically, this release of the nel_full data file includes 54 non-environmental laboratory variables. It also includes four variables confirming the respondent's fasting status prior the phlebotomy component. Of the 59 variables released, four are DVs created during data processing. The *Derived Variable Specifications: Cycle 6 Non-Environmental Lab and Fasted* document is available, and contains information on how these DVs are created/calculated.

A complete list of variables can be found by referring to the non-environmental lab file data dictionary.

NOTE: In cycle 6, some samples had been lost for variable LAB_ALT (alanine aminotransferase) due to freezing in sites 7 and 12.

11.1.9 Fasted subsample (fast_sub)

This file contains data for a subset of 2,432 of the 5,797 CHMS respondents, namely the selected respondents who had fasted for a minimum of 10 hours. The file contains data for six variables: insulin, low density lipoproteins (LDL), triglycerides, apolipoproteins A1 and B, and c-telopeptide of collagen type I. One of the six variables is a DV that is created during data processing. The *Derived Variable Specifications: Cycle 6 Non-Environmental Lab and Fasted* document is available, and contains information on how this DV is created/calculated. The file also contains a weight specific to this sub-population. A complete list of variables can be found by referring to the fasted subsample file data dictionary.

11.1.10 Medication full sample - including dosage derived variables (med_full)

This file contains data for the 4,988 respondents that reported taking any medication in the past month at either the household or the MEC interview (20,374 medications total). The file contains 39 variables, including the data collected as part of the medication blocks at the household and MEC, as well as the ATC and AHFS codes described in Appendix 6 – Medication Classification System. This file also contains data for the 2,949 respondents who reported a medication (6,146 medications in total) containing one of 11 specific ingredients (calcium, folate/folic acid, iodine, iron, magnesium, omega-3, omega-6, vitamin B12, vitamin C, vitamin D, and zinc). These individuals were asked an additional series of questions about the medication (5 variables and one derived variable), and yes/no flags were created to indicate whether a medication containing a given ingredient was reported (one flag overall, and one flag for each ingredient = 12 flags), as well as a derived variable of the estimated daily intake of a given ingredient (one derived variable per ingredient = 11). Finally, the file contains the record identifier as well as MEUFNUM which is used as a unique identifier for each medication reported for a given respondent. There is no weight variable on this file because it can contain more than one observation per person if more than one medication was reported. For more information on how to use the data contained on this file, see Section 12.1.4 – Medication data and derived variables.

NOTE: In cycle 6, more “not stated” values were observed at variable NPI_25B compare to cycle 5. This difference is due to uncompleted transfers of medication data files from the household interview to the MEC (from site 1 to site 3).

11.1.11 Activity monitor data full sample (am_full)

This file contains data for the 5,797 respondents and 92 activity monitor variables as well as the full sample weight. The Derived Variable (DV) Specifications: Cycle 6 Activity Monitor Full Sample and Subsample document is available, and contains information on how these DVs are created/calculated. A complete list of variables can be found by referring to the activity monitor

full sample file data dictionary. See Section 12.2.2.5 and Appendix 8 for more information on using this file.

11.1.12 Activity monitor data subsample (am_sub)

This file contains data for the 4,050 respondents who wore their activity monitor for at least 10 hours per day (5 hours for 3 to 5 year olds) for 4 days or more (3 days or more for 3 to 5 year olds). This file contains 23 variables that were derived based on the variables found on the activity monitor full sample file (am_full). The file also contains a weight specific to this sub-sample. The Derived Variable (DV) Specifications: Cycle 6 Activity Monitor Full Sample and Subsample document is available, and contains information on how these DVs are created/calculated. A complete list of variables can be found by referring to the activity monitor subsample file data dictionary. See Section 12.2.2.5 and Appendix 8 for more information on using this file.

11.1.13 Activity monitor data per minute file (am_permin)

This file includes data for all the respondents who accepted to wear the activity monitor for a 7-day period, who returned the monitor to Head Office and for whom data was successfully downloaded from the device. The activity monitor per minute data file is large, it contains multiple records per respondent going up to 10,080 records for the respondents where the collection interval (or epoch) was set to 1-minute and up to 28,800 for those where the collection interval was set to 15-seconds. It represents a total of 61,330,219 records for the cycle 6 file. Two of the main variables included in this file were extracted from the device using the *Actical* software, one is expressed as a count value per minute (cpm) which represents the physical activity intensity and the second is a step count value which represents the number of steps taken per minute. Date and time variables were added to the file during data processing. A complete list of variables can be found by referring to the activity monitor per minute file data dictionary. See Section 12.2.2.5 and Appendix 8 for more information on the quality of the activity monitor data, and on research being done.

11.1.14 Peripheral quantitative computed tomography (pQCT) data full sample (pqct_full)

This file contains data for the 5786 respondents and includes 37 variables specific to the pQCT scan as well as the full sample weight. A complete list of variables can be found by referring to the pQCT file data dictionary.

11.1.15 Jumping data subsample (Cycle 5 and 6 combined) (C5C6_jump_sub)

The jumping mechanography data collection was stopped after site 11 in cycle 6. To compensate for the loss of data from sites 12-16 in cycle 6, the collected cycle 6 jumping data was combined with the cycle 5 jumping data.

This file contains data for 8773 respondents and includes 17 variables specific to the Multiple Two-Leg Hoping (M2LH) and the Single Two-Leg Jump (S2LJ) tests as well as the cycle 5 and

cycle 6 combined subsample weight. A complete list of variables can be found by referring to the jumping file data dictionary.

NOTE: Data from the sites 12 to 16 of cycle 6 were not included on the file.

11.1.16 Environmental lab data full sample file (el_full)

This file contains data for 5,797 respondents and includes one variable specific to free cotinine measured un serum as well as the full sample weight. Additional information can be found by referring to the environmental lab data full sample file data dictionary.

Note: Free cotinine measured in serum (LAB_SCOT) is analysed in the laboratory using the smoke type derived variable from the household file (SMKDSTY), as the starting point. While the two tests had different limits of detection (LOD) applied at the lab, a researcher should apply the LOD conversion, of 0.005/2, because all the values with a less than LOD result for the higher detection test would have been retested with the lower detection test. For more details, see section 8.6 Analytical range and the Standard operating procedures (SOPs) that is available upon request.

11.1.17 Acrylamide, chromium, and perfluoroalkyl substances (ACP) in blood subsample file (acp_sub)

The file contains data for a subset of 2,519 respondents of the 2,762 selected for the ACP subsample. The file contains 12 variables: acrylamide hemoglobin adduct, glycidamide hemoglobin adduct, chromium, perfluorobutane sulfonate, perfluorodecanoic acid, perfluorohexane sulfonate, perfluorohexanoic acid, perfluoro-n-butyric acid, perfluorononanoic acid, perfluorooctane sulfonate, perfluorooctanoic acid and perfluoroundecanoic acid. The file also contains a record identifier and a weight specific to this subpopulation (wgt_acp).

11.1.18 Trace elements (cadmium, lead, selenium, and mercuries) in blood – environmental lab subsample file (el2_sub)

The file contains valid data for a subset of 4,597 respondents of the 4,918 selected for the trace elements in blood main subsample. The file contains many variables, including cadmium, lead, selenium and mercury (total, inorganic and methyl) in blood.

The file also contains a record identifier and a weight specific to this subpopulation. A complete list of the variables can be found by referring to the trace elements in blood main subsample file data dictionary.

11.1.19 Environmental urine main subsample file (eu_sub)

The file contains valid data for a subset of 2,541 respondents of the 2,631 that were selected for the environmental urine main subsample. The file contains many variables, including phthalate metabolites (23 variables), cadmium (1 variable), boron (1 variable), bisphenol A (1 variable), organophosphate insecticides (6 variables), total ethylene thiourea (1 variable), ortho-phenylphenol (2 variables), pyrethroid pesticides (5 variables), arsenics (5 variables), plasticizers

(11 variables) and parabens (4 variables). Results adjusted for urine creatinine and/or adjusted for specific gravity can be created by the user. See section 8.6 for more details.

The file also contains a record identifier and a weight specific to this subpopulation. A complete list of the variables can be found by referring to the environmental urine main subsample file data dictionary.

11.1.20 Tobacco metabolites subsample file (tob_sub)

The file contains data for a subset of 2,780 respondents of the 2,839 that were selected for the tobacco metabolites subsample. These respondents provided urine and had a valid result for at least one of the tobacco related measures. The file contains 3 variables: total NNAL, free Cotinine, free trans-3-hydroxycotinine. The file also contains a record identifier and a weight specific to this subpopulation.

11.2 Key variables for linking data files

As a result of the large number of files disseminated for cycle 6, it will often be necessary to be able to link these files to have access to a larger pool of information. In particular, users may want to link the subsample files and vision and medication files to the household and clinic full sample files in order to obtain contextual questionnaire and physical measures information for their analysis. Users will also need to link the appropriate bootstrap weights file to each data file in order to produce estimates for the total population (see Section 12.1).

In order to facilitate the linking of two or more files, a variable that uniquely identifies each respondent on each file is required. For the CHMS, the variable to be used is called CLINICID. With this variable, data users are able to join the data from any subsample file to the household and clinic full sample files for a particular respondent.

11.3 Key variables and definitions

Other variables which may be particularly useful for data users are listed below:

CLC_AGE	Respondent's age at the time of the MEC visit
CLC_SEX	Sex of Respondent
DHH_MS	Marital Status of Respondent
EDUDR04	Highest level of education - respondent, 4 levels
EDUDR10	Highest level of education - respondent, 10 levels
PROXY	Proxy or non-proxy interview
WGT_FULL	Weight for the following full sample files: ▪ Household ▪ Clinic

The naming convention used for the weighting variables listed above is the same as the naming convention for the corresponding bootstrap weight files, with the only differences being the suffix in the file name and the fact that the variable names are uppercase and the file names are lowercase.

For example, ".txt" can be removed from the bootstrap file "wgt_full" and then capitalized to obtain the name of the weighting variable on the full sample data files.

11.4 Use of age and sex variables

Age and sex are collected twice during the survey process for the CHMS; first through the household interview, then again during the MEC component. Because the two appointments could be several days apart, it is possible that a respondent's age could be different on the household and MEC files. Each application uses the age and sex information that was collected as the reference for coverage and skips patterns for sections within that application. For that reason, it is important to use the appropriate age and sex variables for the data being analyzed. Incorrect usage of age and sex variables could lead to errors in analyses, due to respondents whose ages changed between the two appointments. For example, a respondent who is 11 at the time of the household interview and 12 at the time of the MEC visit will be in different age groups depending on the data being analyzed (i.e. 6 to 11 for household analysis and 12 to 19 for MEC analysis).

For household variables, DHH_AGE and DHH_SEX should be used; whereas, for MEC variables, CLC_AGE and CLC_SEX should be used. If variables from both components are analyzed, use the age and sex variables for the module containing the most important variables of interest.

11.5 Use of weight variables

The CHMS is a sample survey, which means that the respondents "represent" many other Canadians not included in the survey. For example, a 1% sample would mean that each CHMS respondent represented 100 Canadians. In order that the results of the survey are representative of the population, survey weights were created. These survey weights, when applied to the survey results, enable data users to create estimates for the entire population.

Each respondent record on each data file has a unique survey weight attached to it. In order to produce estimates for a particular characteristic, the data user must sum the weights for each respondent with that characteristic. The total created by that calculation would produce an estimate of that characteristic in the total population. There are various software packages available that will use survey weights to produce estimates (see Section 12.2.4 for more information).

Because of the small sample size of individual cycles of the CHMS, the results should only be used to produce national estimates. Due to the different number of respondents contained within the various files output for the CHMS, each file produced will contain a weight variable. When calculating estimates on linked files (i.e. a subsample file linked to the full sample file), the weight stored on the file containing the variable of interest is to be used.
For information on the use of combined weight files, see Section 12.2.3.

11.6 Variable naming convention

The CHMS naming convention for cycle 6 of the survey follows the same pattern as that used by previous cycles of the CHMS and many other Statistics Canada surveys. The variable name is constructed in a way that allows the data users to easily identify the originating section of the survey, the type of variable, and the survey question that collected the data. The variable names

have also been created in such a way as to identify similar data between different cycles of the CHMS.

Each variable name must adhere to a mandatory requirement which restricts variable names to a maximum of 8 characters for ease of use by analytical software products. As a result, each character of the variable name contains information about the data contained in the variable.

Generally speaking:

Positions 1 - 3: Section name
Position 4: "_" underscore (_= collected data) or Variable type
Positions 5 - 8: Question reference (question number or acronym/short form to represent the concept of the variable)
For example: The variable from question 101 of the household Chronic Conditions section, CCC_101:

Positions 1 - 3: CCC chronic conditions section
Position 4: "_" underscore (_= collected data)
Positions 5 - 7: 101 question number

11.6.1 Position 4: Place Holder or Variable Type

_ Collected variable: A variable that appeared directly on the questionnaire or was collected during the direct measures components

C Coded variable: A variable coded from one or more collected variables using standardized coding systems (e.g., SIC, NAICS, ATC)

D Derived variable (DV): A variable assigned or calculated from one or more collected or coded variables

F Flag variable: A variable calculated by the data collection computer application for later use during the interview, or assigned based on one or more collected variables (similar to a DV, e.g. CCCF1). In the case of the income imputation flag (INCFHHLD), derived in head office after imputation.

G Grouped variable: Collected, coded, or derived variables collapsed into groups

11.6.2 Positions 5-8: Question reference

In general, the fifth to seventh positions follow the variable numbering used on the questionnaire. The letter "Q" for questions read to the respondent or "N" for interviewer-completed questions is removed, and all question numbers are presented in a two-digit format which sometimes requires to remove leading zeroes. For example, question Q01A in the questionnaire becomes simply 01A, question Q15 becomes simply 15, and question Q040BM becomes 40BM. In other cases, an

acronym or short form is used to represent the concept of the variable and an eighth character is sometimes required.

For questions that allow multiple response categories to be selected, the final position in the variable naming sequence is represented by a letter. For this type of question, new variables were created to differentiate between a “yes” and “no” answer for each response category. For example, if multiple responses could be selected for Q2, the new questions would be named Q2A for category 1, Q2B for category 2, Q2C for category 3, etc. If only categories 2 and 3 were selected, then Q2A = No (“2” on the file), Q2B = Yes (“1” on the file), Q2C = Yes (“1” on the file) and Q2D = No (“2” on the file).

To help reconcile the variable names found on the household and clinic full sample data files from those on the household and clinic questionnaires, users can consult the questionnaires, data dictionaries, and other documentation available upon request.

11.7 Access to data files

Access to CHMS data is provided through the RDCs. These research data centres require researchers to submit a research proposal before access to the data is granted. These projects are accepted based on a set of specific rules. When the project is accepted, the researcher is designated as a "deemed employee" of Statistics Canada for the duration of the research, and given access to the data from designated Statistics Canada sites. For more information, please consult the Statistics Canada webpage: www.statcan.gc.ca/eng/rdc/index .

Another means of access to the data files is to provide Statistics Canada with specifications for tabulations. On a fee-for-service basis, these tables are programmed and run against the data files by Statistics Canada employees. This service allows users who do not possess knowledge of tabulation software products or who do not have access to a RDC to get custom results. The results are screened for confidentiality and reliability concerns before release to the client. For more information, contact Health Statistics Division Client Services at 613-951-1746 or by e-mail at statcan.hd-ds.statcan@canada.ca.

Finally, HC and PHAC government employees have access to CHMS data files through a share agreement. At the MEC, respondents provide their consent to “share the information collected during the survey with its partners” who “will keep the information confidential, and use it for statistical purposes only”. Within HC/PHAC, employees who require CHMS information in order to do their work apply for access and are granted approval upon filling out and signing an acknowledgment document and providing a purpose of research. To arrange access, HC employees should contact the Data and Information System team at hc.dais.sc@canada.ca while PHAC employees should contact the Data Coordination and Access Program at phac.dcap-pacd.aspc@canada.ca.

12. Guidelines for tabulation, analysis and release

This chapter provides guidelines to be applied by data users in tabulating, analyzing, publishing, or otherwise releasing any data derived from the survey files.

12.1 Guidelines for tabulation

The sample design used for this survey is not self-weighted. In other words, the sampling weight is not the same for all the persons included in the sample. Even to produce simple estimates, including ordinary statistical tables, the user must employ the appropriate sampling weight. Otherwise, the estimates calculated on the basis of the full sample data files cannot be considered representative of the population observed, and they will not correspond to those of Statistics Canada. For further information on the creation of the survey weights (the sampling weights that have been adjusted for non-response), refer to Chapter 9 of this User Guide. Information on the use of the weight variable is found in Section 11.5 and information on the key variables used for linking data files is found in Section 11.2.

Users should also keep in mind that because of the treatment reserved for weights, some software packages do not yield estimates that exactly match those of Statistics Canada.

12.1.1 Tabulation of categorical and quantitative estimates

There are two main types of point estimates of population characteristics that can be generated from the data files: categorical estimates and quantitative estimates. A brief explanation of each estimate type is given prior to describing how the survey data can be tabulated.

12.1.1.1 Categorical estimates

A categorical estimate is the estimate of the number or percentage of the surveyed population that possess a certain characteristic or fall into some defined category. For example, the proportion of respondents with diabetes and the number of persons with each type of diabetes are both categorical estimates.

Examples of categorical questions:

Do you have diabetes? (CCC_Q51)
1 Yes
2 No

If Yes, were you diagnosed with: (CCC_Q52)
1 … insulin dependent diabetes (Type 1)?
2 … non-insulin dependent diabetes (Type 2)?
3 … gestational diabetes ?

Estimates of the number of people with a certain characteristic can be obtained from the data file by summing the final weights of all records [14] possessing the characteristic of interest. A proportion is calculated as $\hat{X}/\hat{Y}$ using the following steps:

a) Summing the final weights of records having the characteristic of interest for the numerator ($\hat{X}$)
b) Summing the final weights of records having the characteristic of interest for the denominator ($\hat{Y}$)
c) Dividing the numerator estimate by the denominator estimate

For example, to obtain the proportion of respondents 20-79 with Type 2 diabetes, the numerator is obtained by summing the weights (WGT_FULL) for all respondents aged 20 to 79 who answered non-insulin dependent diabetes (Type 2), answer 2 to CCC_Q52. The denominator is the sum of the weights for all 20-79 year old respondents to the survey.

12.1.1.2 Quantitative estimates

A quantitative, or continuous, estimate is an estimate of the total or of the mean, median or other measure of central tendency of quantities based on some or all members of the surveyed population. For example, the average age of first diagnosis with type 2 diabetes is a quantitative estimate.

Example of a quantitative question:

If CCC_Q52 = 1 or 2 then:
How old were you when this was first diagnosed? (CCC_Q53)
|_|_|_| Age in years

For Canadian Health Measures Survey (CHMS) data, two different estimates of the mean are commonly generated for continuous variables: the arithmetic mean and the geometric mean. The arithmetic mean is the simple average of the data and is best suited for variables that are evenly distributed around the mean. The arithmetic mean is calculated by:

a) Multiplying the value of the variable of interest by the final weight and summing this quantity over all records of interest to obtain the numerator ($\hat{X}$)
b) Summing the final weights of records having the characteristic of interest for the denominator ($\hat{Y}$)
c) Dividing the numerator estimate by the denominator estimate.

For example, to obtain the average age when respondents were first diagnosed with Type 2 diabetes, first compute the numerator ($\hat{X}$) by summing the product between the age given in CCC_Q53 and the survey weight (WGT_FULL) for all respondents who answered 2 to question CCC_Q52 (have Type 2 diabetes). Next, sum the value of WGT_FULL for all respondents who answered 2 to question CCC_Q52 to obtain the denominator ($\hat{Y}$). Divide $\hat{X}$ by $\hat{Y}$ to obtain the estimate of the average age of first diagnosis with Type 2 diabetes.

The geometric mean is also a measure of central tendency; however, it is more robust to the presence of extreme values then the arithmetic mean. Thus, the geometric mean provides a better indication of central tendency for data that is highly skewed, meaning the data is unevenly spread towards higher or lower values. The geometric mean is typically used for environmental chemicals. It is calculated by:

a) Computing the natural log transform of the value of the variables of interest and multiply it by the final weight. The numerator ($\widehat{X}$) is then created by summing all of these values.
b) Summing the final weights of records having the characteristic of interest for the denominator ($\widehat{Y}$)
c) Dividing the numerator estimate by the denominator estimate and then calculating the exponential value.

To determine if the arithmetic mean or the geometric mean is a more appropriate measure for a given variable it is best to plot the data with a histogram. If the data has a relatively even distribution of high and low values than the arithmetic mean will be a good indicator of central tendency. In this case, the value of the mean and the median will be very similar. If the data show an uneven spread towards high or low values, then a geometric mean would be more appropriate for the data. In this case, the mean and the median would be quite different.

12.1.2 Imputation of household income

During cycle 6, 83% of CHMS respondents reported their total household income (THI_01).

Non-respondents to total household income were asked a series of questions (THI_Q02 to THI_Q04 that are not included in the data dictionary) in order to determine a household income range. In cycle 6, 77.5% of total household income non-respondents provided a response or partial response for the range; therefore, there is some form of household income information for 96.2% of respondents. A study conducted in 2009 using data from the Canadian Community Health Survey (CCHS), suggested that the income non-respondents had different health characteristics than the respondents[21]. Because of this difference in health profiles, analyses that are based exclusively on income respondents may be biased. As a result, it was decided to impute the total household income for the CHMS cycle 6 release.

To make the imputation as accurate as possible, readily available administration data was used to direct the process. As a first step, all households were linked to household income data from the Socioeconomic Indicators File (SEF). This variable is then used to impute the household income using nearest neighbour imputation. The household income found on the SEF is used as a distance measure to determine which pair of respondent-non-respondent records is the “nearest” within imputation classes. The reported data from the respondent, or donor, is then copied to the non-respondent or recipient. For respondents who provided an income range, a nearest neighbour is selected within the same income range and if possible the same site and household size.

An extra post treatment step was added to cycle 6 to account for an increased number of inconsistencies between the reported personal income and the reported household income. Compared to cycle 4 (the last time income was asked on the questionnaire), there were more cases in cycle 6 where personal income was reported as much higher than the household income. In the

cases identified as having too large a difference, the household income was replaced by the personal income. There were fewer than 50 records where this change occurred.

With the implementation of the imputation process, there is no non-response to total household income (variable THI_01) in cycle 6. An imputation flag THIFIMP4 is included on the file to identify which records have an imputed value for total household income.

It should be noted that total personal income (variable TPI_01) is not imputed. Users should take the appropriate steps when analyzing this variable.

For users interested in conducting analysis focusing specifically on household income, it is highly recommended to rely on other income data sources rather than using the CHMS. Those interested in analysing household income as it relates to health should use derived income variables such as income quintiles rather than the actual income value directly. The imputation process performs well at predicting income quintiles but there is variability in predicting the exact income value.

12.1.3 Other missing data

Missing data are the result of non-response to some or all questions on the survey. Instances of total non-response, where there is no data for a selected respondent, are adjusted for in the calculation of the survey weights. Thus, by using the survey weights to create estimates, the bias that can be introduced by total non-response is reduced. The CHMS data files contain only respondents to the survey; however some variables may be missing for some respondents (partial non-response). There are three main options for analysing a variable with some missing data:

a) Keep the records with missing values in the analysis and report the results for the missing category separately. In this case, careful interpretation of the missing category is necessary.
b) Remove records with missing values from the analysis. This is valid if the missing values are random and do not represent any non-response bias. To test this assumption, compare the respondents with valid data to the respondents with missing data to look for any differences in some key variables. If there are differences, then it is an indication of non-response bias due to eliminating the missing values from the analysis.
c) Impute the missing values to a new value prior to the analysis.

12.1.4 Medication data and derived variables

Derived variables were created for medications containing any of 11 specific ingredients (calcium, folate/folic acid, iodine, omega 3, omega 6, iron, magnesium, vitamin B12, vitamin C, vitamin D, and zinc). The information on the amount of a specific ingredient per one unit of a given product was obtained either from Health Canada's Drug Product Database or from Health Canada's Licensed Natural Health Products Database. This information was used with collection information on the frequency of use for each medication to estimate the average daily intake for each ingredient. The *Derived Variable Specifications: Cycle 6 Medications* document is available, and contains information on how the DVs are created/calculated.

12.2 Guidelines for statistical analysis

12.2.1 Some recommendations for doing analysis with data from the CHMS

The CHMS was designed to provide national baseline prevalence estimates for a variety of health indicators. Due to cost considerations, 16 collection sites from 5 regional strata were chosen; 2 sites from the Atlantic region, 3 sites from the Quebec region, 6 sites from the Ontario region, 3 sites from the Prairies region, and 2 sites from the British Columbia region. Then a sample of individuals of all ages was selected from each chosen site. This design should yield approximately unbiased national prevalence estimates that would have CV's of approximately 16.5% for a prevalence of 10% for each of the 5 age groups (6-11, 12-19, 20-39, 40-59, and 60-79) by sex and for 3-5 year olds of both sexes combined.

While the small number of sampled collection sites can produce national baseline prevalence estimates that meet the above criteria, it has the drawback of leaving at most 11 "degrees of freedom"[15] for variance estimation. Limited degrees of freedom have several consequences for analysis and inference[19]; in particular:

- The variability of variance estimates of estimated quantities needs to be taken into account when doing analyses,
- Estimated covariance matrices of vectors of estimates (such as the vector of estimated coefficients of a model) could be singular or close to singular, thus possibly not invertible[20],
- It may not be possible to calculate some test statistics,
- The usual asymptotic distributions of many test statistics may not hold when there are only a small number of primary sampling units (PSUs), in this case the collection sites in the sample, even when the total sample size is large.

Because of the possible consequences of having a small number of PSUs, a researcher is advised to consider the following recommendations when analyzing CHMS data:

- Produce only national estimates since regional estimates will have even fewer than 11 degrees of freedom because fewer PSUs would be involved in variance estimation. This still allows the researcher to do analyses of many subpopulations such as a single age group or one sex, since all ages and both sexes are in the sample from each collection site. With certain caveats, it is possible to produce estimates at the Ontario or Quebec level by combining data from several cycles[21].
- Avoid fitting models with a large number of coefficients. Limit the number of parameters in a regression model to 10 as one degree of freedom is used to estimate the intercept. A continuous explanatory variable is equivalent to one parameter in the model. The number of parameters used by a categorical variable is the number of categories minus 1. For example, sex has two categories therefore it uses 1 parameter in a regression model.
- Use analytical methods that are less impacted by the limited degrees of freedom, or are conservative. In particular, when testing a hypothesis involving a vector of quantities, Wald statistics and their modifications have been found to be unstable with the exception of the Satterthwaite-adjusted Wald F test. Another choice would be the pseudo-likelihood ratio

test developed by Lumley and Scott (2014)[22]. If neither of these two tests are available, use a set of t-tests, one for each Beta parameter, with a Bonferroni adjustment of p-values.

- As the number of degrees of freedom is at most 11 and may be less for some analyses, consider using a smaller value of alpha to determine if a test is significant. For example, a test could be considered significant if the p-value is less than 0.01 rather than the usual cut-off of 0.05.
- Since bootstrap weights are used for variance estimation with the CHMS, it will likely be necessary to specify the degree of freedom in the software being used because the default degree of freedom is likely to be incorrect.

12.2.2 Data comparability over time

12.2.2.1 Normative scales

To help give some context to the raw data that was collected at the MEC, normative scales were used to create derived variables that assign a category or health rating based on an individual's raw data results compared to the normative scales (e.g. using the calculated body mass index to assign ratings of underweight, normal weight, overweight, or obese). The norms used for cycle 6 data collection were agreed upon at the start of collection, and respondents were informed of how they measured against those norms in the final report delivered to them. In most cases, norms from the previous cycles were used again for cycle 6 in order to maintain as much consistency as possible. As the raw data is also included on the data files, users are free to use norms other than those included on the file.

12.2.2.2 Chemistry and lipid panels

The chemistry and lipid panels were analyzed at Dynacare Laboratories on the Roche Modular in cycles 5 and 6 and at the Health Canada – Nutrition laboratory on the Ortho Clinical Diagnostics Vitros 5,1FS analyzer in previous cycles. There was an instrument change in cycle 6. The first 6 sites were analyzed on a Roche Modular and the remaining sites were analyzed on a Roche Cobas 6000. However the reagents and methodology are the same. The validation data from DYNA determined that while there may be some variability for some analytes, using the Bland-Altman comparisons show that there are no clinical differences between the two instrument methods and that no correction factors would be needed for cycle 6.

NOTE: The only variables not affected by this change are LAB_HBA1 and LAB_CRP even though they were analysed at Dynacare Laboratories.

12.2.2.3 Analytical intervals

The analytical intervals, including the limit of detection (LOD) and the upper limit of the analytical range (ULOD), for certain analytical methods changed between cycles. Although the LOD values do not change by a large margin, the differences should be noted when comparing data from different cycles. A full listing of the analytical ranges including LOD and limits of quantification (LOQ) values is available upon request.

12.2.2.4 Environmental data

Urinary cadmium results from cycles 1 and 2 are not comparable with those from cycle 6 due to a change in analytical reporting based on molybdenum interference. Correction of molybdenum-based interference on cadmium concentrations was performed mathematically using equations derived following the addition of molybdenum to urine samples.

12.2.2.5 Activity monitor data for 3 to 5 year olds

Researchers analyzing activity monitor (accelerometer) data for 3 to 5 year olds are advised to take note of an important methodological difference between cycle 2 and cycles 3, 4, 5 and 6. In cycle 2, all activity monitor data was collected in 60 second epochs and this may not be the optimal approach for 3 to 5 year old children. Data are being collected in shorter epochs (15 seconds) in cycles 3, 4, 5 and 6 (for 3 to 5 year olds only) to align with current research. See Appendix 8 for more information on research being conducted to examine the impact of epoch length on physical activity derived variables as well as assessing adherence to the new preschool physical activity guidelines using accelerometer data.

12.2.2.6 Vitamin D standardization

The methods for analyzing vitamin D changed between cycles. A change in matrix for vitamin D measurement, from plasma to serum, occurred between the cycle 2 and cycle 3 reference periods. Serum has been used in each subsequent cycle. A change in methodology occurred as well between the cycle 4 and cycle 5 reference periods.

Equations are currently being developed to enable data users to adjust all vitamin D values to the international vitamin D standardization program:

Cycle 1 (standardized) = 5.6422 + 0.76091 × Cycle 1 (original)

Cycle 2 (standardized) = 7.22233 + 0.81869 × Cycle 2 (original)

Cycle 3 & Cycle 4 (date yet to be determined)

Cycles 5 & 6 released as standardized.

These equations have not been applied to the original data sets and should be applied by the data user. Unstandardized data should not be compared between cycles. The standard operating procedures (SOPs) are available upon request.

12.2.3 Combining multiple cycles of CHMS data

Combining data from two or more cycles of the CHMS can help researchers gain statistical power, enabling them to analyze and publish high quality results for a greater combination of data variables than they would be able to using data from one cycle alone. Researchers combining data should use the official Statistics Canada combined weight files for the specific combination of cycles being combined (i.e. cycles 1 and 2 OR cycles 2 and 3 OR cycles 1, 2 and 3 OR cycles 3 and 4). An instructions document[21] is available and should be followed closely to determine the file and variable combinations that can be combined and to be aware of any potential analysis limitations. This document will be updated as different cycle 6 files are released.

Using the instructions and correct weights will ensure that the data and methods used in analysing the files are as consistent as possible and that analysis done on data from multiple cycles of the survey is representative of the population.

12.2.4 Software packages available

While many analysis procedures found in statistical packages allow weights to be used, the meaning or definition of the weight in these procedures can differ from what is appropriate in a sample survey framework. As a result, the procedure may produce an estimate that is correct but a variance estimate that is almost meaningless. A software package with procedures for the analysis of survey data should be used instead. A suitable software package for analysing CHMS data should allow the use of sampling weights, bootstrap weights to estimate the variance, and specification of the number of degrees of freedom to use for confidence intervals and significant tests. Examples of available software packages are SUDAAN release 11.0.3 (2018), SAS Institute Inc., version 9.4 (2014), STATA 11.2 and WESVAR 5.1. Table 12.2 gives a comparison of the estimates available in each software package. For more detailed comparison and examples of the code for each program refer to HSMD (2016)[23]. The estimates produced by these software packages may yield slightly different estimates due to differences in the formulas for the standard error.

Table 12.2 Comparison of the procedures available in various software packages

Test	Estimate	Software package			
		SUDAAN 11.0.3	SAS 9.4	STATA 11.2	WESVAR 5.1
Means and Total	Point Estimate	Yes	Yes	Yes	Yes
	Standard Error	Yes	Yes	Yes	Yes
	Confidence Interval	Yes	Yes	Yes	Yes
	Coefficient of Variation	can calculate	Yes	Yes	Yes
Proportions	Point Estimate	Yes	Yes	Yes	Yes
	Standard Error	Yes	Yes	Yes	Yes
	Confidence Interval	Yes	Yes	Yes	Yes
	Coefficient of Variation	can calculate	can calculate	Yes	Yes
Percentiles	Point Estimate	Yes	yes in 9.3 only	Yes	Yes
	Standard Error	Yes	yes in 9.3	No	Yes
	Confidence Interval	Yes	yes in 9.3	No	Yes
	Coefficient of Variation	can calculate	can calculate	No	Yes
Linear Regression	Regression coefficients and standard errors	Yes	Yes	Yes	Yes
	Satterthwaite adjusted statistics	Yes	No	No	No
Logistic Regression	Regression coefficients and standard errors	Yes	Yes	Yes	Yes

12.3 Guidelines for releasing data

12.3.1 Sample size and disclosure control

Before releasing or publishing any estimates from the data files, users must determine the number of sampled respondents having the characteristic of interest. No estimates based on less than 10 respondents can be released to ensure confidentiality of respondents.

Statistics Canada is prohibited by law from releasing any information it collects which could identify any person, business, or organization, unless consent has been given by the respondent or as permitted by the Statistics Act. Various confidentiality rules are applied to all data that are released or published to prevent the publication or disclosure of any information deemed confidential. If necessary, data are suppressed to prevent direct or residual disclosure of identifiable data. The standard table symbol X was used to indicate the data were suppressed to meet the confidentiality requirements of the Statistics Act.

It is Statistics Canada practice to remove personal identifiers from statistical master files when they are no longer required for data processing or other authorized purposes.

12.3.2 Coefficient of variation and confidence intervals

As mentioned in Section 10.2.2, an indicator of the scope of the sampling error as measured by the coefficient of variation (CV) can be produced for each estimate. Table 12.1 gives the Statistics Canada guidelines for releasing estimates based on their CV. Users are strongly advised to adhere to these guidelines, except in the case where estimated proportions approach zero or 100% or if the statistics of interest approach zero. In these instances, the calculated CV is less reliable, therefore confidence intervals would be more helpful.

Table 12.1 Sampling variability guidelines

Type of Estimate	CV (in %)	Guidelines
Acceptable	$0.0 \leq CV \leq 16.6$	Estimates can be considered for general unrestricted release. Requires no special notation.
Marginal	$16.6 < CV \leq 33.3$	Estimates can be considered for general unrestricted release but should be accompanied by a warning cautioning subsequent users of the high sampling variability associated with the estimates. Such estimates should be identified by the letter E (or in some other similar fashion).

Unacceptable	CV > 33.3	Statistics Canada recommends not to release estimates of unacceptable quality. However, if the user chooses to do so then estimates should be flagged with the letter F (or in some other fashion) and the following warning should accompany the estimates: "The user is advised that . . . (specify the data) . . . do not meet Statistics Canada's quality standards for this statistical program. Conclusions based on these data will be unreliable and most likely invalid. These data and any consequent findings should not be published. If the user chooses to publish these data or findings, then this disclaimer must be published with the data."

12.3.3 Rounding guidelines

In order to ensure that estimates for publication or other releases derived from the data files correspond to those produced by Statistics Canada, users are urged to adhere to the following guidelines regarding the rounding of such estimates:

a) Estimates in the main body of a statistical table are to be rounded to the nearest hundred units using the normal rounding technique. In normal rounding, if the first or only digit to be dropped is 0 to 4, the last digit to be retained is not changed. If the first or only digit to be dropped is 5 to 9, the last digit to be retained is raised by one. For example, in normal rounding to the nearest 100, if the last two digits are between 00 and 49, they are changed to 00 and the preceding digit (the hundreds digit) is left unchanged. If the last digits are between 50 and 99 they are changed to 00 and the proceeding digit is increased by 1;

b) Marginal sub-totals and totals in statistical tables are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units using normal rounding;

c) Averages (excluding physical measures or laboratory data), proportions, rates and percentages are to be computed from unrounded components (i.e., numerators and/or denominators) and then are to be rounded themselves to one decimal using normal rounding;

d) Averages from physical measures or laboratory data are to be computed from unrounded components and then are to be rounded themselves to the same number of decimal places as the original data, which reflect the precision of the measuring instrument that was used – in the case of environmental lab data, estimates of average levels (arithmetic or geometric) or percentile distributions should be rounded to two significant digits;

e) Sums and differences of aggregates (or ratios) are to be derived from their corresponding unrounded components and then are to be rounded themselves to the nearest 100 units (or the nearest one decimal) using normal rounding;

f) In instances where, due to technical or other limitations, a rounding technique other than

normal rounding is used, resulting in estimates to be published or otherwise released that differ from corresponding estimates published by Statistics Canada, users are urged to note the reason for such differences in the publication or release document(s);

g) Under no circumstances are unrounded estimates to be published or otherwise released by users. Unrounded estimates imply greater precision than actually exists.

13. References and end notes

1. Mather, Anne. 2020. *Sampling documentation for cycle 6 of the Canadian Health Measures Survey*. Internal document, Statistics Canada, Ottawa, Ontario.

2. Strata are defined according to composition of the household in the 2011 Census, updated with administrative information available at the time of sample selection.

3. Canadian Society for Exercise Physiology. 2013. *Canadian Physical Training for Health (CSEP-PATH)*. Canadian Society for Exercise Physiology (CSEP). Ottawa.

4. National Institutes of Health. 2000. *The Practical Guide to the Identification, Evaluation and Treatment of Overweight and Obesity in Adults.* National Institutes of Health (NIH), Bethesda, MD, USA.

5. Patry-Parisien, Jennifer, Margot Shields and Shirley Bryan. 2012. *Comparison of waist circumference using the World Health Organization and National Institutes of Health protocols*. Health Reports. Vol. 23, no. 3. Statistics Canada Catalogue no. 82-003-XPF.

6. Harv S Vij and Scott Howell. 1998. *Improving the Specific Gravity Adjustment Method for Assessing Urinary Concentrations of Toxic Substances*. American Industrial Hygieve Association Journal. Vol. 59, no. 6, p. 375-380.

7. Xia, Y. et all. 2014. Comparison of creatinine and specific gravity for hydration corrections on measurement of the tobacco-specific nitrosamine 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanol (NNAL) in urine. Journal of clinical laboratory analysis, Vo. 28, no. 5, p. 353-363.

8. United States Environmental Protection Agency (EPA). 2016. *Definition and procedure for the determination of the method detection limit – Revision 2*. Code of Federal Regulation 40, appendix B of part 136, Office of Water, United States Environmental Protection Agency, Washington DC, USA.

9. Pollock, T. et all. 2021. *Trends in environmental chemical concentrations in the Canadian population: Biomonitoring data from the Canadian Health Measures Survey 2007-2017.* Environmental Health Science and Research Bureau, Health Canada, Ottawa, Ontario

10. The five regions are: British Columbia, the Prairies (Alberta, Manitoba, and Saskatchewan), Ontario, Quebec and the Atlantic provinces (Newfoundland and Labrador, Prince Edward Island, Nova Scotia and New Brunswick).

11. The number of sites selected in each region is provided in Table 6.1 in Section 6.3.1 (Sampling of collection sites).

12. Haziza, David, and Jean-François Beaumont. 2007. “On the construction of imputation classes in surveys.” *International Statistical Review*. Vol. 75, no 1. March 2007. p. 25-43.

13. Rao, JNK, CFJ Wu, and Kim Yue. “Some recent work on resampling methods for complex surveys.” *Survey Methodology*. Vol. 18, no 2. December 1992. p. 209-217.

14. Among the dwellings initially selected, some are not within the scope of the survey. These include, for example, vacant or demolished dwellings, non-residential dwellings or dwellings in which all household members are under 3 or over 79 years of age.

15. “Degrees of freedom” is being used as a generic term to reflect the amount of information used to estimate variances and covariance. An approximation often used for the value of “degrees of freedom” is # of PSUs - # of strata. For cycle 6 of the CHMS, the 16 collection sites are the PSUs and there are 5 regions (strata) resulting in 11 degrees of freedom (16-5). This is an approximate estimate of the degrees of freedom and provide only the maximum value.

16. Dion, SM, and S Giroux. 2012. *Cycle 1 of the Canadian Health Measures Survey: Bias Study*. Internal report, Statistics Canada, Ottawa, Ontario.

17. The terms “record” and “variable” are used throughout this document. A record corresponds to a line on the file and represents a respondent. A “variable” corresponds to a column on the file and represents either a question, a measure, a weight or a bootstrap.

18. Sarafin, C. 2009. *CCHS Health Indicators by Income Quintiles*. Internal document, Statistics Canada, Ottawa, Ontario.

19. In particular, confidence intervals and tests of hypotheses.

20. Invertible covariance matrices are needed to perform Wald tests for tests on vectors of parameters.

21. Statistics Canada. 2019. *Instructions for combining multiple cycles of Canadian Health Measures Survey (CHMS) Data*. Internal document, Statistics Canada, Ottawa, Ontario.

22. Lumley, Thomas, and Alastair Scott. 2014. "Tests for Regression Models Fitted to Survey Data." *Australian & New Zealand Journal of Statistics*. Vol. 65, no.1. March 2014. p. 1-14.

23. Household Survey Methods Division (HSMD). 2016. *Comparison of software programs for analysing the CHMS.* Internal document, Statistics Canada, Ottawa, Ontario.

Appendix 1 - Acronyms and Abbreviations

AHFS	American Hospital Formulary Service
ATC	Anatomical Therapeutic Chemical
BMI	Body mass index
BP	Blood pressure
CAI	Computer assisted interviewing
CATI	Computer assisted telephone interview
CBC	Complete blood count
CCHS	Canadian Community Health Survey
CHMS	Canadian Health Measures Survey
CMA	Census Metropolitan Area
CSEP	Canadian Society of Exercise Physiology
CSEP-PATH	CSEP-Canadian Physical Training for Health
CV	Coefficient of variation
DIN	Drug Identification Number
DV	Derived variables
HC	Health Canada
HMS	Health measures specialist
HR	Household response rate
HRGs	Homogeneous response groups
IATA	International Air Transport Association
mCAFT	Modified Aerobic Fitness Test
MEC	Mobile Examination Centre
NAICS	North American Industry Classification System
LOD	Limit of detection
LOQ	Limit of quantification
NHPN	Natural Health Product Number
NIH	National Institutes for Health
NHANES	National Health and Nutrition Examination Survey
NML	National Microbiology Laboratories
NOC-S	National Occupational Classification – Statistics
PHAC	Public Health Agency of Canada
PIA	Privacy Impact Assessment
PSU	Primary sampling units
QA	Quality assurance
QC	Quality control
VOC	Volatile organic compound
RDC	Research Data Centers
SOP	Standard operating procedures
TTY	Teletypewriter (telecommunication device for the hearing impaired)
WHO	World Health Organization

Appendix 2 - List of Canadian Health Measures Survey (CHMS) documents available

Summaries of disseminated products

Plans for dissemination

CHMS Content summary for cycles 1 to 8

- The content summary document is divided into separate tables which list all of the content topics in the survey by age group of respondent. There are tables on the household questionnaire, mobile examination centre (MEC) physical measures, MEC questionnaire, laboratory blood and urine tests, laboratory indoor air sample tests and laboratory tap water sample tests.

CHMS Laboratory analytical ranges for cycles 1 to 6

- The analytical ranges document provides information on analytical ranges (lower limits of detection, limits of quantification, upper analytical ranges) for all laboratory tests. It also provides conversion factors for each analyte. This document is partially available upon request.

CHMS Errata document

- The CHMS Errata document contains a listing of the errors in previously disseminated files. Corrections are provided when possible.

CHMS Data User Guide (each cycle)

- Table of Contents for CHMS Data User Guide.

CHMS Derived Variables (DVs) documentation for cycles 1 to 6

- There are separate DV documents for the following types of DVs: household and mobile examination centre (MEC), medication, activity monitor, hearing, vision, organohalogens in pooled serum, non-environmental laboratory measures, fluoride and volatile organic compounds, and other environmental laboratory measures.

CHMS Data Dictionaries for cycles 1 to 6

- There are separate data dictionaries for each data file. Unrounded data dictionaries are only available to share partners and data users in the Research Data Centers (RDC). Rounded data dictionaries are available upon request.

Instructions for Combining Multiple Cycles of Canadian Health Measures Survey (CHMS) Data

Relationship (DYAD) data files – Documentation (Cycles 1 to 6)

- This document aims to provide recommendations to users of the relationship file. For further information, refer to the document *Using the relationship files and paired respondent data in the Canadian Health Measures Survey: Feasibility study – an update*. The relationship data files - documentation and the feasibility report documents are available upon request.

Supporting documentation for the climate and air quality file (Cycles 1 to 4)

CHMS sampling documentation

Presentations on using CHMS data (Cycles 1 to 5)

- CHMS Data User Workshop

For more information or to obtain copies of the documents in the list above, please contact Statistics Canada's Statistical Information Service (toll-free 1-800-263-1136; 514-283-8300; statcan.infostats-infostats.statcan@canada.ca).

Appendix 3 - List of Collection Sites for Cycle 6

The CHMS collection sites for cycle 6 are as follows:

Ottawa Centre
Lower Sackville / Bedford
Quebec Centre
Richmond Hill
Nanaimo
Mississauga Northwest
Canmore / Banff
Montreal Centre
Vancouver North and West
Edmonton West / St. Albert
Owen Sound
Winnipeg Southwest
Deer Lake / Pasadena
Toronto Centre
Baie-Comeau
London West

Note: As the CHMS was designed to produce national estimates, it is not recommended to do analysis at lower geographic levels as it could result in either extreme sampling variability or unstable estimates of the sampling variability. Possible exceptions are analyses at the provincial level for Ontario or Quebec when data from more than one cycle are combined (more information on combining data from multiple cycles is available upon request).

For more information, please contact Statistics Canada's Statistical Information Service (1-800-263-1136; 514-283-8300; statcan.infostats-infostats.statcan@canada.ca).

Aussi disponible en français.

Statistics Canada Statistique Canada

Canada

Appendix 4 - Pre-testing Guidelines

Pre-testing Guidelines for fasting appointments

GUIDELINES TO FOLLOW BEFORE YOUR APPOINTMENT

❖ During the **12 hours prior** to your clinic appointment:

- do not eat or drink anything other than water (no food, candies, gum, cough lozenges, flavoured water, coffee, alcoholic beverages, etc.)

❖ During the **2 hours prior** to your clinic appointment:

- do not urinate, as you will be asked to **provide a urine sample upon your arrival**
- do not smoke or use other tobacco and nicotine products

❖ On **the day of** your clinic appointment:

- take your medications as usual
- Do not exercise

WHAT TO BRING TO YOUR APPOINTMENT

❖ **All medications** (prescription or over-the-counter), **natural health products**, or **supplements** that you did **not** mention during the household interview

❖ **Inhaler or medication (**if you have a breathing condition; e.g. asthma)

❖ **Clothes and footwear appropriate for exercise** (e.g., short-sleeved top, flat indoor footwear, shorts or loose fitting clothing that can be rolled up above the knee)

❖ **Provincial health insurance card** (health card)

❖ **Information of two contact persons** (name, address and phone number)

❖ **Eyewear prescription** (if you wear eye glasses, contact lenses, or both)

❖ **Eyeglasses you wear for distance viewing** (contact lenses need to be removed one hour prior to your appointment)

❖ **Contact lenses container and solution** (only if you were unable to remove your contact lenses one hour prior to your clinic visit)

Pre-testing Guidelines for non-fasting appointments

GUIDELINES TO FOLLOW BEFORE YOUR APPOINTMENT

❖ During the **6 hours prior** to your clinic appointment:

- do not drink any alcoholic beverages

❖ During the **2 hours prior** to your clinic appointment:

- do not urinate, as you will be asked to **provide a urine sample upon your arrival**
- do not consume caffeinated products (e.g., chocolate, coffee, tea, soft drinks, or energy drinks)
- do not smoke or use other tobacco and nicotine products

❖ On **the day** of your clinic appointment:

- take your medications as usual
- do not exercise

WHAT TO BRING TO YOUR APPOINTMENT

❖ **All medications** (prescription or over-the-counter), **natural health products**, or **supplements** that you did **not** mention during the household interview

❖ **Inhaler or medication** (if you have a breathing condition; e.g. asthma)

❖ **Clothes and footwear appropriate for exercise** (e.g. short-sleeved top, flat indoor footwear, shorts or loose fitting clothing that can be rolled up above the knee)

❖ **Provincial health insurance card** (health card)

❖ **Information of two contact persons** (name, address and phone number)

❖ **Eyewear prescription** (if you wear eye glasses, contact lenses, or both)

❖ **Eyeglasses you wear for distance viewing** (contact lenses need to be removed one hour prior to your appointment)

❖ **Contact lenses container and solution** (only if you were unable to remove your contact lenses one hour prior to your clinic visit).

Appendix 5 - Exclusion Criteria

The exclusion criteria for the physical measures were separated into automatic application exclusions and staff decision exclusions. These can be defined as the following:

Automatic application exclusion criteria (Hard Coded) include exclusions that are made automatically by the application based on previously asked questions in order to prevent the respondent from completing certain physical components (e.g., a respondent who is pregnant is automatically screened out of the waist circumference measurement).

Staff decision exclusion criteria is based on the staff assessing if the respondent's condition will negatively impact the data quality of the physical measure or could compromise the safety of the respondent (e.g., the respondent has difficulty keeping is balance while standing; therefore, the staff decision would be to exclude the respondent from the jumping mechanography and the mCAFT).

Urine Collection 1 & 2

Non-Eligible	Hard Coded	HMS-Coded
N/A	- Respondent is in a wheel-chair and has a catheter	**Absolute Exclusion:** - Respondent has a catheter

Anthropometry

Non-Eligible	Hard Coded	HMS-Coded
Height		
N/A	N/A	- Acute/Chronic condition, respondent unable to stand unassisted (e.g. injury to both legs)
Weight		
N/A	N/A	- Respondent has a fiberglass/plaster cast (cannot be removed for the measurement)
Waist Circumference		
N/A	- Respondent is pregnant	- Respondent is wheelchair-bound and the landmarks cannot be identified correctly - Acute/Chronic condition, respondent unable to stand unassisted (e.g. injury to both legs) - Colostomy bag if it interferes with the measurement
Neck Circumference		
N/A	N/A	- Acute/Chronic condition, respondent unable to perform the test (e.g. tracheotomy)

Blood Pressure

Non-Eligible	Hard Coded	HMS-Coded
- Respondent is less than 6 years old	- Double mastectomy, including lumpectomy	**Absolute Exclusion:** - Presence of either of the following on **both** arms: edema, severe skin condition, gauze dressings, open sores or wounds, casts, paralysis, a-v shunts - Double arm amputee **Right Arm Exclusions:** - Presence of either of the following on the right arm: edema, severe skin condition, gauze dressings, open sores or wounds, cast, paralysis, a-v shunt - Right mastectomy - Right arm amputee **HMS Decision:** Right Arm Exclusions - Presence of either of the following on the right arm: edema, severe skin condition, gauze dressings, open sores or wounds, cast, paralysis, a-v shunt, amputation

Phlebotomy

Non-Eligible	Hard Coded	Phlebotomist-Coded
N/A	- Respondents who have a clotting condition such as hemophilia or von Willebrand's disease - Respondent who had chemotherapy in the past 4 weeks - Respondents who had a double mastectomy	- Presence of either of the following on both arms and no other available venipuncture sites: rashes, gauze dressings, casts, edema, paralysis, tubes, open sores or wounds, withered arms or limbs missing, damaged, sclerosed or occluded veins, allergies to cleansing reagents, burned or scarred tissue, shunt, or IV on both arms.

Activity Monitor

Non-Eligible	Hard Coded	HMS-Coded
N/A	N/A	- Respondent is wheelchair-bound as a result of an acute/chronic condition

Grip strength

Non-Eligible	Hard Coded	HMS-Coded
- Respondent <6 years old	- Respondent has been diagnosed with an aneurysm - Respondent reports currently taking prescribed medications for aneurysm - Respondent has experienced a stroke	- Respondent reported drinking alcohol within the past 6 hours and the perceived level of intoxication would prevent him/her from accurately or safely performing the test - Acute/Chronic condition, respondent unable to perform the test with either hand - Bone or a joint problem that prevents the respondent from performing the test with either hand - Respondent has been told by a health professional that they should not perform certain types of physical activity - Respondent specified another reason for which they should not do physical activity - HMS identifies that respondent is currently taking prescribed medications for aneurysm - Any other reason why the respondent should not perform the test

Modified Canadian Aerobic Fitness Test (mCAFT)

Non-Eligible	Hard Coded	HMS-Coded
- Respondent <8 years old or >69 years old - Appointment is a home visit	- Respondent >12 weeks pregnant - Respondent takes a quick relief medication for a breathing condition and does not have it with them - Respondent feels pain in their chest when taking part in physical activity - Within the last month, respondent has felt pain in their chest when not taking part in physical activity - Within the last year, respondent has lost balance or consciousness during or after exercise or at random - Respondent is over 14 years old and has a high resting systolic blood pressure (> 144 mm Hg), or high resting diastolic blood pressure (> 94 mm Hg), or high resting heart rate (> 99 bpm) - Respondent takes HR or BP altering medication - Respondent has extreme difficulty breathing at rest	- Respondent had a large meal and is unable to perform the test - Respondent reported drinking alcohol within the past 6 hours and the perceived level of intoxication would prevent him/her from accurately or safely performing the test - Acute/Chronic condition, respondent unable to perform the test - If the respondent has lost balance or consciousness under other circumstances, within the last 12 months - Bone or a joint problem that prevents the respondent from performing the test - Respondent has been told by a health professional that they should not perform certain types of physical activity - Respondent specified another reason for which they should not do physical activity - Respondent mentions that they are wearing a pacemaker - Any other reason why the respondent should not perform the test

Sit and Reach

Non-Eligible	Hard Coded	HMS-Coded
- Respondent <6 years old or >69 years old - Appointment is a home visit	- Respondent >12 weeks pregnant	- Respondent reported drinking alcohol within the past 6 hours and the perceived level of intoxication would prevent him/her from accurately or safely performing the test - Acute/Chronic condition, respondent unable to perform the test - Bone or a joint problem that prevents the respondent from performing the test - Respondent has been told by a health professional that they should not perform certain types of physical activity - Respondent specified another reason for which they should not do physical activity - Any other reason why the respondent should not perform the test

Peripheral Quantitative Computed Tomography (pQCT)

Non-Eligible	Hard Coded	HMS-Coded
- Respondent <6 years old - Appointment is a home visit	- Respondent is pregnant - Risk of pregnancy Right/Left and Both Leg Exclusions - Respondent has broken, fractured, cracked or chipped their leg(s) below the knee joint(s), including the ankle(s) in the last 12 months - Respondent has metal in their leg(s) anywhere below the knee(s), including the ankle(s), for example, screws, rods, plates, shrapnel or pellets	- Respondent reported drinking alcohol within the past 6 hours and the perceived level of intoxication would prevent him/her from accurately or safely performing the test - Acute/Chronic condition, respondent unable to perform the test - Any other reason why the respondent should not perform the test

Jumping mechanography

Non-Eligible	Hard Coded	HMS-Coded
- Respondent <6 years old - Appointment is a home visit	- Respondent >12 weeks pregnant - Respondent has been diagnosed with an aneurysm - Respondent reports currently taking prescribed medications for aneurysm - Respondent feels pain in their chest when taking part in physical activity - Within the last month, respondent has felt pain in their chest when not taking part in physical activity - Within the last year, respondent has lost balance or consciousness during or after exercise or at random - Within the last 12 weeks, respondent has had an eye injury, an injection of medicine into an eye, or eye surgery. - Respondent's blood pressure is indicative of severe hypertension, as assessed by the HMS during the Blood Pressure component.	- Respondent reported drinking alcohol within the past 6 hours and the perceived level of intoxication would prevent him/her from accurately or safely performing the test - Acute/Chronic condition, respondent unable to perform the test - Within the last year, the respondent has lost balance or consciousness under other circumstances - Bone or a joint problem that prevents the respondent from performing the test - Respondent has been told by a health professional that they should not perform certain types of physical activity - Respondent specified another reason for which they should not do physical activity - Any other reason why the respondent should not perform the test

Visual Acuity

Non-Eligible	Hard Coded	HMS-Coded
- Respondent <6 years old - Respondent is aged 20-39 years without reported Type 1 or Type 2 diabetes - Appointment is a home visit	Right/Left and Both Eye Exclusions - Within the last 12 weeks, respondent has had an eye injury, an injection of medicine into an eye, or eye surgery. - Within the past year, respondent has received corneal transplant surgery on the eye(s) - Respondent has an/two artificial eye(s) - Respondent is blind in one/both eyes - Respondent has an eye infection	- Respondent reported drinking alcohol within the past 6 hours and the perceived level of intoxication would prevent him/her from accurately or safely performing the test - Acute/Chronic condition, respondent unable to perform the test - Any other reason why the respondent should not perform the test

Intraocular pressure

Non-Eligible	Hard Coded	HMS-Coded
- Respondent <19 years old - Respondent is aged 20-39 years without reported Type 1 or Type 2 diabetes - Appointment is a home visit	Right/Left and Both Eye Exclusions - Within the last 12 weeks, respondent has had an eye injury, an injection of medicine into an eye, or eye surgery. - Within the past year, respondent has received corneal transplant surgery on the eye(s) - Respondent has an/two artificial eye(s) - Respondent is blind in one/both eyes - Respondent has an eye infection	- Respondent reported drinking alcohol within the past 6 hours and the perceived level of intoxication would prevent him/her from accurately or safely performing the test - Acute/Chronic condition, respondent unable to perform the test - Any other reason why the respondent should not perform the test

Appendix 6 - Medication Classification Systems

- **Anatomical Therapeutic Chemical (ATC) classification system:**
  The Anatomical Therapeutic Chemical (ATC) classification system was developed by the World Health Organization (WHO). It classifies pharmaceutical products according to the organ or system on which they act and their chemical, pharmacological, and therapeutic properties. An ATC code was assigned to each medication using the Drug Identification Number (DIN) and Health Canada's ATC coding system. When a medication had more than one indication, the ATC code was decided based on the main indication of the medication. The main indication was determined by Health Canada reviewers using the Product Monograph (a factual document on a drug product that describes the properties, claims, indications, and conditions of use of the drug product). In cases where it was still not clear from the Product Monograph which ATC code to assign, Health Canada contacted the ATC group at the WHO Collaborating Centre for Drug Statistics Methodology in Norway for clarification. Sometimes, a medication was not assigned an ATC code if the DIN was missing or if the DIN did not exist in the Health Canada drug database; for example, with experimental drugs. In these cases the ATC code appears as "not stated" on the file. The classification system is only available in English. For more information on Anatomical Therapeutical Chemical System please refer to: http://www.whocc.no/atc_ddd_index/.

- **American Hospital Formulary Service (AHFS) classification system:**
  The American Hospital Formulary Service (AHFS) classification system is published by the American Society of Health-System Pharmacists to describe the mode of action of pharmaceutical products, including vitamin and mineral supplements. The classification system is only available in English. For more information on the American Hospital Formulary Service (AHFS) classification system, please refer to: http://www.ahfsdruginformation.com/.

Appendix 7 - Response Rates

Appendix 7A - CHMS Cycle 6 Full Sample Response Rates by age group and sex

Sex	Age Group	Households (Hhlds) where one person was selected						Households (Hhlds) where two persons were selected					
		# resp. Hhlds (R1)	# persons selected (PS1)	# respondents to the quest. (Q1)	Hhld quest. response rate (PQ1)	# participants at the MEC (C1)	MEC resp. rate (PC1)	# resp. Hhlds (R2)	# persons selected (PS2)	# respondents to the quest. (Q2)	Hhld quest. response rate (PQ2)	# participants at the MEC (C2)	MEC resp. rate (PC2)
Both Sexes	3 to 5	3	3	3	100	2	66.7	975	975	896	91.9	574	64.1
	6 to 11	3	3	3	100	2	66.7	1,584	1,584	1,435	90.6	1,012	70.5
	12 to 19	1,045	1,045	886	84.8	634	71.6	539	545	491	90.1	369	75.2
	20 to 39	783	783	685	87.5	457	66.7	1,081	1,081	922	85.3	624	67.7
	40 to 59	833	833	757	90.9	539	71.2	870	870	762	87.6	571	74.9
	60 to 79	1,501	1,501	1,374	91.5	962	70	83	83	72	86.7	51	70.8
	3 to 79	**4,168**	**4,168**	**3,708**	**89**	**2,596**	**70**	**2,569**	**5,138**	**4,578**	**89.1**	**3,201**	**69.9**
	6 to 79	**4,165**	**4,165**	**3,705**	**89**	**2,594**	**70**	**2,569**	**4,163**	**3,682**	**88.4**	**2,627**	**71.3**
Males	6 to 11	2	2	2	100	1	50	769	769	711	92.5	506	71.2
	12 to 19	526	526	453	86.1	320	70.6	257	257	232	90.3	176	75.9
	20 to 39	457	457	390	85.3	258	66.2	531	531	440	82.9	304	69.1
	40 to 59	403	403	359	89.1	248	69.1	467	467	401	85.9	304	75.8
	60 to 79	727	727	652	89.7	471	72.2	39	39	34	87.2	27	79.4
	6 to 79	**2,115**	**2,115**	**1,856**	**87.8**	**1,298**	**69.9**	**1,698**	**2,063**	**1,818**	**88.1**	**1,317**	**72.4**
Females	6 to 11	1	1	1	100	1	100	815	815	724	88.8	506	69.9
	12 to 19	519	519	433	83.4	314	72.5	286	288	259	89.9	193	74.5
	20 to 39	326	326	295	90.5	199	67.5	550	550	482	87.6	320	66.4
	40 to 59	430	430	398	92.6	291	73.1	403	403	361	89.6	267	74
	60 to 79	774	774	722	93.3	491	68	44	44	38	86.4	24	63.2
	6 to 79	**2,050**	**2,050**	**1,849**	**90.2**	**1,296**	**70.1**	**1,678**	**2,100**	**1,864**	**88.8**	**1,310**	**70.3**

Appendix 7A - CHMS Cycle 6 Full Sample Response Rates by age group and sex, continued

Sex	Age Group	Overall Combined Response Rate *	Response rate not adjusted for one or two persons selected				
			# persons selected	# respondents to the questionnaire	HHLD questionnaire response rate	# participants at the MEC	MEC response rate
Both Sexes	3 to 5	43.4	978	899	91.9	576	64.1
	6 to 11	47.1	1,587	1,438	90.6	1,014	70.5
	12 to 19	46.5	1,590	1,377	86.6	1,003	72.8
	20 to 39	42.7	1,864	1,607	86.2	1,081	67.3
	40 to 59	48	1,703	1,519	89.2	1,110	73.1
	60 to 79	47.1	1,584	1,446	91.3	1,013	70.1
	3 to 79	**45.9**	**9,306**	**8,286**	**89**	**5,797**	**70**
	6 to 79	**46.1**	**8,328**	**7,387**	**88.7**	**5,221**	**70.7**
Males	6 to 11	48.5	771	713	92.5	507	71.1
	12 to 19	46.7	783	685	87.5	496	72.4
	20 to 39	41.9	988	830	84	562	67.7
	40 to 59	46.8	870	760	87.4	552	72.6
	60 to 79	47.9	766	686	89.6	498	72.6
	6 to 79	**46**	**4,178**	**3,674**	**87.9**	**2,615**	**71.2**
Females	6 to 11	45.8	816	725	88.8	507	69.9
	12 to 19	46.3	807	692	85.7	507	73.3
	20 to 39	43.7	876	777	88.7	519	66.8
	40 to 59	49.4	833	759	91.1	558	73.5
	60 to 79	46.4	818	760	92.9	515	67.8
	6 to 79	**46.3**	**4,150**	**3,713**	**89.5**	**2,606**	**70.2**

* The number of households within the scope of the survey was 9,143. This number cannot be broken down by age group or sex because no information is known about the households that did not respond. The overall household response rate is 73.7% (6,737/9,143). This rate is used to calculate all of the combined response rates by age group and sex.

Appendix 7B - CHMS Combined Cycles 5 and 6 (sites 1 to 10) Vision Subsample Response Rates by age group and sex

Sex	Age group	Full sample		Vision subsample	
		# participants at the MEC	Overall combined response rate (%)	# participants with valid RTP or FDT data	Overall combined response rate (%)
Both Sexes	20 to 39	1,712	44.5	20	44.5
	40 to 59	1,758	49.1	1,711	48.5
	60 to 79	1,655	48.2	1,582	47.5
	20 to 79	**5,125**	**47.2**	**3,313**	**46.5**
Males	20 to 39	858	41.9	11	41.9
	40 to 59	867	47.4	843	46.9
	60 to 79	828	49.6	791	48.9
	20 to 79	**2,553**	**46**	**1,645**	**45.4**
Females	20 to 39	854	47.4	9	47.4
	40 to 59	891	50.9	868	50.2
	60 to 79	827	46.8	791	46.1
	20 to 79	**2,572**	**48.4**	**1,668**	**47.6**

Appendix 7C - CHMS Cycle 6 Blood Draw and Urine Response Rates by age group and sex

Sex	Age Group	Households where one person was selected					Households where two persons were selected					Blood draw combined response rate (BCOMBRR)	Urine combined response rate (UCOMBRR)
		# participants at the MEC (C1)	# persons who provided blood (B1)	Blood draw response rate (BC1)	# persons who provided urine (U1)	Urine response rate (UC1)	# participants at the MEC (C2)	# persons who provided blood (B2)	Blood draw response rate (BC2)	# persons who provided urine (U2)	Urine response rate (UC2)		
Both Sexes	3 to 5	2	2	100	2	100	574	508	88.5	522	90.9	38.4	39.5
	6 to 11	2	2	100	2	100	1012	890	87.9	986	97.4	41.4	45.9
	12 to 19	634	608	95.9	619	97.6	369	333	90.2	358	97	43.6	45.3
	20 to 39	457	447	97.8	450	98.5	624	613	98.2	619	99.2	41.9	42.3
	40 to 59	539	524	97.2	534	99.1	571	561	98.2	564	98.8	46.9	47.5
	60 to 79	962	934	97.1	945	98.2	51	49	96.1	50	98	45.7	46.3
	3 to 79	**2,596**	**2,517**	**97**	**2,552**	**98.3**	**3,201**	**2,954**	**92.3**	**3,099**	**96.8**	**43.7**	**44.9**
	6 to 79	**2,594**	**2,515**	**97**	**2,550**	**98.3**	**2,627**	**2,446**	**93.1**	**2,577**	**98.1**	**44**	**45.3**
Males	6 to 11	1	1	100	1	100	506	453	89.5	498	98.4	43.4	47.7
	12 to 19	320	308	96.3	314	98.1	176	161	91.5	173	98.3	44.1	45.8
	20 to 39	258	254	98.4	257	99.6	304	301	99	303	99.7	41.4	41.8
	40 to 59	248	244	98.4	244	98.4	304	300	98.7	300	98.7	46.1	46.1
	60 to 79	471	464	98.5	464	98.5	27	26	96.3	26	96.3	47.1	47.1
	6 to 79	**1,298**	**1,271**	**97.9**	**1,280**	**98.6**	**1,317**	**1,241**	**94.2**	**1,300**	**98.7**	**44.3**	**45.4**
Females	6 to 11	1	1	100	1	100	506	437	86.4	488	96.4	39.6	44.2
	12 to 19	314	300	95.5	305	97.1	193	172	89.1	185	95.9	43.1	44.7
	20 to 39	199	193	97	193	97	320	312	97.5	316	98.8	42.5	42.8
	40 to 59	291	280	96.2	290	99.7	267	261	97.8	264	98.9	47.9	49
	60 to 79	491	470	95.7	481	98	24	23	95.8	24	100	44.4	45.5
	6 to 79	**1,296**	**1,244**	**96**	**1,270**	**98**	**1,310**	**1,205**	**92**	**1,277**	**97.5**	**43.6**	**45.3**

Appendix 7C: CHMS Cycle 6 Blood Draw and Urine Response Rates by age group and sex, continued

Sex	Age Group	Response rate not adjusted for one or two persons selected				
		# participants at the MEC	# persons who provided blood	Blood draw response rate	# persons who provided urine	Urine response rate
Both Sexes	3 to 5	576	510	88.5	524	91
	6 to 11	1,014	892	88	988	97.4
	12 to 19	1,003	941	93.8	977	97.4
	20 to 39	1,081	1,060	98.1	1,069	98.9
	40 to 59	1,110	1,085	97.7	1,098	98.9
	60 to 79	1,013	983	97	995	98.2
	3 to 79	**5,797**	**5,471**	**94.4**	**5,651**	**97.5**
	6 to 79	**5,221**	**4,961**	**95**	**5,127**	**98.2**
Males	6 to 11	507	454	89.5	499	98.4
	12 to 19	496	469	94.6	487	98.2
	20 to 39	562	555	98.8	560	99.6
	40 to 59	552	544	98.6	544	98.6
	60 to 79	498	490	98.4	490	98.4
	6 to 79	**2,615**	**2,512**	**96.1**	**2,580**	**98.7**
Females	6 to 11	507	438	86.4	489	96.4
	12 to 19	507	472	93.1	490	96.6
	20 to 39	519	505	97.3	509	98.1
	40 to 59	558	541	97	554	99.3
	60 to 79	515	493	95.7	505	98.1
	6 to 79	**2,606**	**2,449**	**94**	**2,547**	**97.7**

Appendix 7D - CHMS Cycle 6 Fasted Subsample Response Rates by age group and sex

Sex	Age Group	Households where 1 person was selected to be fasted								Households where 2 persons were selected to be fasted								Fasted Combined Response Rate
		Numb of resp. Hhlds (RR1)	# pers. select. for fasted subsample (SPS1)	# resp. to the quest. (SQ1)	HHLD quest. resp. rate (SSQ1)	# part. at the MEC (SC1)	MEC resp. rate (SSC1)	# of fasted resp. (SF1)	Fasted resp. rate (SSF1)	Numb of resp. Hhlds (RR2)	# pers. select. for fasted subsample (SPS2)	# resp. to the quest. (SQ2)	HHLD quest. resp. (SSQ2)	# part. at the MEC (SC2)	MEC resp. rate (SSC2)	# of fasted resp. (SF2)	Fasted resp. rate (SSF2)	(FSCOMBRR)
Both Sexes	6 to 11	2	2	2	100.0	2	100.0	2	100.0	951	951	851	89.5	582	68.4	392	67.4	30.5
	12 to 19	687	687	596	86.8	428	71.8	345	80.6	272	276	243	88.0	179	73.7	116	64.8	35.3
	20 to 39	823	823	717	87.1	480	66.9	406	84.6	272	272	219	80.5	154	70.3	129	83.8	36.0
	40 to 59	657	657	589	89.6	409	69.4	347	84.8	381	381	333	87.4	238	71.5	188	79.0	38.0
	60 to 79	894	894	809	90.5	545	67.4	487	89.4	36	36	30	83.3	26	86.7	20	76.9	40.2
	6 to 79	**3,063**	**3,063**	**2,713**	**88.6**	**1,864**	**68.7**	**1,587**	**85.1**	**958**	**1,916**	**1,676**	**87.5**	**1,179**	**70.3**	**845**	**71.7**	**36.8**
Males	6 to 11	1	1	1	100.0	1	100.0	1	100.0	473	473	432	91.3	298	69.0	216	72.5	33.7
	12 to 19	360	360	322	89.4	227	70.5	182	80.2	128	128	111	86.7	82	73.9	52	63.4	35.3
	20 to 39	467	467	395	84.6	261	66.1	225	86.2	119	119	90	75.6	65	72.2	53	81.5	35.0
	40 to 59	336	336	300	89.3	208	69.3	182	87.5	197	197	167	84.8	121	72.5	102	84.3	39.3
	60 to 79	422	422	374	88.6	256	68.4	236	92.2	21	21	17	81.0	16	94.1	14	87.5	41.6
	6 to 79	**1,586**	**1,586**	**1,392**	**87.8**	**953**	**68.5**	**826**	**86.7**	**711**	**938**	**817**	**87.1**	**582**	**71.2**	**437**	**75.1**	**37.1**
Females	6 to 11	1	1	1	100.0	1	100.0	1	100.0	478	478	419	87.7	284	67.8	176	62.0	27.2
	12 to 19	327	327	274	83.8	201	73.4	163	81.1	147	148	132	89.2	97	73.5	64	66.0	35.2
	20 to 39	356	356	322	90.4	219	68.0	181	82.6	153	153	129	84.3	89	69.0	76	85.4	37.2
	40 to 59	321	321	289	90.0	201	69.6	165	82.1	184	184	166	90.2	117	70.5	86	73.5	36.6
	60 to 79	472	472	435	92.2	289	66.4	251	86.9	15	15	13	86.7	10	76.9	6	60.0	38.9
	6 to 79	**1,477**	**1,477**	**1,321**	**89.4**	**911**	**69.0**	**761**	**83.5**	**731**	**978**	**859**	**87.8**	**597**	**69.5**	**408**	**68.3**	**35.6**

Appendix 7D - CHMS Cycle 6 Fasted Subsample Response Rates by age group and sex, continued

Sex	Age Group	Response rate not adjusted for one or two persons selected						
		# persons selected for fasted subsample	# respondents to the questionnaire	HHLD questionnaire response rate	# participants at the MEC	MEC response rate	# fasted respondents	Fasted response rate
Both Sexes	06 to 11	953	853	89.5	584	68.5	394	67.5
	12 to 19	963	839	87.1	607	72.3	461	75.9
	20 to 39	1,095	936	85.5	634	67.7	535	84.4
	40 to 59	1,038	922	88.8	647	70.2	535	82.7
	60 to 79	930	839	90.2	571	68.1	507	88.8
	06 to 79	**4,979**	**4,389**	**88.2**	**3,043**	**69.3**	**2,432**	**79.9**
Males	06 to 11	474	433	91.4	299	69.1	217	72.6
	12 to 19	488	433	88.7	309	71.4	234	75.7
	20 to 39	586	485	82.8	326	67.2	278	85.3
	40 to 59	533	467	87.6	329	70.4	284	86.3
	60 to 79	443	391	88.3	272	69.6	250	91.9
	06 to 79	**2,524**	**2,209**	**87.5**	**1,535**	**69.5**	**1,263**	**82.3**
Females	06 to 11	479	420	87.7	285	67.9	177	62.1
	12 to 19	475	406	85.5	298	73.4	227	76.2
	20 to 39	509	451	88.6	308	68.3	257	83.4
	40 to 59	505	455	90.1	318	69.9	251	78.9
	60 to 79	487	448	92.0	299	66.7	257	86.0
	06 to 79	**2,455**	**2,180**	**88.8**	**1,508**	**69.2**	**1,169**	**77.5**

Appendix 7E - CHMS Cycle 6 Activity Monitor Response Rates by age group and sex

Sex	Age Group	Households where one person was selected			Households where two persons were selected				Response rate not adjusted for one or two persons selected		
		# participants at the MEC offered an activity monitor (OF1)	# pers. who returned the monitor with the required # of days of valid entries (V1)	Activity monitor response rate (AM1)	# participants at the MEC offered an activity monitor (OF2)	# pers. who returned the monitor with the required # of days of valid entries (V2)	Activity monitor response rate (AM2)	Activity monitor combined response rate (AMCBRR)	# MEC respondents offered an activity monitor	# pers. who returned the monitor with the required # of days of valid entries	Activity monitor response rate
Both Sexes	3 to 5	2	2	100.0	571	419	73.4	31.9	573	421	73.5
	6 to 11	2	1	50.0	1,008	782	77.6	36.5	1,010	783	77.5
	12 to 19	633	331	52.3	366	223	60.9	25.8	999	554	55.5
	20 to 39	452	277	61.3	619	458	74.0	29.3	1,071	735	68.6
	40 to 59	538	408	75.8	568	459	80.8	37.7	1,106	867	78.4
	60 to 79	954	656	68.8	51	34	66.7	32.4	1,005	690	68.7
	3 to 79	**2,581**	**1,675**	**64.9**	**3,183**	**2,375**	**74.6**	**31.5**	**5,764**	**4,050**	**70.3**
	6 to 79	**2,579**	**1,673**	**64.9**	**2,612**	**1,956**	**74.9**	**31.7**	**5,191**	**3,629**	**69.9**
Males	6 to 11	1	0	0.0	503	396	78.7	38.1	504	396	78.6
	12 to 19	320	160	50.0	174	99	56.9	24.5	494	259	52.4
	20 to 39	256	153	59.8	301	222	73.8	28.2	557	375	67.3
	40 to 59	248	186	75.0	302	241	79.8	36.3	550	427	77.6
	60 to 79	468	321	68.6	27	18	66.7	32.8	495	339	68.5
	6 to 79	**1,293**	**820**	**63.4**	**1,307**	**976**	**74.7**	**31.6**	**2,600**	**1,796**	**69.1**
Females	6 to 11	1	1	100.0	505	386	76.4	35.0	506	387	76.5
	12 to 19	313	171	54.6	192	124	64.6	27.0	505	295	58.4
	20 to 39	196	124	63.3	318	236	74.2	30.6	514	360	70.0
	40 to 59	290	222	76.6	266	218	82.0	39.1	556	440	79.1
	60 to 79	486	335	68.9	24	16	66.7	31.9	510	351	68.8
	6 to 79	**1,286**	**853**	**66.3**	**1,305**	**980**	**75.1**	**32.5**	**2,591**	**1,833**	**70.7**

Appendix 7F - CHMS combined Cycles 5 and 6 (sites 1 to 11) Jumping Subsample Response Rates by age group and sex

Sex	Age group	Full sample		Jumping subsample	
		# participants at the MEC	Overall combined response rate (%)	# participants with one valid JFM or HFM data	Overall combined response rate (%)
Both Sexes	6 to 11	1,724	48.8	1,591	47.5
	12 to 19	1,753	46.8	1,508	45.8
	20 to 39	1,771	44.2	1,484	42.8
	40 to 59	1,812	48.9	1,487	47.4
	60 to 79	1,713	48.0	1,125	46.2
	6 to 79	**8,773**	**47.1**	**7,195**	**45.7**
Males	6 to 11	858	49.3	786	47.9
	12 to 19	875	47.1	763	46.5
	20 to 39	879	41.6	752	40.1
	40 to 59	896	47.3	758	45.9
	60 to 79	852	49.6	589	47.6
	6 to 79	**4,360**	**46.7**	**3,648**	**45.4**
Females	6 to 11	866	48.3	805	47.1
	12 to 19	878	46.6	745	45.3
	20 to 39	892	47.0	732	45.9
	40 to 59	916	50.5	729	48.9
	60 to 79	861	46.6	536	44.8
	6 to 79	**4,413**	**47.7**	**3,547**	**46.3**

Appendix 7G - CHMS Cycle 6 Acrylamide, Chromium and Perfluoroalkyl Substances (ACP) in Blood Subsample Response Rates by age group and sex

Sex	Age Group	Households where one person was selected				Households where two persons were selected				ACP subsample combined response rate (CHCBRR)
		# participants at the MEC (C1)	# persons selected for the ACP subsample (CHS1)	# persons selected with valid laboratory results (CHR1)	ACP subsample response rate (CH1)	# participants at the MEC (C2)	# persons selected for the ACP subsample (CHS2)	# persons selected with valid laboratory results (CHR2)	ACP subsample response rate (CH2)	
Both Sexes	3 to 5	2	2	2	100.00	574	574	482	83.97	36.47
	6 to 11	2	2	2	100.00	1,012	590	504	85.42	40.23
	12 to 19	634	342	324	94.74	369	204	184	90.20	43.26
	20 to 39	457	139	135	97.12	624	201	196	97.51	41.60
	40 to 59	539	168	165	98.21	571	184	181	98.37	47.21
	60 to 79	962	342	331	96.78	51	14	13	92.86	45.51
	3 to 79	**2,596**	**995**	**959**	**96.38**	**3,201**	**1,767**	**1,560**	**88.29**	**42.82**
	6 to 79	**2,594**	**993**	**957**	**96.37**	**2,627**	**1,193**	**1,078**	**90.36**	**43.38**
Males	6 to 11	1	1	1	100.00	506	296	254	85.81	41.59
	12 to 19	320	167	158	94.61	176	108	98	90.74	43.52
	20 to 39	258	74	71	95.95	304	94	94	100.00	41.13
	40 to 59	248	71	70	98.59	304	102	101	99.02	46.20
	60 to 79	471	164	160	97.56	27	6	5	83.33	46.37
	6 to 79	**1,298**	**477**	**460**	**96.44**	**1,317**	**606**	**552**	**91.09**	**43.27**
Females	6 to 11	1	1	1	100.00	506	294	250	85.03	38.94
	12 to 19	314	175	166	94.86	193	96	86	89.58	42.98
	20 to 39	199	65	64	98.46	320	107	102	95.33	42.14
	40 to 59	291	97	95	97.94	267	82	80	97.56	48.25
	60 to 79	491	178	171	96.07	24	8	8	100.00	44.65
	6 to 79	**1,296**	**516**	**497**	**96.32**	**1,310**	**587**	**526**	**89.61**	**43.21**

Appendix 7G – CHMS Cycle 6 Acrylamide, Chromium and Perfluoroalkyles Substances (ACP) in Blood Subsample Response Rates by age group and sex, continued

Sex	Age Group	Response rate not adjusted for one or two persons selected			
		# participants at the MEC	**# persons selected for the ACP subsample**	**# persons selected with valid laboratory results**	**ACP subsample response rate**
Both Sexes	3 to 5	576	576	484	84.03
	6 to 11	1,014	592	506	85.47
	12 to 19	1,003	546	508	93.04
	20 to 39	1,081	340	331	97.35
	40 to 59	1,110	352	346	98.30
	60 to 79	1,013	356	344	96.63
	3 to 79	**5,797**	**2,762**	**2,519**	**91.20**
	6 to 79	**5,221**	**2,186**	**2,035**	**93.09**
Males	6 to 11	507	297	255	85.86
	12 to 19	496	275	256	93.09
	20 to 39	562	168	165	98.21
	40 to 59	552	173	171	98.84
	60 to 79	498	170	165	97.06
	6 to 79	**2,615**	**1,083**	**1,012**	**93.44**
Females	6 to 11	507	295	251	85.08
	12 to 19	507	271	252	92.99
	20 to 39	519	172	166	96.51
	40 to 59	558	179	175	97.77
	60 to 79	515	186	179	96.24
	6 to 79	**2,606**	**1,103**	**1,023**	**92.75**

Appendix 7H - CHMS Cycle 6 Trace Elements (Cadmium, Lead, Selenium, and Mercuries) in Blood Subsample Response Rates by age group and sex

Sex	Age Group	Households where one person was selected				Households where two persons were selected				Trace elements subsample combined response rate (EUCBRR)
		# participants at the MEC (C1)	# persons selected for the trace elements subsample (SC1)	#persons selected with valid laboratory results (SU1)	Trace elements subsample response rate (SSU1)	# participants at the MEC (C2)	# persons selected for the trace elements subsample (SC2)	#persons selected with valid laboratory results (SU2)	Trace elements subsample response rate (SSU2)	
Both Sexes	3 to 5	2	2	2	100.0	574	574	480	83.6	36.3
	6 to 11	2	2	2	100.0	1,012	590	498	84.4	39.8
	12 to 19	634	342	322	94.2	369	204	183	89.7	43.0
	20 to 39	457	457	445	97.4	624	624	608	97.4	41.6
	40 to 59	539	539	524	97.2	571	571	559	97.9	46.9
	60 to 79	962	962	926	96.3	51	51	48	94.1	45.3
	3 to 79	**2,596**	**2,304**	**2,221**	**96.4**	**3,201**	**2,614**	**2,376**	**90.9**	**43.3**
	6 to 79	**2,594**	**2,302**	**2,219**	**96.4**	**2,627**	**2,040**	**1,896**	**92.9**	**43.8**
Males	6 to 11	1	1	1	100.0	506	296	251	84.8	41.1
	12 to 19	320	167	157	94.0	176	108	97	89.8	43.2
	20 to 39	258	258	252	97.7	304	304	300	98.7	41.2
	40 to 59	248	248	244	98.4	304	304	299	98.4	46.0
	60 to 79	471	471	461	97.9	27	27	26	96.3	46.8
	6 to 79	**1,298**	**1,145**	**1,115**	**97.4**	**1,317**	**1,039**	**973**	**93.6**	**44.0**
Females	6 to 11	1	1	1	100.0	506	294	247	84.0	38.5
	12 to 19	314	175	165	94.3	193	96	86	89.6	42.8
	20 to 39	199	199	193	97.0	320	320	308	96.3	42.1
	40 to 59	291	291	280	96.2	267	267	260	97.4	47.8
	60 to 79	491	491	465	94.7	24	24	22	91.7	43.9
	6 to 79	**1,296**	**1,157**	**1,104**	**95.4**	**1,310**	**1,001**	**923**	**92.2**	**43.5**

Appendix 7H - CHMS Cycle 6 Trace Elements (Cadmium, Lead, Selenium, and Mercuries) in Blood Subsample Response Rates by age group and sex, continued

Sex	Age Group	Response rate not adjusted for one or two persons selected			
		# participants at the MEC	# persons selected for the trace elements subsample	# persons selected with valid laboratory results	Trace elements subsample response rate
Both Sexes	3 to 5	576	576	482	83.7
	6 to 11	1,014	592	500	84.5
	12 to 19	1,003	546	505	92.5
	20 to 39	1,081	1,081	1,053	97.4
	40 to 59	1,110	1,110	1,083	97.6
	60 to 79	1,013	1,013	974	96.2
	3 to 79	**5,797**	**4,918**	**4,597**	**93.5**
	6 to 79	**5,221**	**4,342**	**4,115**	**94.8**
Males	6 to 11	507	297	252	84.8
	12 to 19	496	275	254	92.4
	20 to 39	562	562	552	98.2
	40 to 59	552	552	543	98.4
	60 to 79	498	498	487	97.8
	6 to 79	**2,615**	**2,184**	**2,088**	**95.6**
Females	6 to 11	507	295	248	84.1
	12 to 19	507	271	251	92.6
	20 to 39	519	519	501	96.5
	40 to 59	558	558	540	96.8
	60 to 79	515	515	487	94.6
	6 to 79	**2,606**	**2,158**	**2,027**	**93.9**

Appendix 7I - CHMS Cycle 6 Environmental Urine Main Subsample Response Rates by age group and sex

Sex	Age Group	Households where one person was selected				Households where two persons were selected				Env. urine main subsample combined response rate (EUCBRR)
		# participants at the MEC (C1)	# persons selected for the env. urine main subsample (SC1)	#persons selected with valid laboratory results (SU1)	Env. urine main subsample response rate (SSU1)	# participants at the MEC (C2)	# persons selected for the env. urine main subsample (SC2)	#persons selected with valid laboratory results (SU2)	Env. urine main subsample response rate (SSU2)	
Both Sexes	3 to 5	2	2	2	100.0	574	567	516	91.0	39.5
	6 to 11	2	2	2	100.0	1,012	510	498	97.6	46.0
	12 to 19	634	322	314	97.5	369	197	191	97.0	45.2
	20 to 39	457	138	137	99.3	624	197	195	99.0	42.4
	40 to 59	539	162	161	99.4	571	184	182	98.9	47.6
	60 to 79	962	336	331	98.5	51	14	12	85.7	46.1
	3 to 79	**2,596**	**962**	**947**	**98.4**	**3,201**	**1,669**	**1,594**	**95.5**	**44.7**
	6 to 79	**2,594**	**960**	**945**	**98.4**	**2,627**	**1,102**	**1,078**	**97.8**	**45.3**
Males	6 to 11	1	1	1	100.0	506	252	249	98.8	47.9
	12 to 19	320	152	148	97.4	176	103	103	100.0	45.9
	20 to 39	258	73	73	100.0	304	92	92	100.0	41.9
	40 to 59	248	71	70	98.6	304	104	102	98.1	46.0
	60 to 79	471	163	162	99.4	27	6	5	83.3	47.2
	6 to 79	**1,298**	**460**	**454**	**98.7**	**1,317**	**557**	**551**	**98.9**	**45.5**
Females	6 to 11	1	1	1	100.0	506	258	249	96.5	44.2
	12 to 19	314	170	166	97.6	193	94	88	93.6	44.5
	20 to 39	199	65	64	98.5	320	105	103	98.1	42.9
	40 to 59	291	91	91	100.0	267	80	80	100.0	49.4
	60 to 79	491	173	169	97.7	24	8	7	87.5	45.1
	6 to 79	**1,296**	**500**	**491**	**98.2**	**1,310**	**545**	**527**	**96.7**	**45.2**

Appendix 7I - CHMS Cycle 6 Environmental Urine Main Subsample Response Rates by age group and sex, continued

Sex	Age Group	Response rate not adjusted for one or two persons selected			
		# participants at the MEC	# persons selected for the env. urine main subsample	# persons selected with valid laboratory results	Env. urine main subsample response rate
Both Sexes	3 to 5	576	569	518	91.0
	6 to 11	1,014	512	500	97.7
	12 to 19	1,003	519	505	97.3
	20 to 39	1,081	335	332	99.1
	40 to 59	1,110	346	343	99.1
	60 to 79	1,013	350	343	98.0
	3 to 79	**5,797**	**2,631**	**2,541**	**96.6**
	6 to 79	**5,221**	**2,062**	**2,023**	**98.1**
Males	6 to 11	507	253	250	98.8
	12 to 19	496	255	251	98.4
	20 to 39	562	165	165	100.0
	40 to 59	552	175	172	98.3
	60 to 79	498	169	167	98.8
	6 to 79	**2,615**	**1,017**	**1,005**	**98.8**
Females	6 to 11	507	259	250	96.5
	12 to 19	507	264	254	96.2
	20 to 39	519	170	167	98.2
	40 to 59	558	171	171	100.0
	60 to 79	515	181	176	97.2
	6 to 79	**2,606**	**1,045**	**1,018**	**97.4**

Appendix 7J - CHMS Cycle 6 Tobacco Metabolites Subsample Response Rates by age group and sex

Sex	Age Group	Households where one person was selected					Households where two persons were selected					Tobacco subsample combined response rate (TOBCRR)
		Number of questionnaire respondents (Q1)	Number of respondents at the MEC (C1)	Number of respondents selected for the subsample (TOBS1)	Number of persons who provided urine (with valid data) (TOBR1)	Tobacco subsample response rate (with valid data) (TOB1)	Number of questionnaire respondents (Q2)	Number of respondents at the MEC (C2)	Number of respondents selected for the subsample (TOBS2)	Number of persons who provided urine (with valid data) (TOBR2)	Tobacco subsample response rate (with valid data) (TOB2)	
Both Sexes	6 to 11	3	2	2	2	100.0	1,435	1,012	513	497	96.9	45.6
	12 to 19	886	634	619	604	97.6	491	369	357	346	96.9	45.2
	20 to 39	685	457	246	241	98.0	922	624	303	301	99.3	42.2
	40 to 59	757	539	244	242	99.2	762	571	214	211	98.6	47.5
	60 to 79	1,374	962	327	322	98.5	72	51	14	14	100.0	46.4
	6 to 79	**3,705**	**2,594**	**1,438**	**1,411**	**98.1**	**3,682**	**2,627**	**1,401**	**1,369**	**97.7**	**45.2**
Males	6 to 11	2	1	1	1	100.0	711	506	250	247	98.8	47.9
	12 to 19	453	320	314	308	98.1	232	176	168	165	98.2	45.8
	20 to 39	390	258	157	156	99.4	440	304	165	165	100.0	41.8
	40 to 59	359	248	131	129	98.5	401	304	129	127	98.4	46.0
	60 to 79	652	471	159	157	98.7	34	27	8	8	100.0	47.3
	6 to 79	**1,856**	**1,298**	**762**	**751**	**98.6**	**1,818**	**1,317**	**720**	**712**	**98.9**	**45.4**
Females	6 to 11	1	1	1	1	100.0	724	506	263	250	95.1	43.5
	12 to 19	433	314	305	296	97.0	259	193	189	181	95.8	44.7
	20 to 39	295	199	89	85	95.5	482	320	138	136	98.6	42.5
	40 to 59	398	291	113	113	100.0	361	267	85	84	98.8	49.1
	60 to 79	722	491	168	165	98.2	38	24	6	6	100.0	45.6
	6 to 79	**1,849**	**1,296**	**676**	**660**	**97.6**	**1,864**	**1,310**	**681**	**657**	**96.5**	**45.0**

Appendix 7J - CHMS Cycle 6 Tobacco Metabolites Subsample Response Rates by age group and sex, continued

Sex	Age Group	Response rates not adjusted for one or two persons				
		Number of questionnaire respondents	Number of respondents at the MEC	Number of respondents selected for the subsample	Number of persons who provided urine (with valid data)	Urine tobacco subsample response rate (with valid data)
Both Sexes	6 to 11	1,438	1,014	515	499	96.9
	12 to 19	1,377	1,003	976	950	97.3
	20 to 39	1,607	1,081	549	542	98.7
	40 to 59	1,519	1,110	458	453	98.9
	60 to 79	1,446	1,013	341	336	98.5
	6 to 79	**7,387**	**5,221**	**2,839**	**2,780**	**97.9**
Males	6 to 11	713	507	251	248	98.8
	12 to 19	685	496	482	473	98.1
	20 to 39	830	562	322	321	99.7
	40 to 59	760	552	260	256	98.5
	60 to 79	686	498	167	165	98.8
	6 to 79	**3,674**	**2,615**	**1,482**	**1,463**	**98.7**
Females	6 to 11	725	507	264	251	95.1
	12 to 19	692	507	494	477	96.6
	20 to 39	777	519	227	221	97.4
	40 to 59	759	558	198	197	99.5
	60 to 79	760	515	174	171	98.3
	6 to 79	**3,713**	**2,606**	**1,357**	**1,317**	**97.1**

Appendix 8 - Activity Monitor Research

In cycle 2 of the Canadian Health Measures Survey (CHMS), activity monitor data was collected in 60 second epochs for all respondents. Emerging research has demonstrated that epochs shorter than 60 seconds may more accurately capture the sporadic and intermittent physical activity that is typical in preschool aged children. This led the CHMS to collect activity monitor data in 15 second epochs for 3 to 5 year olds in cycles 3,4,5, and 6. In a comparative study using two Actical accelerometers (one collecting data in 15-seconds epoch and the other one in 60-seconds epoch) in children aged 3 to 5 (Colley et al. Health Reports 2014), it was shown that counts per minute and counts per 15-seconds epoch were highly correlated leading to correction equations. These equations can be used to compare the results and/or to combine the data from both cycles. However, the study also showed that the correlation was low for step counts collected in different epoch length. Therefore it is not recommended to compare steps per day in cycle 2 with cycles 3,4,5, or 6 nor to combine the step data.

Further research is also needed to understand how to assess the new preschool physical activity guidelines using accelerometer data. The new guidelines state that "preschoolers should accumulate at least 180 minutes spent in a variety of physical activities spread throughout the day, of which at least 60 minutes is energetic play" (Tremblay, et al., 2017). Intensity cut-points have been published for moderate and vigorous physical activity in this age group for the Actical accelerometer; however, more work is needed to understand what the bottom cut-off should be (i.e. the transition between sedentary and light). Another gap is the lack of a daily step count target for this age group. Based on CHMS data from cycle 1, a daily step target of 12,000 steps was published in 2012 for children and youth aged 6 to 19 (Colley, Janssen et al., MSSE 2012). Similar analyses need to be done for 3 to 5 year old children to find an appropriate daily step target.

References (Activity Monitor Research):

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Tremblay, et al. 2017. "Canadian 24-Hour Movement Guidelines for the Early Years (0-4 years): An Integration of Physical Activity, Sedentary Behavior, and Sleep". *BMC Public Health*. Vol. 17(Suppl 5):874.